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### Methods for forming three-dimensional memory devices

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#### Abstract

A method for forming a three-dimensional (3D) memory device is disclosed. An array of NAND memory strings is formed on a first substrate. A first semiconductor layer is formed above the array of NAND memory strings. The first semiconductor layer includes single crystalline silicon. A first transistor is formed on the first semiconductor layer. A second semiconductor layer is formed above the first transistor. The second semiconductor layer includes single crystalline silicon. A second transistor is formed on the second semiconductor layer.

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**Inventors:** Yang; Yuancheng (Wuhan, CN), Zhou; Wenxi (Wuhan, CN), Xia; Zhiliang (Wuhan, CN), Liu; Wei (Wuhan, CN)

**Applicant:** YANGTZE MEMORY TECHNOLOGIES CO., LTD. (Wuhan, CN)

**Family ID:** 84689828

**Assignee:** YANGTZE MEMORY TECHNOLOGIES CO., LTD. (Wuhan, CN)

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*Primary Examiner:* Dang; Phuc T

*Attorney, Agent or Firm:* BAYES PLLC

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/480,821 filed on Sep. 21, 2021, issued as U.S. Pat. No. 12,020,750, which is a continuation of International Application No. PCT/CN2021/103610, filed on Jun. 30, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” both of which are hereby incorporated by reference in their entireties. This application is also related to U.S. application Ser. No. 17/480,852, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/480,897, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/480,931, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/480,949, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/480,975, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/480,998, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/481,020, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/481,040, filed on Sep. 21, 2021, entitled “THREE-DIMENSIONAL

MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” all of which are hereby incorporated by reference in their entireties.

## BACKGROUND

(1) The present disclosure relates to memory devices and fabrication methods thereof.

(2) Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit.

(3) A three-dimensional (3D) memory architecture can address the density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral circuits for facilitating operations of the memory array.

## SUMMARY

(4) In one aspect, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes an array of memory cells and a first semiconductor layer in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a first peripheral circuit of the array of memory cells including a first transistor, and a second semiconductor layer in contact with the first transistor. A third semiconductor structure includes a second peripheral circuit of the array of memory cells including a second transistor, and a third semiconductor layer in contact with the second transistor. The second semiconductor layer is between the first bonding interface and the first peripheral circuit. The third semiconductor layer is between the second bonding interface and the second peripheral circuit.

(5) In another aspect, a system includes a memory device configured to store data. The memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes an array of memory cells and a first semiconductor layer in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a first peripheral circuit of the array of memory cells including a first transistor, and a second semiconductor layer in contact with the first transistor. A third semiconductor structure includes a second peripheral circuit of the array of memory cells including a second transistor, and a third semiconductor layer in contact with the second transistor. The second semiconductor layer is between the first bonding interface and the first peripheral circuit. The third semiconductor layer is between the second bonding interface and the second peripheral circuit. The system also includes a memory controller coupled to the memory device and configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit.

(6) In still another aspect, a method for forming a 3D memory device is disclosed. An array of NAND memory strings is formed on a first substrate. A first semiconductor layer is formed above the array of NAND memory strings. The first semiconductor layer includes single crystalline silicon. A first transistor is formed on the first semiconductor layer. A second semiconductor layer is formed above the first transistor. The second semiconductor layer includes single crystalline silicon. A second transistor is formed on the second semiconductor layer.

(7) In yet another aspect, a method for forming a 3D memory device is disclosed. An array of NAND memory strings is formed on a first substrate. A first transistor is formed on a second substrate. A second transistor is formed on a third substrate. The first substrate and second substrate

are bonded in a face-to-back manner. The second substrate and the third substrate are bonded in a face-to-back manner.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.
- (2) FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device, according to some aspects of the present disclosure.
- (3) FIG. 1B illustrates a schematic view of a cross-section of another 3D memory device, according to some aspects of the present disclosure.
- (4) FIG. 1C illustrates a schematic view of a cross-section of still another 3D memory device, according to some aspects of the present disclosure.
- (5) FIG. 1D illustrates a schematic view of a cross-section of yet another 3D memory device, according to some aspects of the present disclosure.
- (6) FIG. 2 illustrates a schematic circuit diagram of a memory device including peripheral circuits, according to some aspects of the present disclosure.
- (7) FIG. 3 illustrates a block diagram of a memory device including a memory cell array and peripheral circuits, according to some aspects of the present disclosure.
- (8) FIG. 4A illustrates a block diagram of peripheral circuits provided with various voltages, according to some aspects of the present disclosure.
- (9) FIG. 4B illustrates a schematic diagram of peripheral circuits provided with various voltages arranged in separate semiconductor structures, according to some aspects of the present disclosure.
- (10) FIGS. 5A and 5B illustrate a perspective view and a side view, respectively, of a planar transistor, according to some aspects of the present disclosure.
- (11) FIGS. 6A and 6B illustrate a perspective view and a side view, respectively, of a 3D transistor, according to some aspects of the present disclosure.
- (12) FIG. 7 illustrates a circuit diagram of a word line driver and a page buffer, according to some aspects of the present disclosure.
- (13) FIGS. 8A-8C illustrate side views of various NAND memory strings in 3D memory devices, according to various aspects of the present disclosure.
- (14) FIGS. 9A and 9B illustrate schematic views of cross-sections of 3D memory devices having three stacked semiconductor structures, according to various aspects of the present disclosure.
- (15) FIGS. 10A and 10B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 9A and 9B, according to various aspects of the present disclosure.
- (16) FIGS. 11A-11C illustrate side views of various examples of the 3D memory devices in FIGS. 10A and 10B, according to various aspects of the present disclosure.
- (17) FIGS. 12A-12H illustrate a fabrication process for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure.
- (18) FIGS. 13A-13H illustrate another fabrication process for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure.
- (19) FIG. 14 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure.
- (20) FIG. 15 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure.
- (21) FIGS. 16A and 16B illustrate schematic views of cross-sections of the 3D memory devices in

FIGS. 9A and 9B, according to various aspects of the present disclosure.

(22) FIGS. 17A-17C illustrate side views of various examples of the 3D memory devices in FIGS. 16A and 16B, according to various aspects of the present disclosure.

(23) FIGS. 18A-18F illustrate a fabrication process for forming the 3D memory devices in FIGS. 16A and 16B, according to some aspects of the present disclosure.

(24) FIGS. 19A-19F illustrate another fabrication process for forming the 3D memory devices in FIGS. 16A and 16B, according to some aspects of the present disclosure.

(25) FIG. 20 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 16A and 16B, according to some aspects of the present disclosure.

(26) FIG. 21 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 16A and 16B, according to some aspects of the present disclosure.

(27) FIGS. 22A and 22B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 9A and 9B, according to various aspects of the present disclosure.

(28) FIGS. 23A-23C illustrate side views of various examples of the 3D memory devices in FIGS. 16A and 16B, according to various aspects of the present disclosure.

(29) FIGS. 24A-24F illustrate a fabrication process for forming the 3D memory devices in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(30) FIGS. 25A-25F illustrate another fabrication process for forming the 3D memory devices in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(31) FIG. 26 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(32) FIG. 27 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(33) FIGS. 28A and 28B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 9A and 9B, according to various aspects of the present disclosure.

(34) FIGS. 29A and 29B illustrate side views of various examples of the 3D memory devices in FIGS. 28A and 28B, according to various aspects of the present disclosure.

(35) FIGS. 30A-30F illustrate a fabrication process for forming the 3D memory devices in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(36) FIGS. 31A-31F illustrate another fabrication process for forming the 3D memory devices in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(37) FIG. 32 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(38) FIG. 33 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(39) FIGS. 34A and 34B illustrate schematic views of cross-sections of 3D memory devices having three stacked semiconductor structures, according to various aspects of the present disclosure.

(40) FIGS. 35A and 35B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 34A and 34B, according to some aspects of the present disclosure.

(41) FIGS. 36A and 36B illustrate side views of various examples of the 3D memory devices in FIGS. 35A and 35B, according to various aspects of the present disclosure.

(42) FIGS. 37A-37G illustrate a fabrication process for forming the 3D memory device in FIGS. 35A and 35B, according to some aspects of the present disclosure.

(43) FIG. 38 illustrates a flowchart of a method for forming the 3D memory device in FIGS. 35A and 35B, according to some aspects of the present disclosure.

(44) FIGS. 39A and 39B illustrate schematic views of cross-sections of 3D memory devices having two stacked semiconductor structures, according to various aspects of the present disclosure.

(45) FIGS. 40A and 40B illustrate side views of various examples of the 3D memory devices in FIGS. 39A and 39B, according to various aspects of the present disclosure.

(46) FIGS. 41A-41E illustrate a fabrication process for forming the 3D memory devices in FIGS.

**39A and 39B**, according to some aspects of the present disclosure.

(47) FIGS. **42A-42I** illustrate another fabrication process for forming the 3D memory devices in FIGS. **39A** and **39B**, according to some aspects of the present disclosure.

(48) FIG. **43** illustrates a flowchart of a method for forming the 3D memory devices in FIGS. **39A** and **39B**, according to some aspects of the present disclosure.

(49) FIGS. **44A** and **44B** illustrate schematic views of cross-sections of 3D memory devices having two stacked semiconductor structures, according to some aspects of the present disclosure.

(50) FIGS. **45A** and **45B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **44A** and **44B**, according to some various of the present disclosure.

(51) FIGS. **46A-46G** illustrate a fabrication process for forming the 3D memory devices in FIGS. **44A** and **44B**, according to some aspects of the present disclosure.

(52) FIG. **47** illustrates a flowchart of a method for forming the 3D memory devices in FIGS. **44A** and **44B**, according to some aspects of the present disclosure.

(53) FIGS. **48A-48D** illustrate a fabrication process of transfer bonding, according to some aspects of the present disclosure.

(54) FIGS. **49A-49D** illustrate another fabrication process of transfer bonding, according to some aspects of the present disclosure.

(55) FIG. **50** illustrates a block diagram of an exemplary system having a memory device, according to some aspects of the present disclosure.

(56) FIG. **51A** illustrates a diagram of an exemplary memory card having a memory device, according to some aspects of the present disclosure.

(57) FIG. **51B** illustrates a diagram of an exemplary solid-state drive (SSD) having a memory device, according to some aspects of the present disclosure.

(58) The present disclosure will be described with reference to the accompanying drawings.

#### DETAILED DESCRIPTION

(59) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the present disclosure can also be employed in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, and modified with one another and in ways not specifically depicted in the drawings, such that these combinations, adjustments, and modifications are within the scope of the present disclosure.

(60) In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures, or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(61) It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

(62) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are



intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(63) As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layers thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductors and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

(64) With the development of 3D memory devices, such as 3D NAND Flash memory devices, the more stacked layers (e.g., more word lines and the resulting more memory cells) require more peripheral circuits (and the components, e.g., transistors, forming the peripheral circuits) for operating the 3D memory devices. For example, the number and/or size of page buffers needs to increase to match the increased number of memory cells. In another example, the number of string drivers in the word line driver is proportional to the number of word lines in the 3D NAND Flash memory. Thus, the continuous increase of the word lines also increases the area occupied by the word line driver, as well as the complexity of metal routings, sometimes even the number of metal layers. Moreover, in some 3D memory devices in which the memory cell array and peripheral circuits are fabricated on different substrates and bonded together, the continuous increase of peripheral circuits' areas makes it the bottleneck for reducing the total chip size since the memory cell array can be scaled up vertically by increasing the number of levels instead of increasing the planar size.

(65) Thus, it is desirable to reduce the planar areas occupied by the peripheral circuits of the 3D memory devices with the increased numbers of peripheral circuits and the transistors thereof. However, scaling down the transistor size of the peripheral circuits following the advanced complementary metal-oxide-semiconductor (CMOS) technology node trend used for the logic devices would cause a significant cost increase and higher leakage current, which are undesirable for memory devices. Moreover, because the 3D NAND Flash memory devices require a relatively high voltage (e.g., above 5 V) in certain memory operations, such as program and erase, unlike logic devices, which can reduce its working voltage as the CMOS technology node advances, the voltage provided to the memory peripheral circuits cannot be reduced. As a result, scaling down the memory peripheral circuit sizes by following the trend for advancing the CMOS technology nodes, like the normal logic devices, becomes infeasible.

(66) To address one or more of the aforementioned issues, the present disclosure introduces various solutions in which the peripheral circuits of a memory device are disposed in different planes (levels, tiers) in the vertical direction, i.e., stacked over one another, to reduce the planar chip size of the peripheral circuits, as well as the total chip size of the memory device. In some implementations, the memory cell array (e.g., NAND memory strings), the memory peripheral circuits provided with a relatively high voltage (e.g., above 5 V), and the memory peripheral circuits provided with a relatively low voltage (e.g., below 1.3 V) are disposed in different planes in the vertical direction, i.e., stacked over one another, to further reduce the chip size. The 3D memory device architectures and fabrication processes disclosed in the present disclosure can be easily scaled up vertically to stack more peripheral circuits in different planes to further reduce the chip size.

(67) The peripheral circuits can be separated into different planes in the vertical direction based on different performance requirements, for example, the voltages applied to the transistors thereof, which affect the dimensions of the transistors (e.g., gate dielectric thickness), dimensions of the substrates in which the transistors are formed (e.g., substrate thickness), and thermal budgets (e.g., the interconnect material). Thus, peripheral circuits with different dimension requirements (e.g., gate dielectric thickness and substrate thickness) and thermal budgets can be fabricated in different processes to reduce the design and process constraints from each other, thereby improving the device performance and fabrication complexity.

(68) According to some aspects of the present disclosure, the memory cell array and various peripheral circuits with different performance and dimension requirements can be fabricated in parallel on different substrates and then stacked over one another using various joining technologies, such as hybrid bonding, transfer bonding, etc. As a result, the fabrication cycle of the memory device can be further reduced. Moreover, since the thermal budgets of the different devices become independent to each other, interconnect materials with desirable electric performance but low thermal budget, such as copper, can be used in interconnecting the memory cells and transistors of the peripheral circuits, thereby further improving the device performance. Bonding technologies can introduce additional benefits as well. In some implementations, hybrid bonding in a face-to-face manner achieves millions of parallel short interconnects between the bonded semiconductor structures to increase the throughput and input/output (I/O) speed of the memory devices. In some implementations, transfer bonding re-uses a single wafer to transfer thin semiconductor layers thereof onto different memory devices for forming transistors thereon, which can reduce the cost of the memory devices.

(69) The 3D memory device architectures and fabrication processes disclosed in the present disclosure have the flexibility to allow various substrate materials suitable for different memory cell array designs, such as NAND memory strings suitable for gate-induced drain leakage (GIDL) erase operations or P-type bulk erase operations. In some implementations, single crystalline silicon (a.k.a. single-crystal silicon or monocrystalline silicon) with superior carrier electronic properties—the lack of grain boundaries allows better charge carrier flow and prevents electron recombination—is used as the substrate material of the NAND memory string array to achieve faster memory operations. In some implementations, polysilicon (a.k.a. polycrystalline silicon) is used as the substrate material of the NAND memory string array for GIDL erase operations.

(70) The 3D memory device architectures and fabrication processes disclosed in the present disclosure also have the flexibility to allow various device pad-out schemes to meet different needs and different designs of the memory cell array. In some implementations, the pad-out interconnect layer is formed from the side of the semiconductor structure that has the peripheral circuits to shorten the interconnect distance between the pad-out interconnect layer and the transistors of the peripheral circuits to reduce the parasitic capacitance from the interconnects and improve the electric performance. In some implementations, the pad-out interconnect layer is formed on a thinned substrate in which the memory cell array is formed to enable inter-layer vias (LLVs, e.g., submicron-level) for pad-out interconnects with high I/O throughput and low fabrication complicity.

(71) FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device **100**, according to some aspects of the present disclosure. 3D memory device **100** represents an example of a bonded chip. In some implementations, at least some of the components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) are formed separately on different substrates in parallel and then jointed to form a bonded chip (a process referred to herein as a “parallel process”). In some implementations, at least one semiconductor layer is attached onto another semiconductor structure using transferring bonding, then some of the components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) are formed on the attached semiconductor layer (a process referred to herein as a “series process”). It is understood that in

some examples, the components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) may be formed by a hybrid process that combines the parallel process and the series process.

(72) It is noted that x- and y-axes are added in FIG. **1A** to further illustrate the spatial relationships of the components of a semiconductor device. A substrate of a semiconductor device, e.g., 3D memory device **100**, includes two lateral surfaces (e.g., a top surface and a bottom surface) extending laterally in the x-direction (the lateral direction or width direction). As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of a semiconductor device is determined relative to the substrate of the semiconductor device in the y-direction (the vertical direction or thickness direction) when the substrate is positioned in the lowest plane of the semiconductor device in the y-direction. The same notion for describing the spatial relationships is applied throughout the present disclosure.

(73) 3D memory device **100** can include a first semiconductor structure **102** including an array of memory cells (also referred to herein as a “memory cell array”). In some implementations, the memory cell array includes an array of NAND Flash memory cells. For case of description, a NAND Flash memory cell array may be used as an example for describing the memory cell array in the present disclosure. But it is understood that the memory cell array is not limited to NAND Flash memory cell array and may include any other suitable types of memory cell arrays, such as NOR Flash memory cell array, phase change memory (PCM) cell array, resistive memory cell array, magnetic memory cell array, spin transfer torque (STT) memory cell array, to name a few.

(74) First semiconductor structure **102** can be a NAND Flash memory device in which memory cells are provided in the form of an array of 3D NAND memory strings and/or an array of two-dimensional (2D) NAND memory cells. NAND memory cells can be organized into pages or fingers, which are then organized into blocks in which each NAND memory cell is coupled to a separate line called a bit line (BL). All cells with the same vertical position in the NAND memory cell can be coupled through the control gates by a word line (WL). In some implementations, a memory plane contains a certain number of blocks that are coupled through the same bit line. First semiconductor structure **102** can include one or more memory planes, and the peripheral circuits that are needed to perform all the read/program (write)/erase operations can be included in a second semiconductor structure **104** and a third semiconductor structure **106**.

(75) In some implementations, the array of NAND memory cells is an array of 2D NAND memory cells, each of which includes a floating-gate transistor. The array of 2D NAND memory cells includes a plurality of 2D NAND memory strings, each of which includes a plurality of memory cells connected in series (resembling a NAND gate) and two select transistors, according to some implementations. Each 2D NAND memory string is arranged in the same plane (i.e., referring to herein a flat, two-dimensional (2D) surface, different from the term “memory plane” in the present discourse) on the substrate, according to some implementations. In some implementations, the array of NAND memory cells is an array of 3D NAND memory strings, each of which extends vertically above the substrate (in 3D) through a stack structure, e.g., a memory stack. Depending on the 3D NAND technology (e.g., the number of layers/tiers in the memory stack), a 3D NAND memory string typically includes a certain number of NAND memory cells, each of which includes a floating-gate transistor or a charge-trap transistor.

(76) As shown in FIG. **1A**, 3D memory device **100** can also include a second semiconductor structure **104** and a third semiconductor structure **106** each including some of the peripheral circuits of the memory cell array in first semiconductor structure **102**. That is, the peripheral circuits of the memory cell array can be separated into at least two other semiconductor structures (e.g., **104** and **106** in FIG. **1A**). The peripheral circuits (a.k.a. control and sensing circuits) can include any suitable digital, analog, and/or mixed-signal circuits used for facilitating the operations of the memory cell array. For example, the peripheral circuits can include one or more of a page buffer, a decoder (e.g., a row decoder and a column decoder), a sense amplifier, a driver (e.g., a word line

driver), an I/O circuit, a charge pump, a voltage source or generator, a current or voltage reference, any portions (e.g., a sub-circuit) of the functional circuits mentioned above, or any active or passive components of the circuit (e.g., transistors, diodes, resistors, or capacitors). The peripheral circuits in second and third semiconductor structures **104** and **106** can use CMOS technology, e.g., which can be implemented with logic processes in any suitable technology nodes.

(77) As shown in FIG. 1A, first, second, and third semiconductor structures **102**, **104**, and **106** are stacked over one another in different planes, according to some implementations. As a result, the memory cell array in first semiconductor structure **102**, the peripheral circuits in second semiconductor structure **104**, and the peripheral circuits in third semiconductor structure **106** can be stacked over one another in different planes to reduce the planar size of 3D memory device **100**, compared with memory devices in which all the peripheral circuits are disposed in the same plane.

(78) As shown in FIG. 1A, 3D memory device **100** further includes a first bonding interface **103** vertically between first semiconductor structure **102** and second semiconductor structure **104**, as well as a second bonding interface **105** vertically between second semiconductor structure **104** and third semiconductor structure **106**. First and second bonding interface **103** or **105** can be an interface between two semiconductor structures formed by any suitable bonding technologies as described below in detail, such as hybrid bonding, anodic bonding, fusion bonding, transfer bonding, adhesive bonding, eutectic bonding, to name a few. In some implementations as shown in FIG. 1A, second semiconductor structure **104** is bonded to other two semiconductor structures **102** and **106** on opposite sides thereof. That is, second semiconductor structure **104** can be vertically between first and third semiconductor structures **102** and **106**.

(79) In some implementations, each of second and third semiconductor structures **104** and **106** does not include any memory cell. In other words, each of second and third semiconductor structures **104** and **106** only includes peripheral circuits, but not the memory cell array, according to some implementations. As a result, the memory cell array can be only included in first semiconductor structure **102**, but not second or third semiconductor structure **104** or **106**. Further, the number of semiconductor structures including peripheral circuits can be different from the number of semiconductor structures including memory cell array. In some implementations, the number of semiconductor structures including peripheral circuits is larger than the number of semiconductor structures including memory cell array. For example, as shown in FIG. 1A, the number of semiconductor structures including peripheral circuits is 2 (i.e., **104** and **106**), while the number of semiconductor structures including memory cell array is 1 (i.e., **102**).

(80) It is understood that the relative positions of stacked first, second, and third semiconductor structures **102**, **104**, and **106** are not limited and may vary in different examples. FIG. 1B illustrates a schematic view of a cross-section of another exemplary 3D memory device **101**, according to some implementations. Different from 3D memory device **100** in FIG. 1A in which second semiconductor structure **104** including some of the peripheral circuits is vertically between first semiconductor structure **102** including the memory cell array and third semiconductor structure **106** including some of the peripheral circuits, in 3D memory device **101** in FIG. 1B, first semiconductor structure **102** including the memory cell array is between second and third semiconductor structures **104** and **106** each including some of the peripheral circuits. Nevertheless, first bonding interface **103** can still be formed vertically between first and second semiconductor structures **102** and **104** in 3D memory device **101**. Instead of having a second bonding interface **105** vertically between second and third semiconductor structures **104** and **106**, 3D memory device **100** can include a third bonding interface **107** vertically between first and third semiconductor structures **102** and **106**. Similar to first and second bonding interfaces **103** and **105**, third bonding interface **107** can be an interface between two semiconductor structures formed by any suitable bonding technologies as described below in detail, such as hybrid bonding, anodic bonding, fusion bonding, transfer bonding, adhesive bonding, eutectic bonding, to name a few. In some implementations as shown in FIG. 1B, first semiconductor structure **102** is bonded to other two

semiconductor structures **104** and **106** on opposite sides thereof.

(81) As described below in detail, some or all of first, second, and third semiconductor structures **102**, **104**, and **106** can be fabricated separately (and in parallel in some implementations) by the parallel process, such that the thermal budget of fabricating one of first, second, and third semiconductor structures **102**, **104**, and **106** does not limit the processes of fabricating another one of first, second, and third semiconductor structures **102**, **104**, and **106**. Moreover, a large number of interconnects (e.g., bonding contacts and/or inter-layer vias (ILVs)/through substrate vias (TSVs)) can be formed across bonding interfaces **103**, **105**, and **107** to make direct, short-distance (e.g., micron- or submicron-level) electrical connections between adjacent semiconductor structures **102**, **104**, and **106**, as opposed to the long-distance (e.g., millimeter or centimeter-level) chip-to-chip data bus on the circuit board, such as printed circuit board (PCB), thereby eliminating chip interface delay and achieving high-speed I/O throughput with reduced power consumption. Data transfer among the memory cell array and the different peripheral circuits in different semiconductor structures **102**, **104**, and **106** can be performed through the interconnects (e.g., bonding contacts and/or ILVs/TSVs) across bonding interfaces **103**, **105**, and **107**. By vertically integrating first, second, and third semiconductor structures **102**, **104**, and **106**, the chip size can be reduced, and the memory cell density can be increased.

(82) It is also understood that the number of bonding interfaces in a 3D memory device is not limited and may vary in different examples. FIG. 1C illustrates a schematic view of a cross-section of still another exemplary 3D memory device **120**, according to some implementations. Similar to 3D memory devices **100** and **101**, the memory cell array and at least two portions of the peripheral circuits can be stacked over one another in different planes in 3D memory device **120**. However, different from 3D memory devices **100** and **101** that include two bonding interfaces **103** and **105** or **103** and **107**, 3D memory device **120** includes a single bonding interface **109** vertically between first semiconductor structure **102** in which the memory array is disposed and a fourth semiconductor structure **108** in which the two separate portions of the peripheral circuits are disposed, according to some implementations. That is, the two vertically separated portions of the peripheral circuits are not separated by bonding interface(s) as a result of a bonding process, but instead, are disposed on opposite sides of a same semiconductor layer **112** (e.g., a thinned silicon substrate) in fourth semiconductor structure **108**. Depending on the thickness of semiconductor layer **112**, interconnects (e.g., ILVs in the submicron-level or TSVs in the micron- or tens micron-level) can be formed through semiconductor layer **112** to make direct, short-distance (e.g., submicron- to tens micron-levels) electrical connections between the different portions of the peripheral circuits on opposite sides of semiconductor layer **112** in fourth semiconductor structure **108**.

(83) It is further understood that the types of devices disposed on opposite sides of semiconductor layer **112** are not limited and may vary in different examples. FIG. 1D illustrates a schematic view of a cross-section of yet another exemplary 3D memory device **121**, according to some implementations. Similar to 3D memory devices **100**, **101**, and **120**, the memory cell array and at least two portions of the peripheral circuits can be stacked over one another in different planes in 3D memory device **121**. Different from 3D memory device **120** in FIG. 1C in which both peripheral circuits are formed on opposite sides of semiconductor layer **112**, in 3D memory device **121**, the memory cell array and some of the peripheral circuits are formed on opposite sides of semiconductor layer **112** in a fifth semiconductor structure **110**. That is, 3D memory device **121** can include a single bonding interface **111** vertically between second semiconductor structure **104** (or third semiconductor structure **106**) having some of the peripheral circuits and fifth semiconductor structure **110** in which the memory cell array and some of the peripheral circuits are disposed, according to some implementations. Similar to 3D memory device **120**, depending on the thickness of semiconductor layer **112**, interconnects (e.g., ILVs in the submicron-level or TSVs in the micron- or tens micron-level) can be formed through semiconductor layer **112** to make direct,

short-distance (e.g., submicron- to tens micron-levels) electrical connections between some of the peripheral circuits and the memory cell array on opposite sides of semiconductor layer **112** in fifth semiconductor structure **110**. It is understood that the numbers of stacked semiconductor structures in 3D memory devices **100**, **101**, **120**, and **121** are not limited by the examples shown in FIGS. **1A-1D**, and additional semiconductor structure(s) may be further stacked above, below, or between semiconductor structures shown in FIGS. **1A-1D** in the vertical direction.

(84) FIG. **2** illustrates a schematic circuit diagram of a memory device **200** including peripheral circuits, according to some aspects of the present disclosure. Memory device **200** can include a memory cell array **201** and peripheral circuits **202** coupled to memory cell array **201**. 3D memory devices **100**, **101**, **120**, and **121** may be examples of memory device **200** in which memory cell array **201** and at least two portions of peripheral circuits **202** may be included in various stacked semiconductor structures **102**, **104**, **106**, **108**, and **110**. Memory cell array **201** can be a NAND Flash memory cell array in which memory cells **206** are provided in the form of an array of NAND memory strings **208** each extending vertically above a substrate (not shown). In some implementations, each NAND memory string **208** includes a plurality of memory cells **206** coupled in series and stacked vertically. Each memory cell **206** can hold a continuous, analog value, such as an electrical voltage or charge, that depends on the number of electrons trapped within a region of memory cell **206**. Each memory cell **206** can be either a floating gate type of memory cell including a floating-gate transistor or a charge trap type of memory cell including a charge-trap transistor.

(85) In some implementations, each memory cell **206** is a single-level cell (SLC) that has two possible memory states and thus, can store one bit of data. For example, the first memory state “0” can correspond to a first range of voltages, and the second memory state “1” can correspond to a second range of voltages. In some implementations, each memory cell **206** is a multi-level cell (MLC) that is capable of storing more than a single bit of data in more than four memory states. For example, the MLC can store two bits per cell, three bits per cell (also known as triple-level cell (TLC)), or four bits per cell (also known as a quad-level cell (QLC)). Each MLC can be programmed to assume a range of possible nominal storage values. In one example, if each MLC stores two bits of data, then the MLC can be programmed to assume one of three possible programming levels from an erased state by writing one of three possible nominal storage values to the cell. A fourth nominal storage value can be used for the erased state.

(86) As shown in FIG. **2**, each NAND memory string **208** can include a source select gate (SSG) transistor **210** at its source end and a drain select gate (DSG) transistor **212** at its drain end. SSG transistor **210** and DSG transistor **212** can be configured to activate selected NAND memory strings **208** (columns of the array) during read and program operations. In some implementations, SSG transistors **210** of NAND memory strings **208** in the same block **204** are coupled through a same source line (SL) **214**, e.g., a common SL, for example, to the ground. DSG transistor **212** of each NAND memory string **208** is coupled to a respective bit line **216** from which data can be read or programmed via an output bus (not shown), according to some implementations. In some implementations, each NAND memory string **208** is configured to be selected or deselected by applying a select voltage (e.g., above the threshold voltage of DSG transistor **212**) or a deselect voltage (e.g., 0 V) to respective DSG transistor **212** through one or more DSG lines **213** and/or by applying a select voltage (e.g., above the threshold voltage of SSG transistor **210**) or a deselect voltage (e.g., 0 V) to respective SSG transistor **210** through one or more SSG lines **215**.

(87) As shown in FIG. **2**, NAND memory strings **208** can be organized into multiple blocks **204**, each of which can have a common source line **214**. In some implementations, each block **204** is the basic data unit for erase operations, i.e., all memory cells **206** on the same block **204** are erased at the same time. Memory cells **206** of adjacent NAND memory strings **208** can be coupled through word lines **218** that select which row of memory cells **206** is affected by read and program operations. In some implementations, each word line **218** is coupled to a page **220** of memory cells

**206**, which is the basic data unit for program and read operations. The size of one page **220** in bits can correspond to the number of NAND memory strings **208** coupled by word line **218** in one block **204**. Each word line **218** can include a plurality of control gates (gate electrodes) at each memory cell **206** in respective page **220** and a gate line coupling the control gates.

(88) FIGS. **8A-8C** illustrate side views of various NAND memory strings **208** in 3D memory devices, according to various aspects of the present disclosure. As shown in FIG. **8A**, NAND memory string **208** can extend vertically through a memory stack **804** above a substrate **802**. Substrate **802** can be a semiconductor layer including silicon (e.g., single crystalline silicon, c-silicon), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), or any other suitable semiconductor materials. In some implementations, substrate **802** includes single crystalline silicon.

(89) Memory stack **804** can include interleaved gate conductive layers **806** and dielectric layers **808**. The number of the pairs of gate conductive layers **806** and dielectric layers **808** in memory stack **804** can determine the number of memory cells **206** in memory cell array **201**. Gate conductive layer **806** can include conductive materials including, but not limited to, tungsten (W), cobalt (Co), copper (Cu), aluminum (Al), polysilicon, doped silicon, silicides, or any combination thereof. In some implementations, each gate conductive layer **806** includes a metal layer, such as a tungsten layer. In some implementations, each gate conductive layer **806** includes a doped polysilicon layer. Each gate conductive layer **806** can include control gates surrounding the memory cells, the gates of DSG transistors **212**, or the gates of SSG transistors **210**, and can extend laterally as DSG line **213** at the top of memory stack **804**, SSG line **215** at the bottom of memory stack **804**, or word line **218** between DSG line **213** and SSG line **215**.

(90) As shown in FIG. **8A**, NAND memory string **208** includes a channel structure **812A** extending vertically through memory stack **804**. In some implementations, channel structure **812A** includes a channel hole filled with semiconductor material(s) (e.g., as a semiconductor channel **820**) and dielectric material(s) (e.g., as a memory film **818**). In some implementations, semiconductor channel **820** includes silicon, such as polysilicon. In some implementations, memory film **818** is a composite dielectric layer including a tunneling layer **826**, a storage layer **824** (also known as a “charge trap/storage layer”), and a blocking layer **822**. Channel structure **812A** can have a cylinder shape (e.g., a pillar shape). Semiconductor channel **820**, tunneling layer **826**, storage layer **824**, blocking layer **822** are arranged radially from the center toward the outer surface of the pillar in this order, according to some implementations. Tunneling layer **826** can include silicon oxide, silicon oxynitride, or any combination thereof. Storage layer **824** can include silicon nitride, silicon oxynitride, silicon, or any combination thereof. Blocking layer **822** can include silicon oxide, silicon oxynitride, high dielectric constant (high-k) dielectrics, or any combination thereof. In one example, memory film **818** may include a composite layer of silicon oxide/silicon oxynitride/silicon oxide (ONO). Channel structure **812A** can further include a channel plug **816** on the drain end of NAND memory string **208**. Channel plug **816** can include polysilicon and be in contact with semiconductor channel **820**.

(91) As shown in FIG. **8A**, NAND memory string **208** can further include a semiconductor plug **814** on the source end thereof, which is in contact with semiconductor channel **820** of channel structure **812A**. Semiconductor plug **814**, also known as selective epitaxial growth (SEG), can be selectively grown from substrate **802** and thus, has the same material as substrate **802**, such as single crystalline silicon. Channel structure **812A** in contact with semiconductor plug **814** on the source end of NAND memory string **208** (e.g., at the bottom of NAND memory string **208** shown in FIG. **8A**, a.k.a. a bottom plug) is referred to herein as a “bottom plug channel structure” **812A**.

(92) As shown in FIG. **8A**, a slit structure **828A** can extend vertically through memory stack **804** and be in contact with substrate **802**. Slit structure **828A** can include a source contact **830** having conductive materials, such as polysilicon, metals, metal compounds (e.g., titanium nitride (TiN), tantalum nitride (TaN), etc.), or silicides, as well as a well **832** (e.g., a P-well and/or an N-well) in

substrate **802**. In some implementations, source contact **830** and well **832** of slit structure **828A**, part of substrate **802** between slit structure **828A** and channel structure **812A**, and semiconductor plug **814** function as parts of source line **214** coupled to the source of NAND memory string **208**, for example, for applying an erase voltage to the source of NAND memory string **208** during erase operations.

(93) Different from bottom plug channel structure **812A** in FIG. **8A**, as shown in FIG. **8B**, NAND memory string **208** includes a sidewall plug channel structure **812B** and is free of semiconductor plug **814** on the source end thereof, according to some implementations. Instead, a sidewall semiconductor layer **803** vertically between substrate **802** and memory stack **804** can be in contact with the sidewall of semiconductor channel **820** of channel structures **812B**. Sidewall semiconductor layer **803** can include semiconductor materials, such as polysilicon. Also different from slit structure **828A** in FIG. **8A**, as shown in FIG. **8B**, a slit structure **828B** does not include well **832**, and source contact **830** of slit structure **828B** is in contact with sidewall semiconductor layer **803**, according to some implementations. In some implementations, source contact **830** of slit structure **828B** and sidewall semiconductor layer **803** collectively function as parts of source line **214** coupled to the source of NAND memory string **208**, for example, for applying an erase voltage to the source of NAND memory string **208** during erase operations.

(94) As shown in FIG. **8C**, in some implementations, substrate **802** (e.g., having single crystalline silicon) is replaced with a semiconductor layer **805** in contact with semiconductor channel **820** of a bottom open channel structure **812C** on the source end of NAND memory string **208**. Parts of memory film **818** of channel structure **812C** on the source end can be removed to expose semiconductor channel **820** to contact semiconductor layer **805**. In some implementations, part of semiconductor channel **820** on the source end of NAND memory string **208** is doped to form a doped region **834** that is in contact with semiconductor layer **805**. Semiconductor layer **805** can include semiconductor materials, such as polysilicon. In some implementations, semiconductor layer **805** includes N-type doped polysilicon to enable GILD erase operations. Also different from slit structures **828A** and **828B** in FIGS. **8A** and **8B**, as shown in FIG. **8C**, a slit structure **828C** does not include source contact **830** and thus, does not function as part of source line **214**, according to some implementations. Instead, source contacts (not shown) may be formed on an opposite side of semiconductor layer **805** with respect to channel structure **812C**, such that the source contacts and parts of semiconductor layer **805** may function as parts of source line **214** coupled to the source of NAND memory string **208**, for example, for applying an erase voltage to the source of NAND memory string **208** during erase operations.

(95) Referring to FIG. **2**, peripheral circuits **202** can be coupled to memory cell array **201** through bit lines **216**, word lines **218**, source lines **214**, SSG lines **215**, and DSG lines **213**. As described above, peripheral circuits **202** can include any suitable circuits for facilitating the operations of memory cell array **201** by applying and sensing voltage signals and/or current signals through bit lines **216** to and from each target memory cell **206** through word lines **218**, source lines **214**, SSG lines **215**, and DSG lines **213**. Peripheral circuits **202** can include various types of peripheral circuits formed using CMOS technologies. For example, FIG. **3** illustrates some exemplary peripheral circuits **202** including a page buffer **304**, a column decoder/bit line driver **306**, a row decoder/word line driver **308**, a voltage generator **310**, control logic **312**, registers **314**, an interface (I/F) **316**, and a data bus **318**. It is understood that in some examples, additional peripheral circuits **202** may be included as well.

(96) Page buffer **304** can be configured to buffer data read from or programmed to memory cell array **201** according to the control signals of control logic **312**. In one example, page buffer **304** may store one page of program data (write data) to be programmed into one page **220** of memory cell array **201**. In another example, page buffer **304** also performs program verify operations to ensure that the data has been properly programmed into memory cells **206** coupled to selected word lines **218**.



- (97) Row decoder/word line driver **308** can be configured to be controlled by control logic **312** and select block **204** of memory cell array **201** and a word line **218** of selected block **204**. Row decoder/word line driver **308** can be further configured to drive memory cell array **201**. For example, row decoder/word line driver **308** may drive memory cells **206** coupled to the selected word line **218** using a word line voltage generated from voltage generator **310**.
- (98) Column decoder/bit line driver **306** can be configured to be controlled by control logic **312** and select one or more 3D NAND memory strings **208** by applying bit line voltages generated from voltage generator **310**. For example, column decoder/bit line driver **306** may apply column signals for selecting a set of N bits of data from page buffer **304** to be outputted in a read operation.
- (99) Control logic **312** can be coupled to each peripheral circuit **202** and configured to control operations of peripheral circuits **202**. Registers **314** can be coupled to control logic **312** and include status registers, command registers, and address registers for storing status information, command operation codes (OP codes), and command addresses for controlling the operations of each peripheral circuit **202**.
- (100) Interface **316** can be coupled to control logic **312** and configured to interface memory cell array **201** with a memory controller (not shown). In some implementations, interface **316** acts as a control buffer to buffer and relay control commands received from the memory controller and/or a host (not shown) to control logic **312** and status information received from control logic **312** to the memory controller and/or the host. Interface **316** can also be coupled to page buffer **304** and column decoder/bit line driver **306** via data bus **318** and act as an I/O interface and a data buffer to buffer and relay the program data received from the memory controller and/or the host to page buffer **304** and the read data from page buffer **304** to the memory controller and/or the host. In some implementations, interface **316** and data bus **318** are parts of an I/O circuit of peripheral circuits **202**.
- (101) Voltage generator **310** can be configured to be controlled by control logic **312** and generate the word line voltages (e.g., read voltage, program voltage, pass voltage, local voltage, and verification voltage) and the bit line voltages to be supplied to memory cell array **201**. In some implementations, voltage generator **310** is part of a voltage source that provides voltages at various levels of different peripheral circuits **202** as described below in detail. Consistent with the scope of the present disclosure, in some implementations, the voltages provided by voltage generator **310**, for example, to row decoder/word line driver **308**, column decoder/bit line driver **306**, and page buffer **304** are above certain levels that are sufficient to perform the memory operations. For example, the voltages provided to the page buffer circuits in page buffer **304** and/or the logic circuits in control logic **312** may be between 1.3 V and 5 V, such as 3.3 V, and the voltages provided to the driving circuits in row decoder/word line driver **308** and/or column decoder/bit line driver **306** may be between 5 V and 30 V.
- (102) Different from logic devices (e.g., microprocessors), memory devices, such as 3D NAND Flash memory, require a wide range of voltages to be supplied to different memory peripheral circuits. For example, FIG. 4A illustrates a block diagram of peripheral circuits provided with various voltages, according to some aspects of the present disclosure. In some implementations, a memory device (e.g., memory device **200**) includes a low low voltage (LLV) source **401**, a low voltage (LV) source **403**, and a high voltage (HV) source **405**, each of which is configured to provide a voltage at a respective level (Vdd1, Vdd2, or Vdd3). For example,  $V_{dd3} > V_{dd2} > V_{dd1}$ . Each voltage source **401**, **403**, or **405** can receive a voltage input at a suitable level from an external power source (e.g., a battery). Each voltage source **401**, **403**, or **405** can also include voltage converters and/or voltage regulators to convert the external voltage input to the respective level (Vdd1, Vdd2, or Vdd3) and maintain and output the voltage at the respective level (Vdd1, Vdd2, or Vdd3) through a corresponding power rail. In some implementations, voltage generator **310** of memory device **200** is part of voltage sources **401**, **403**, and **405**.
- (103) In some implementations, LLV source **401** is configured to provide a voltage below 1.3 V,

such as between 0.9 V and 1.2 V (e.g., 0.9 V, 0.95 V, 1 V, 1.05 V, 1.1 V, 1.15 V, 1.2 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). In one example, the voltage is 1.2 V. In some implementations, LV source **403** is configured to provide a voltage between 1.3 V and 3.3 V (e.g., 1.3 V, 0.1.4 V, 1.5 V, 1.6 V, 1.7 V, 1.8 V, 1.9 V, 2 V, 2.1 V, 2.2 V, 2.3 V, 2.4 V, 2.5 V, 2.6 V, 2.7 V, 2.8 V, 2.9 V, 3 V, 3.1 V, 3.2 V, 3.3 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). In one example, the voltage is 3.3 V. In some implementations, HV source **405** is configured to provide a voltage greater than 3.3 V, such as between 5 V and 30 V (e.g., 5 V, 6 V, 7 V, 8 V, 9 V, 10 V, 11 V, 12 V, 13 V, 14 V, 15 V, 16 V, 17 V, 18 V, 19 V, 20 V, 21 V, 22 V, 23 V, 24 V, 25 V, 26 V, 27 V, 28 V, 29 V, 30 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the voltage ranges described above with respect to HV source **405**, LV source **403**, and LLV source **401** are for illustrative purposes and non-limiting, and any other suitable voltage ranges may be provided by HV source **405**, LV source **403**, and LLV source **401**.

(104) Based on their suitable voltage levels (Vdd1, Vdd2, or Vdd3), the memory peripheral circuits (e.g., peripheral circuits **202**) can be categories into LLV circuits **402**, LV circuits **404**, and HV circuits **406**, which can be coupled to LLV source **401**, LV source **403**, and HV source **405**, respectively. In some implementations, HV circuits **406** includes one or more driving circuits that are coupled to the memory cell array (e.g., memory cell array **201**) through word lines, bit lines, SSG lines, DSG lines, source lines, etc., and configured to drive the memory cell array by applying a voltage at a suitable level to the word lines, bit lines, SSG lines, DSG lines, source lines, etc., when performing memory operations (e.g., read, program, or erase). In one example, HV circuit **406** may include word line driving circuits (e.g., in row decoder/word line driver **308**) that are coupled to word lines and apply a program voltage (Vprog) or a pass voltage (Vpass) in the range of, for example, 5 V and 30 V, to the word lines during program operations. In another example, HV circuit **406** may include bit line driving circuits (e.g., in column decoder/bit line driver **306**) that are coupled to bit lines and apply an erase voltage (Veras) in the range of, for example, 5 V and 30 V, to bit lines during erase operations. In some implementations, LV circuits **404** include page buffer circuits (e.g., in latches of page buffer **304**) and are configured to buffer the data read from or programmed to the memory cell array. For example, the page buffer may be provided with a voltage of, for example, 3.3 V, by LV source **403**. LV circuits **404** can also include logic circuits (e.g., in control logic **312**). In some implementations, LLV circuits **402** include an I/O circuit (e.g., in interface **316** and/or data bus **318**) configured to interface the memory cell array with a memory controller. For example, the I/O circuit may be provided with a voltage of, for example, 1.2 V, by LLV source **401**.

(105) As described above, to reduce the total area occupied by the memory peripheral circuits, peripheral circuits **202** can be separately formed in different planes based on different performance requirements, such as the applied voltages. For example, FIG. 4B illustrates a schematic diagram of peripheral circuits provided with various voltages arranged in separate semiconductor structures, according to some aspects of the present disclosure. In some implementations, LLV circuits **402** and HV circuits **406** are separated, for example, in semiconductor structures **408** and **410**, respectively, due to their significant difference in voltages and the resulting difference in device dimensions, such as different semiconductor layer (e.g., substrate or thinned substrate) thicknesses and different gate dielectric thicknesses. In one example, the thickness of the semiconductor layer (e.g., a substrate or a thinned substrate) in which HV circuits **406** are formed in semiconductor structure **410** may be larger than the thickness of the semiconductor layer (e.g., a substrate or a thinned substrate) in which LLV circuits **402** are formed in semiconductor structure **408**. In another example, the thickness of the gate dielectric of transistors forming HV circuits **406** may be larger than the thickness of the gate dielectric of transistors forming LLV circuits **402**. For example, the thickness difference may be at least 5-fold. It is understood that stacked LLV circuits **402** and HV

circuits **406** in different planes may be formed in two semiconductor structure **408** or **410** separated by bonding interface(s) (e.g., in FIGS. **1A** and **1B**) or on opposite sides of a semiconductor layer (e.g., in FIGS. **1C** and **1D**).

(106) LV circuits **404** can be formed in either semiconductor structure **408** or **410**, or in another semiconductor, i.e., in the same plane as LLV circuits **402** or HV circuits **406**, or a different plane from LLV circuits **402** and HV circuits **406**. As shown in FIG. **4B**, in some implementations, some of LV circuits **404** are formed in semiconductor structure **408**, i.e., in the same plane as LLV circuits **402**, while some of LV circuits **404** are formed in semiconductor structure **410**, i.e., in the same plane as HV circuits **406**. That is, LV circuits **404** can be separated into different planes as well. The thickness of the gate dielectric of transistors forming LV circuits **404** in semiconductor structure **408** can be the same as the thickness of the gate dielectric of transistors forming LV circuits **404** in semiconductor structure **410**, for example, when the same voltage is applied to LV circuits **404** in different semiconductor structures **408** and **410**. In some implementations, the same voltage is applied to both LV circuits **404** in semiconductor structure **408** and the LV circuits **404** in semiconductor structure **410**, such that the voltage applied to HV circuits **406** in semiconductor structure **410** is higher than the voltage applied to LV circuits **404** in semiconductor structure **408** or **410**, which is in turn higher than the voltage applied to LLV circuits **402** in semiconductor structure **408**. Moreover, since the voltage applied to LV circuits **404** is between the voltages applied to HV circuits **406** and LLV circuits **402**, the thickness of the gate dielectric of transistors forming LV circuits **404** is between the thickness of the gate dielectric of transistors forming HV circuits **406** and the thickness of the gate dielectric of transistors forming LLV circuits **402**, according to some implementations. For example, the gate dielectric thickness of transistors forming LV circuits **404** may be larger than the gate dielectric thickness of transistors forming LLV circuits **402**, but smaller than the gate dielectric thickness of transistors forming HV circuits **406**.

(107) Based on the different performance requirements (e.g., associated with different applied voltages), peripheral circuits **202** can be separated into at least two stacked semiconductor structures **408** and **410** in different planes. In some implementations, the I/O circuits in interface **316** and/or data bus **318** (as LLV circuits **402**) and logic circuits in control logic **312** (as part of LV circuits) are disposed in semiconductor structure **408**, while the page buffer circuits in page buffer **304** and driving circuits in row decoder/word line driver **308** and column decoder/bit line driver **306** are disposed in semiconductor structure **410**. For example, FIG. **7** illustrates a circuit diagram of word line driver **308** and page buffer **304**, according to some aspects of the present disclosure. In some implementations, page buffer **304** includes a plurality of page buffer circuits **702** each coupled to one NAND memory string **208** via a respective bit line **216**. That is, memory device **200** can include bit lines **216** respectively coupled to NAND memory strings **208**, and page buffer **304** can include page buffer circuits **702** respectively coupled to bit lines **216** and NAND memory strings **208**. Each page buffer circuit **702** can include one or more latches, switches, supplies, nodes (e.g., data nodes and I/O nodes), current mirrors, verify logic, sense circuits, etc. In some implementations, each page buffer circuit **702** is configured to store sensing data corresponding to read data, which is received from a respective bit line **216**, and output the stored sensing data to at the time of the read operation; each page buffer circuit **702** is also configured to store program data and output the stored program data to a respective bit line **216** at the time of the program operation.

(108) In some implementations, word line driver **308** includes a plurality of string drivers **704** (a.k.a. driving circuits) respectively coupled to word lines **218**. Word line driver **308** can also include a plurality of local word lines **706** (LWLs) respectively coupled to string drivers **704**. Each string driver **704** can include a gate coupled to a decoder (not shown), a source/drain coupled to a respective local word line **706**, and another source/drain coupled to a respective word line **218**. In some memory operations, the decoder can select certain string drivers **704**, for example, by applying a voltage signal greater than the threshold voltage of string drivers **704**, and a voltage (e.g., program voltage, pass voltage, or erase voltage) to each local word line **706**, such that the

voltage is applied by each selected string driver **704** to a respective word line **218**. In contrast, the decoder can also deselect certain string drivers **704**, for example, by applying a voltage signal smaller than the threshold voltage of string drivers **704**, such that each deselected string driver **704** floats a respective word line **218** during the memory operation.

(109) In some implementations, page buffer circuits **702** include parts of LV circuits **404** disposed in semiconductor structures **408** and/or **410**. In one example, since the number of page buffer circuits **702** increases as the number of bit numbers increases, which may occupy a large area for memory devices with large numbers of memory cells, page buffer circuits **702** may be split to semiconductor structures **408** and **410**. In some implementations, string drivers **704** include parts of HV circuits **406** disposed in semiconductor structure **410**.

(110) Consistent with the scope of the present disclosure, each peripheral circuit **202** can include a plurality of transistors as the basic building units thereof. The transistors can be metal-oxide-semiconductor field-effect-transistors (MOSFETs) in 2D (2D transistors, a.k.a. planar transistors) or 3D (3D transistors). For example, FIGS. 5A and 5B illustrate a perspective view and a side view, respectively, of a planar transistor **500**, according to some aspects of the present disclosure, and FIGS. 6A and 6B illustrate a perspective view and a side view, respectively, of a 3D transistor **600**, according to some aspects of the present disclosure. FIG. 5B illustrates the side view of the cross-section of planar transistor **500** in FIG. 5A in the BB plane, and FIG. 6B illustrates the side view of the cross-section of 3D transistor **600** in FIG. 6A in the BB plane.

(111) As shown in FIGS. 5A and 5B, planar transistor **500** can be a MOSFET on a substrate **502**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, SOI, or any other suitable materials. Trench isolations **503**, such as shallow trench isolations (STI), can be formed in substrate **502** and between adjacent planar transistors **500** to reduce current leakage. Trench isolations **503** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high dielectric constant (high-k) dielectrics (e.g., aluminum oxide, hafnium oxide, zirconium oxide, etc.). In some implementations, high-k dielectric materials include any dielectrics having a dielectric constant, or k-value, higher than that of silicon nitride ( $k > 7$ ). In some implementations, trench isolation **503** includes silicon oxide.

(112) As shown in FIGS. 5A and 5B, planar transistor **500** can also include a gate structure **508** on substrate **502**. In some implementations, gate structure **508** is on the top surface of substrate **502**. As shown in FIG. 5B, gate structure **508** can include a gate dielectric **507** on substrate **502**, i.e., above and in contact with the top surface of substrate **502**. Gate structure **508** can also include a gate electrode **509** on gate dielectric **507**, i.e., above and in contact with gate dielectric **507**. Gate dielectric **507** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. In some implementations, gate dielectric **507** includes silicon oxide, i.e., a gate oxide. Gate electrode **509** can include any suitable conductive materials, such as polysilicon, metals (e.g., W, Cu, Al, etc.), metal compounds (e.g., TiN, TaN, etc.), or silicides. In some implementations, gate electrode **509** includes doped polysilicon, i.e., a gate poly.

(113) As shown in FIG. 5A, planar transistor **500** can further include a pair of a source and a drain **506** in substrate **502**. Source and drain **506** can be doped with any suitable P-type dopants, such as boron (B) or Gallium (Ga), or any suitable N-type dopants, such as phosphorus (P) or arsenic (As). Source and drain **506** can be separated by gate structure **508** in the plan view. In other words, gate structure **508** is formed between source and drain **506** in the plan view, according to some implementations. The channel of planar transistor **500** in substrate **502** can be formed laterally between source and drain **506** under gate structure **508** when a gate voltage applied to gate electrode **509** of gate structure **508** is above the threshold voltage of planar transistor **500**. As shown in FIGS. 5A and 5B, gate structure **508** can be above and in contact with the top surface of the part of substrate **502** in which the channel can be formed (the active region). That is, gate structure **508** is in contact with only one side of the active region, i.e., in the plane of the top surface of substrate **502**, according to some implementations. It is understood, although not shown

in FIGS. 5A and 5B, planar transistor **500** may include additional components, such as wells and spacers.

(114) As shown in FIGS. 6A and 6B, 3D transistor **600** can be a MOSFET on a substrate **602**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, silicon on insulator SOI, or any other suitable materials. In some implementations, substrate **602** includes single crystalline silicon. Trench isolations **603**, such as STI, can be formed in substrate **602** and between adjacent 3D transistors **600** to reduce current leakage. Trench isolations **603** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics (e.g., aluminum oxide, hafnium oxide, zirconium oxide, etc.). In some implementations, trench isolation **603** includes silicon oxide.

(115) As shown in FIGS. 6A and 6B, different from planar transistor **500**, 3D transistor **600** can further include a 3D semiconductor body **604** above substrate **602**. That is, in some implementations, 3D semiconductor body **604** at least partially extends above the top surface of substrate **602** to expose not only the top surface, but also the two side surfaces, of 3D semiconductor body **604**. As shown in FIGS. 6A and 6B, for example, 3D semiconductor body **604** may be in a 3D structure, which is also known as a “fin,” to expose three sides thereof. 3D semiconductor body **604** is formed from substrate **602** and thus, has the same semiconductor material as substrate **602**, according to some implementations. In some implementations, 3D semiconductor body **604** includes single crystalline silicon. Since the channels can be formed in 3D semiconductor body **604**, as opposed to substrate **602**, 3D semiconductor body **604** may be viewed as the active region for 3D transistor **600**.

(116) As shown in FIGS. 6A and 6B, 3D transistor **600** can also include a gate structure **608** on substrate **602**. Different from planar transistors **500** in which gate structure **508** is in contact with only one side of the active region, i.e., in the plane of the top surface of substrate **502**, gate structure **608** of 3D transistor **600** can be in contact with a plurality of sides of the active region, i.e., in multiple planes of the top surface and side surfaces of the 3D semiconductor body **604**. In other words, the active region of 3D transistor **600**, i.e., 3D semiconductor body **604**, can be at least partially surrounded by gate structure **608**.

(117) Gate structure **608** can include a gate dielectric **607** over 3D semiconductor body **604**, e.g., in contact with the top surface and two side surfaces of 3D semiconductor body **604**. Gate structure **608** can also include a gate electrode **609** over and in contact with gate dielectric **607**. Gate dielectric **607** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. In some implementations, gate dielectric **607** includes silicon oxide, i.e., a gate oxide. Gate electrode **609** can include any suitable conductive materials, such as polysilicon, metals (e.g., W, Cu, Al, etc.), metal compounds (e.g., TiN, TaN, etc.), or silicides. In some implementations, gate electrode **609** includes doped polysilicon, i.e., a gate poly.

(118) As shown in FIG. 6A, 3D transistor **600** can further include a pair of a source and a drain **606** in 3D semiconductor body **604**. Source and drain **606** can be doped with any suitable P-type dopants, such as B or Ga, or any suitable N-type dopants, such as P or Ar. Source and drain **606** can be separated by gate structure **608** in the plan view. In other words, gate structure **608** is formed between source and drain **606** in the plan view, according to some implementations. As a result, multiple channels of 3D transistor **600** in 3D semiconductor body **604** can be formed laterally between source and drain **606** surrounded by gate structure **608** when a gate voltage applied to gate electrode **609** of gate structure **608** is above the threshold voltage of 3D transistor **600**. Different from planar transistor **500** in which only a single channel can be formed on the top surface of substrate **502**, multiple channels can be formed on the top surface and side surfaces of 3D semiconductor body **604** in 3D transistor **600**. In some implementations, 3D transistor **600** includes a multi-gate transistor. It is understood, although not shown in FIGS. 6A, and 6B, 3D transistor **600** may include additional components, such as wells, spacers, and stressors (a.k.a. strain elements) at source and drain **606**.

(119) It is further understood that FIGS. 6A and 6B illustrate one example of 3D transistors that can be used in memory peripheral circuits, and any other suitable 3D multi-gate transistors may be used in memory peripheral circuits as well, including, for example, a gate all around (GAA) silicon on nothing (SON) transistor, a multiple independent gate FET (MIGET), a trigate FET, a  $\Pi$ -gate FET, and a  $\Omega$ -FET, a quadruple gate FET, a cylindrical FET, or a multi-bridge/stacked nanowire FET.

(120) Regardless of planar transistor **500** or 3D transistor **600**, each transistor a memory peripheral circuit can include a gate dielectric (e.g., gate dielectrics **507** and **607**) having a thickness T (gate dielectric thickness, e.g., shown in FIGS. 5B and 6B). The gate dielectric thickness T of a transistor can be designed to accommodate the voltage applied to the transistor. For example, referring to FIGS. 4A and 4B, the gate dielectric thickness of transistors in HV circuits **406** (e.g., driving circuits such as string drivers **704**) may be larger than the gate dielectric thickness of transistors in LV circuits **404** (e.g., page buffer circuits **702** or logic circuits in control logic **312**), which may be in turn larger than the gate dielectric thickness of transistors in LLV circuits **402** (e.g., I/O circuits in interface **316** and data bus **318**). In some implementations, the difference between the gate dielectric thickness of transistors in HV circuits **406** and the dielectric thickness of transistors in LLV circuits **402** is at least 5-fold, such as between 5-fold and 50-fold. For example, the gate dielectric thickness of transistors in HV circuits **406** may be at least 5 times larger than the gate dielectric thickness of transistors in LLV circuits **402**.

(121) In some implementations, the dielectric thickness of transistors in LLV circuits **402** is between 2 nm and 4 nm (e.g., 2 nm, 2.1 nm, 2.2 nm, 2.3 nm, 2.4 nm, 2.5 nm, 2.6 nm, 2.7 nm, 2.8 nm, 2.9 nm, 3 nm, 3.1 nm, 3.2 nm, 3.3 nm, 3.4 nm, 3.5 nm, 3.6 nm, 3.7 nm, 3.8 nm, 3.9 nm, 4 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the LLV voltage range applied to LLV circuits **402**, as described above in detail, such as below 1.3 V (e.g., 1.2 V). In some implementations, the dielectric thickness of transistors in LV circuits **404** is between 4 nm and 10 nm (e.g., 4 nm, 4.5 nm, 5 nm, 5.5 nm, 6 nm, 6.5 nm, 7 nm, 7.5 nm, 8 nm, 8.5 nm, 9 nm, 9.5 nm, 10 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the LV voltage range applied to LV circuits **404**, as described above in detail, such as between 1.3 V and 3.3 V (e.g., 3.3 V). In some implementations, the dielectric thickness of transistors in HV circuits **406** is between 20 nm and 100 nm (e.g., 20 nm, 21 nm, 22 nm, 23 nm, 24 nm, 25 nm, 26 nm, 27 nm, 28 nm, 29 nm, 30 nm, 31 nm, 32 nm, 33 nm, 34 nm, 35 nm, 36 nm, 37 nm, 38 nm, 39 nm, 40 nm, 45 nm, 50 nm, 55 nm, 60 nm, 65 nm, 70 nm, 75 nm, 80 nm, 85 nm, 90 nm, 95 nm, 100 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the HV voltage range applied to HV circuits **406**, as described above in detail, such as greater than 3.3 V (e.g., between 5 V and 30 V).

(122) FIGS. 9A and 9B illustrate schematic views of cross-sections of 3D memory devices **900** and **901** having three stacked semiconductor structures, according to various aspects of the present disclosure. 3D memory devices **900** and **901** may be examples of 3D memory device **100** in FIG. 1A in which second semiconductor structure **104** including some of the peripheral circuits is disposed vertically between first semiconductor structure **102** including the memory cell array and third semiconductor structure **106** including some of the peripheral circuits. In other words, as shown in FIGS. 9A and 9B, first semiconductor structure **102** including the memory cell array of 3D memory devices **900** and **901** is disposed on one side of 3D memory devices **900** and **901**, third semiconductor structure **106** including some of the peripheral circuits is disposed on another side of 3D memory devices **900** and **901**, and second semiconductor structure **104** including some of the peripheral circuits is disposed in the intermediate of 3D memory devices **900** and **901** (i.e., between 3D memory devices **900** and **901**) in the vertical direction, according to some implementations. Second and third semiconductor structures **104** and **106** each including peripheral circuits can be

immediately adjacent to one another in three stacked semiconductor structures **102**, **104**, and **106**. (123) The above-mentioned arrangement of first, second, and third semiconductor structures **102**, **104**, and **106**, where first semiconductor structure **102** is on one side of 3D memory devices **900** and **901**, are described below in detail with respect to various examples, such as in FIGS. **10A**, **10B**, **16A**, **16B**, **22A**, **22B**, **28A**, and **28B**. The above-mentioned arrangement of first, second, and third semiconductor structures **102**, **104**, and **106** can simplify the fabrication process by using the substrate of first semiconductor structure **102** on which the memory cell array is formed as the base substrate to provide the support for processes, such as thinning, bonding, contact formation, etc. applied to second semiconductor structure **104** and/or third semiconductor structure **106** without the need of introducing another handle substrate (carrier wafer). Moreover, the electrical connections between the memory cell array and the peripheral circuits in each of second and third semiconductor structures **104** and **106** can be formed without penetrating the substrate of first semiconductor structure **102** on which the memory cell array is formed, thereby reducing the wiring length and complexity. Furthermore, in some implementations, by arranging the first semiconductor structure **102** having the memory cell array on one side of 3D memory devices **900** and **901**, the substrate (e.g., a silicon substrate having single crystalline silicon) of first semiconductor structure **102** on which the memory cell array is formed is able to be relatively easily replaced with a semiconductor layer having a different material (e.g., a polysilicon layer), which is suitable for certain channel structures (e.g., bottom open channel structure **812C**) of “charge trap” type of NAND memory strings or “floating gate” type of NAND memory strings. (124) Moreover, as shown in FIGS. **9A** and **9B**, 3D memory device **900** or **901** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **9A**, third semiconductor structure **106** including some of the peripheral circuits on one side of 3D memory device **900** may include the pad-out interconnect layer **902**, such that 3D memory device **900** may be pad-out from the peripheral circuit side to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **900**. In another example shown in FIG. **9B**, first semiconductor structure **102** including the memory cell array on another side of 3D memory device **901** may include pad-out interconnect layer **902**, such that 3D memory device **901** may be pad-out from the memory cell array side. (125) FIGS. **10A** and **10B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **9A** and **9B**, according to various aspects of the present disclosure. 3D memory devices **1000** and **1001** may be examples of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **10A**, 3D memory device **1000** can include stacked first, second, and third semiconductor structures **102**, **104**, and **106**. In some implementations, first semiconductor structure **102** on one side of 3D memory device **1000** includes a semiconductor layer **1002**, a bonding layer **1008**, and a memory cell array vertically between semiconductor layer **1002** and bonding layer **1008**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer **1002** (e.g., as shown in FIGS. **8A-8C**). Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**). Bonding layer **1008** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail. (126) In some implementations, second semiconductor structure **104** in the intermediate of 3D memory device **1000** (i.e., between first and third semiconductor structures **102** and **106**) includes a semiconductor layer **1004**, a bonding layer **1010**, and some of the peripheral circuits of the memory

cell array that are vertically between semiconductor layer **1004** and bonding layer **1010**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Similar to bonding layer **1008** in first semiconductor structure **102**, bonding layer **1010** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. That is, bonding layers **1008** and **1010** can be disposed on opposite sides of bonding interface **103**, and the bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(127) In some implementations, third semiconductor structure **106** on another side of 3D memory device **1000** includes a semiconductor layer **1006** and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1006** and semiconductor layer **1006**. The transistors (e.g., planar transistors **500** and/or 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1006**. Semiconductor layer **1006** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1006** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. It is understood that different from bonding interface **103** between first and second semiconductor structures **102** and **104**, which is between bonding layers **1008** and **1010** and results from hybrid bonding, bonding interface **105** between second and third semiconductor structures **104** and **106** may result from transfer bonding, as described below in detail, and thus, may not be formed between two bonding layers. That is, third semiconductor structure **106** of 3D memory device **1000** in FIG. **10A** does not include a bonding layer with bonding contacts, according to some implementations. As a result, instead of bonding contacts, through contacts (e.g., ILVs/TSVs) across bonding interface **105** and through semiconductor layer **1004** vertically between second and third semiconductor structures **104** and **106** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **104** and **106**.

(128) It is understood that in some examples, second and third semiconductor structures **104** and **106** may also include bonding layers **1012** and **1014**, respectively, disposed on opposite sides of bonding interface **105**, as shown in FIG. **10B**. In FIG. **10B**, second semiconductor structure **104** of a 3D memory device **1001** can include two bonding layers **1010** and **1012** on two sides thereof, and bonding layer **1012** can be disposed vertically between semiconductor layer **1004** and bonding interface **105**. Third semiconductor structure **106** of 3D memory device **1001** can include bonding layer **1014** disposed vertically between bonding interface **105** and the peripheral circuits thereof. Each bonding layer **1012** and **1014** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. The bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, bonding contacts across bonding interface **105** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1004** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **104** and **106**.



(129) As shown in FIGS. **10A** and **10B**, since third and second semiconductor structures **106** and **104** are bonded in a face-to-back manner (e.g., each semiconductor layer **1006** or **1004** being disposed on the top side of respective third or second semiconductor structure **106** or **104** in FIGS. **10A** and **10B**), the transistors in third and second semiconductor structures **106** and **104** are disposed toward the same direction (e.g., the negative y-direction in FIG. **10A**), according to some implementations. In some implementations, the transistors of the peripheral circuits in third semiconductor structure **106** are disposed vertically between bonding interface **105** and semiconductor layer **1006**, and the transistors of the peripheral circuits in second semiconductor structure **104** are disposed vertically between bonding interface **103** and semiconductor layer **1004**. Moreover, since first and second semiconductor structures **102** and **104** are bonded in a face-to-face manner (e.g., semiconductor layer **1002** being disposed on the bottom side of first semiconductor structure **102**, while semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104** in FIGS. **10A** and **10B**), the transistors of peripheral circuits in third and second semiconductor structures **106** and **104** are disposed toward the same direction, facing the memory cell array in first semiconductor structure **102**, according to some implementations. It is understood that pad-out interconnect layer **902** in FIG. **9A** or **9B** is omitted from 3D memory devices **1000** and **1001** in FIGS. **10A** and **10B** for case of illustration and may be included in 3D memory devices **1000** and **1001** as described above with respect to FIGS. **9A** and **9B**.

(130) As described above, second and third semiconductor structures **104** and **106** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and third semiconductor structure **106** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1006** and **1004** in third and second semiconductor structures **106** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1006** in third semiconductor structure **106** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in third and second semiconductor structures **106** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in third semiconductor structure **106** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**. The thicker gate dielectric can sustain a higher working voltage applied to the transistors in third semiconductor structure **106** than the transistors in second semiconductor structure **104** to avoid break down during high voltage operations.

(131) As shown in FIGS. **10A** and **10B**, the peripheral circuits in second semiconductor structure **104** and/or the peripheral circuits in third semiconductor structures **106** can be disposed between bonding interface **103** and semiconductor layer **1006** of third semiconductor structure **106**. The peripheral circuits in second semiconductor structure **104** and/or the peripheral circuits in third semiconductor structures **106** can also be disposed between the memory cell array in first semiconductor structure **102** and semiconductor layer **1006** of third semiconductor structure **106**.

(132) FIGS. **11A-11C** illustrate side views of various examples of 3D memory devices **1000** and **1001** in FIGS. **10A** and **10B**, according to various aspects of the present disclosure. As shown in FIG. **11A**, as one example of 3D memory devices **1000** and **1001** in FIGS. **10A** and **10B**, 3D memory device **1100** is a bonded chip including first semiconductor structure **102**, second semiconductor structure **104**, and third semiconductor structure **106**, which are stacked over one

another in different planes in the vertical direction (e.g., the y-direction in FIG. 11A), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, and second and third semiconductor structures **104** and **106** are bonded at bonding interface **105** therebetween, according to some implementations.

(133) As shown in FIG. 11A, third semiconductor structure **106** can include semiconductor layer **1006** having semiconductor materials. In some implementations, semiconductor layer **1006** is a silicon substrate having single crystalline silicon. Third semiconductor structure **106** can also include a device layer **1102** above and in contact with semiconductor layer **1006**. In some implementations, device layer **1102** includes a first peripheral circuit **1104** and a second peripheral circuit **1106**. First peripheral circuit **1104** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **1106** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **1104** includes a plurality of transistors **1108** in contact with semiconductor layer **1006**, and second peripheral circuit **1106** includes a plurality of transistors **1110** in contact with semiconductor layer **1006**. Transistors **1108** and **1110** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1108** or **1110** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1108** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1110** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **1108** than transistor **1110**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1108** and **1110**) can be formed on or in semiconductor layer **1006** as well.

(134) In some implementations, third semiconductor structure **106** further includes an interconnect layer **1112** above device layer **1102** to transfer electrical signals to and from peripheral circuits **1106** and **1104**. As shown in FIG. 11A, interconnect layer **1112** can be vertically between bonding interface **105** and device layer **1102** (including transistors **1108** and **1110** of peripheral circuits **1104** and **1106**). Interconnect layer **1112** can include a plurality of interconnects (also referred to herein as “contacts”), including lateral lines and vias. As used herein, the term “interconnects” can broadly include any suitable types of interconnects, such as middle-end-of-line (MEOL) interconnects and back-end-of-line (BEOL) interconnects. The interconnects in interconnect layer **1112** can be coupled to transistors **1108** and **1110** of peripheral circuits **1104** and **1106** in device layer **1102**. Interconnect layer **1112** can further include one or more interlayer dielectric (ILD) layers (also known as “intermetal dielectric (IMD) layers”) in which the lateral lines and vias can form. That is, interconnect layer **1112** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1102** are coupled to one another through the interconnects in interconnect layer **1112**. For example, peripheral circuit **1104** may be coupled to peripheral circuit **1106** through interconnect layer **1112**. The interconnects in interconnect layer **1112** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1112** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low dielectric constant (low-k) dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1112** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(135) Second semiconductor structure **104** can be bonded on top of third semiconductor structure **106** in a back-to-face manner at bonding interface **105**. Second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of third semiconductor structure **106** by transfer

bonding. In some implementations, bonding interface **105** is disposed vertically between interconnect layer **1112** and semiconductor layer **1004** as a result of transfer bonding, which transfers semiconductor layer **1004** from another substrate and bonds semiconductor layer **1004** onto third semiconductor structure **106** as described below in detail. In some implementations, bonding interface **105** is the place at which interconnect layer **1112** and semiconductor layer **1004** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of interconnect layer **1112** of third semiconductor structure **106** and the bottom surface of semiconductor layer **1004** of second semiconductor structure **104**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1004** and/or between bonding interface **105** and interconnect layer **1112** to facilitate the transfer bonding of semiconductor layer **1004** onto interconnect layer **1112**. Thus, it is understood that bonding interface **105** may include the surfaces of the dielectric layer(s) in some examples.

(136) Second semiconductor structure **104** can include a device layer **1114** above and in contact with semiconductor layer **1004**. In some implementations, device layer **1114** includes a third peripheral circuit **1116** and a fourth peripheral circuit **1118**. Third peripheral circuit **1116** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **1118** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **1116** includes a plurality of transistors **1120**, and fourth peripheral circuit **1118** includes a plurality of transistors **1122** as well. Transistors **1120** and **1122** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1120** or **1122** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1120** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **1122** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **1120** than transistor **1122**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1120** and **1122**) can be formed on or in semiconductor layer **1004** as well.

(137) Moreover, the different voltages applied to different transistors **1120**, **1122**, **1108**, and **1110** in second and third semiconductor structures **104** and **106** can lead to differences of device dimensions between second and third semiconductor structures **104** and **106**. In some implementations, the thickness of the gate dielectric of transistor **1108** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1120** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **1108** than transistor **1120**. In some implementations, the thickness of the gate dielectric of transistor **1122** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **1110** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **1122** and transistor **1110**. In some implementations, the thickness of semiconductor layer **1006** in which transistor **1108** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **1120** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **1108** than transistor **1120**.

(138) As shown in FIG. **11A**, second semiconductor structure **104** can further include an interconnect layer **1126** above device layer **1114** to transfer electrical signals to and from peripheral circuits **1116** and **1118**. As shown in FIG. **11A**, interconnect layer **1126** can be vertically between bonding interface **103** and device layer **1114** (including transistors **1120** and **1122** of peripheral circuits **1116** and **1118**). Interconnect layer **1126** can include a plurality of interconnects coupled to transistors **1120** and **1122** of peripheral circuits **1116** and **1118** in device layer **1114**. Interconnect layer **1126** can further include one or more ILD layers in which the interconnects can form. That is, interconnect layer **1126** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1114** are coupled to one another through the interconnects in interconnect layer **1126**. For example, peripheral circuit **1116** may be coupled to

peripheral circuit **1118** through interconnect layer **1126**. The interconnects in interconnect layer **1126** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1126** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(139) In some implementations, the interconnects in interconnect layer **1126** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **1126** can occur after the high-temperature processes in forming device layers **1114** and **1102** in second and third semiconductor structures **104** and **106**, as well as being separated from the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **1126** having Cu can become feasible.

(140) As shown in FIG. **11A**, second semiconductor structure **104** can further include one or more contacts **1124** extending vertically through semiconductor layer **1004**. Contact **1124** can extend vertically further through bonding interface **105** to be in contact with the interconnects in interconnect layer **1112**. In some implementations, contact **1124** couples the interconnects in interconnect layer **1126** to the interconnects in interconnect layer **1112** to make an electrical connection across bonding interface **105** between second and third semiconductor structures **104** and **106**. Contact **1124** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1124** includes W. In some implementations, contact **1124** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **1124** can be an ILV having a depth (in the vertical direction) in the submicron-level (e.g., between 10 nm and 1  $\mu$ m), or a TSV having a depth (in the vertical direction) in the micron- or tens micron-level (e.g., between 1  $\mu$ m and 100  $\mu$ m).

(141) As shown in FIG. **11A**, second semiconductor structure **104** can further include a bonding layer **1010** at bonding interface **103** and above and in contact with interconnect layer **1126**. Bonding layer **1010** can include a plurality of bonding contacts **1011** and dielectrics electrically isolating bonding contacts **1011**. Bonding contacts **1011** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, bonding contacts **1011** of bonding layer **1010** include Cu. The remaining area of bonding layer **1010** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1011** and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously.

(142) As shown in FIG. **11A**, first semiconductor structure **102** can further include a bonding layer **1008** at bonding interface **103**, e.g., on the opposite side of bonding interface **103** with respect to bonding layer **1010** in second semiconductor structure **104**. Bonding layer **1008** can include a plurality of bonding contacts **1009** and dielectrics electrically isolating bonding contacts **1009**. Bonding contacts **1009** can include conductive materials, such as Cu. The remaining area of bonding layer **1008** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1009** and surrounding dielectrics in bonding layer **1008** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1010** of second semiconductor structure **104** and the bottom surface of bonding layer **1008** of first semiconductor structure **102**.

(143) Although not shown in FIG. 11A, it is understood that in some examples, similar to bonding interface **103**, bonding interface **105** may result from hybrid bonding and thus, be disposed vertically between two bonding layers (e.g., bonding layers **1012** and **1014** of 3D memory device **1001** in FIG. 10B) each including bonding contacts in second and third semiconductor structures **104** and **106**, respectively.

(144) As shown in FIG. 11A, first semiconductor structure **102** can further include an interconnect layer **1128** above bonding layer **1008** to transfer electrical signals. Interconnect layer **1128** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1128** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **1128** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **1128** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1128** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(145) As shown in FIG. 11A, first semiconductor structure **102** can include a memory cell array, such as an array of NAND memory strings **208** above interconnect layer **1128**. In some implementations, interconnect layer **1128** is vertically between NAND memory strings **208** and bonding interface **103**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **1127**. Memory stack **1127** may be an example of memory stack **804** in FIGS. 8A-8C, and the conductive layer and dielectric layer in memory stack **1127** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **1127** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The adhesive layer can include conductive materials, such as titanium nitride (TiN), which can improve the adhesiveness between the gate electrode and the gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **1127**.

(146) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. 8A-8C. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(147) As shown in FIG. 11A, first semiconductor structure **102** can further include a semiconductor layer **1002** disposed above memory stack **1127** and in contact with the sources of NAND memory strings **208**. In some implementations, NAND memory strings **208** are disposed vertically between bonding interface **103** and semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **1727** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(148) As shown in FIG. 11A, first semiconductor structure **102** can further include a pad-out interconnect layer **902** above and in contact with semiconductor layer **1002**. In some implementations, semiconductor layer **1002** is disposed vertically between pad-out interconnect layer **902** and NAND memory strings **208**. Pad-out interconnect layer **902** can include

interconnects, e.g., contact pads **1132**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **1128** can be formed on opposite sides of semiconductor layer **1002**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **1100** and external devices, e.g., for pad-out purposes.

(149) As shown in FIG. **11A**, first semiconductor structure **102** can further include one or more contacts **1130** extending vertically through semiconductor layer **1002**. In some implementations, contact **1130** couples the interconnects in interconnect layer **1128** to contact pads **1132** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **1130** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1130** includes W. In some implementations, contact **1130** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **1130** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(150) As a result, peripheral circuits **1104**, **1106**, **1116**, and **1118** in third and second semiconductor structures **106** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **1112**, **1126**, and **1128**, bonding layers **1008** and **1010**, as well as contacts **1124**. Moreover, peripheral circuits **1104**, **1106**, **1116**, and **1118** and NAND memory strings **208** in 3D memory device **1100** can be further coupled to external devices through contacts **1130** and pad-out interconnect layer **902**.

(151) It is understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIG. **11A** and may be any other suitable semiconductor materials. For example, as shown in FIG. **11B**, a 3D memory device **1101** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **1101** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **1101** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1100** and **1101** are not repeated for ease of description.

(152) It is also understood that the pad-out of 3D memory devices is not limited to from first semiconductor structure **102** having NAND memory strings **208** as shown in FIGS. **11A** and **11B** (corresponding to FIG. **9B**) and may be from third semiconductor structure **106** having peripheral circuit **1104** (corresponding to FIG. **9A**). For example, as shown in FIG. **11C**, a 3D memory device **1103** may include pad-out interconnect layer **902** in third semiconductor structure **106**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1006** of third semiconductor structure **106** on which transistors **1108** of peripheral circuit **1104** are formed. In some implementations, third semiconductor structure **106** further includes one or more contacts **1134** extending vertically through semiconductor layer **1006**. In some implementations, contact **1134** couples the interconnects in interconnect layer **1112** in third semiconductor structure **106** to contact pads **1132** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1006**. Contact **1134** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1134** includes W. In some implementations, contact **1134** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1006**. Depending on the thickness of semiconductor layer **1006**, contact **1134** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the

micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1100** and **1103** are not repeated for case of description.

(153) It is further understood that in some examples, similar to bonding interface **103**, bonding interface **105** may result from hybrid bonding and thus, be disposed vertically between two bonding layers each including bonding contacts in second and third semiconductor structures **104** and **106**, respectively. For example, as shown in FIG. **11C**, 3D memory device **1103** may include bonding layers **1012** and **1014** in second and third semiconductor structures **104** and **106**, respectively, at bonding interface **105**, i.e., on opposite sides of bonding interface **105**. Bonding layer **1012** or **1014** can include a plurality of bonding contacts **1013** or **1015** and dielectrics electrically isolating bonding contacts **1013** or **1015**. Bonding contacts **1013** and **1015** can include conductive materials, such as Cu. The remaining area of bonding layer **1012** or **1014** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1013** or **1015** and surrounding dielectrics in bonding layer **1012** or **1014** can be used for hybrid bonding. In some implementations, bonding interface **105** is the place at which bonding layers **1012** and **1014** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of bonding layer **1014** of third semiconductor structure **106** and the bottom surface of bonding layer **1012** of second semiconductor structure **104**. Contact **1124** can be coupled to bonding contacts **1013**, and interconnect layer **1112** can be coupled to bonding contacts **1015**.

(154) FIGS. **12A-12H** illustrate a fabrication process for forming the 3D memory devices in FIGS. **10A** and **10B**, according to some aspects of the present disclosure. FIG. **14** illustrates a flowchart of a method **1400** for forming the 3D memory devices in FIGS. **10A** and **10B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **12A-12H** and **14** include 3D memory devices **1100**, **1101**, and **1103** depicted in FIGS. **11A-11C**. FIGS. **12A-12H** and **14** will be described together. It is understood that the operations shown in method **1400** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **14**. For example, operation **1402** may be performed after operation **1408** or in parallel with operations **1404-1408**.

(155) Referring to FIG. **14**, method **1400** starts at operation **1402**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(156) As illustrated in FIG. **12D**, a stack structure, such as a memory stack **1226** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **1224**. To form memory stack **1226**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **1224**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or any combination thereof. Memory stack **1226** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1226** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer (e.g., thermally grown local oxidation of silicon (LOCOS)) including silicon oxide is formed between memory stack **1226** and silicon substrate **1224**.

(157) As illustrated in FIG. 12D, NAND memory strings **1228** are formed above silicon substrate **1224**, each of which extends vertically through memory stack **1226** to be in contact with silicon substrate **1224**. In some implementations, fabrication processes to form NAND memory string **1228** include forming a channel hole through memory stack **1226** (or the dielectric stack) and into silicon substrate **1224** using dry etching/and or wet etching, such as deep reactive-ion etching (DRIE), followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1228** may vary depending on the types of channel structures of NAND memory strings **1228** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. 8A-8C) and thus, are not elaborated for ease of description.

(158) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. 12D, an interconnect layer **1230** is formed above memory stack **1226** and NAND memory strings **1228**. Interconnect layer **1230** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1228**. In some implementations, interconnect layer **1230** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1230** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, chemical mechanical polishing (CMP), wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 12D can be collectively referred to as interconnect layer **1230**.

(159) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. 12D, a bonding layer **1232** is formed above interconnect layer **1230**. Bonding layer **1232** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1230** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1230** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(160) Method **1400** proceeds to operation **1404**, as illustrated in FIG. 14, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 12A, a plurality of transistors **1204** and **1206** are formed on a silicon substrate **1202**. Transistors **1204** and **1206** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1202** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1204** and **1206**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1202** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1204** is different from the thickness of gate dielectric of transistor **1206**, for example, by depositing a thicker silicon



oxide film in the region of transistor **1204** than the region of transistor **1206**, or by etching back part of the silicon oxide film deposited in the region of transistor **1206**. It is understood that the details of fabricating transistors **1204** and **1206** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(161) In some implementations, an interconnect layer **1208** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 12A, an interconnect layer **1208** can be formed above transistors **1204** and **1206**. Interconnect layer **1208** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1204** and **1206**. In some implementations, interconnect layer **1208** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1208** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 12A can be collectively referred to as interconnect layer **1208**. In some implementations, the interconnects in interconnect layer **1208** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(162) Method **1400** proceeds to operation **1406**, as illustrated in FIG. 14, in which a semiconductor layer is formed above the first transistor. The semiconductor layer can include single crystalline silicon. In some implementations, to form the semiconductor layer, another substrate and the second substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(163) As illustrated in FIG. 12B, a semiconductor layer **1210**, such as a single crystalline silicon layer, is formed above interconnect layer **1208** and transistors **1204** and **1206**. Semiconductor layer **1210** can be attached above interconnect layer **1208** to form a bonding interface **1212** vertically between semiconductor layer **1210** and interconnect layer **1208**. The lateral dimensions (e.g., the dimension in the x-direction) of semiconductor layer **1210** are the same as those of silicon substrate **1202** or silicon substrate **1224**, according to some implementations. In some implementations, to form semiconductor layer **1210**, another silicon substrate (not shown in FIG. 12B) and silicon substrate **1202** are bonded in a face-to-face manner (i.e., having the components formed on silicon substrate **1202**, such as transistors **1204** and **1206**, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface **1212**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **1210** attached above interconnect layer **1208**. The same “face-to-face” manner as described above is applied throughout the present disclosure in describing other figures.

(164) FIGS. 48A-48D illustrate a fabrication process of transfer bonding, according to some aspects of the present disclosure. As illustrated in FIG. 48A, a function layer **4804** can be formed on a base substrate **4802**. Function layer **4804** can include device layers, interconnect layers, and/or any suitable layers disclosed herein, such as transistors **1204** and **1206** and interconnect layer **1208** in FIG. 12B. A transfer substrate **4806**, such as a silicon substrate having single crystalline silicon, is provided. In some implementations, transfer substrate **4806** is a single crystalline silicon substrate. As illustrated in FIG. 48B, transfer substrate **4806** and base substrate **4802** (and function layer **4804** formed thereon) can be bonded in a face-to-face manner using any suitable substrate/wafer bonding processes including, for example, anodic bonding and fusion (direct) bonding, thereby forming a bonding interface **4810** between transfer substrate **4806** and base

substrate **4802**. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **4810**. For example, silicon oxide layers may be formed on the top surfaces of both transfer substrate **4806** and function layer **4804** to allow SiO.sub.2—SiO.sub.2 bonding using fusion bonding. Or silicon oxide layer may be formed only on function layer **4804** to allow SiO.sub.2—Si bonding using anodic bonding or fusion bonding. In some implementations in which a silicon oxide layer is formed on transfer substrate **4806** (e.g., shown in FIG. **48B**), transfer substrate **4806** can be flipped upside, such that the silicon oxide layer on transfer substrate **4806** faces down toward base substrate **4802** before the bonding.

(165) As illustrated in FIG. **48C**, a cut layer **4812** can be formed in transfer substrate **4806**, for example, using ion implantation. In some implementations, light elements, such as hydrogen ions, are implanted into transfer substrate **4806** to a desired depth, for example, by controlling the energy of the ion impanation process, to form cut layer **4812**. As illustrated in FIG. **48D**, transfer substrate **4806** can be thinned to leave only a semiconductor layer **4814** vertically between cut layer **4812** and bonding interface **4810**. In some implementations, transfer substrate **4806** is split at cut layer **4812** by applying a mechanical force to transfer substrate **4806**, i.e., peeling off the remainder of transfer substrate **4806** from semiconductor layer **4814**. It is understood that transfer substrate **4806** may be split at cut layer **4812** by any suitable means, not limited to mechanical force alone, such as thermal means, acoustic means, optical means, etc., or any combination thereof. As a result, semiconductor layer **4814** can be transferred from transfer substrate **4806** and bonded onto base substrate **4802** (and function layer **4804**) using a transfer bonding process. In some implementations, a planarization process, such as chemical mechanical polishing (CMP), is performed on semiconductor layer **4812** to polish and smooth the top surface of semiconductor layer **4812** and adjust the thickness of semiconductor layer **4812**. Semiconductor layer **4814** thus can have the same material as transfer substrate **4806**, such as single crystalline silicon. The thickness of semiconductor layer **4814** can be determined by the depth of cut layer **4812**, for example, by adjusting the implantation energy, and/or by the planarization process. Moreover, the remainder of transfer substrate **4806** can be re-used in the same manner to form semiconductor layers bonded onto other base substrates, thereby reducing the material cost of the transfer bonding process.

(166) FIGS. **49A-49D** illustrate another fabrication process of transfer bonding, according to some aspects of the present disclosure. As illustrated in FIG. **49A**, function layer **4804** can be formed on base substrate **4802**. Function layer **4804** can include device layers, interconnect layers, and/or any suitable layers disclosed herein, such as transistors **1204** and **1206** and interconnect layer **1208** in FIG. **12B**. An SOI substrate **4902**, including a base/handle layer **4904**, a buried oxide layer (BOX) **4906**, and a device layer **4908**, can be flipped upside down facing toward base substrate **4802**. As illustrated in FIG. **49B**, SOI substrate **4902** and base substrate **4802** (and function layer **4804** formed thereon) can be bonded in a face-to-face manner using any suitable substrate/wafer bonding processes including, for example, anodic bonding and fusion (direct) bonding, thereby forming a bonding interface **4912** between SOI substrate **4902** and base substrate **4802**. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **4912**. For example, silicon oxide layers may be formed on the top surfaces of both SOI substrate **4902** and function layer **4804** to allow SiO.sub.2—SiO.sub.2 bonding using fusion bonding. Or silicon oxide layer may be

formed only on function layer **4804** to allow SiO<sub>2</sub>—Si bonding using anodic bonding or fusion bonding.

(167) As illustrated in FIGS. **49C** and **49D**, SOI substrate **4902** (shown in FIG. **49B**) can be thinned by sequentially removing base/handle layer **4904** and buried oxide layer **4906**, for example, using wet/dry etching and/or CMP processes, to leave only device layer **4908** (as a semiconductor layer) at bonding interface **4912**. As a result, device layer **4908** can be transferred from SOI substrate **4902** and bonded onto base substrate **4802** (and function layer **4804**) as a semiconductor layer using another transfer bonding process. The transferred semiconductor layer thus can have the same material as device layer **4908**, such as single crystalline silicon. The thickness of the semiconductor layer can be the same as the thickness of device layer **4908**. It is understood that in some examples, device layer **4908** may be further thinned using wet/dry etching and/or CMP processes, such that the transferred semiconductor layer may be thinned than device layer **4908**.

(168) Referring to FIG. **14**, method **1400** proceeds to operation **1408**, in which a second transistor is formed on the semiconductor layer. As illustrated in FIG. **12C**, a plurality of transistors **1214** and **1216** are formed on semiconductor layer **1210** having single crystalline silicon. Transistors **1214** and **1216** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **1210** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1214** and **1216**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **1210** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1214** is different from the thickness of gate dielectric of transistor **1216**, for example, by depositing a thicker silicon oxide film in the region of transistor **1214** than the region of transistor **1216**, or by etching back part of the silicon oxide film deposited in the region of transistor **1216**. It is understood that the details of fabricating transistors **1214** and **1216** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(169) In some implementations, an interconnect layer **1220** is formed above the transistor on the semiconductor layer. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **12C**, an interconnect layer **1220** can be formed above transistors **1214** and **1216**. Interconnect layer **1220** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1214** and **1216**. In some implementations, interconnect layer **1220** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1220** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **12C** can be collectively referred to as interconnect layer **1220**. Different from interconnect layer **1208**, in some implementations, the interconnects in interconnect layer **1220** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **1220** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **1220**.

(170) In some implementations, a contact through the semiconductor layer is formed. As illustrated in FIG. **12C**, one or more contacts **1218** each extending vertically through semiconductor layer **1210** is formed. Contacts **1218** can couple the interconnects in interconnect layers **1220** and **1208**.

Contacts **1218** can be formed by first patterning contact holes through semiconductor layer **1210** and bonding interface **1212** to be in contact with the interconnects in interconnect layer **1208** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(171) In some implementations, a second bonding layer is formed above the interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **12D**, a bonding layer **1222** is formed above interconnect layer **1220**. Bonding layer **1222** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1220** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1220** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor. For example, the adhesion layer may improve the adhesiveness of the conductor to avoid defects, the barrier layer may prevent metal ion (e.g., Cu ions) diffusing from the conductor into other structures to cause contamination, and the seed layer may facilitate the deposition of the conductor (e.g., Cu) in the contact holes to improve the deposition quality and speed.

(172) Method **1400** proceeds to operation **1410**, as illustrated in FIG. **14**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(173) As illustrated in FIG. **12E**, silicon substrate **1224** and components formed thereon (e.g., memory stack **1226** and NAND memory strings **1228** formed therethrough) are flipped upside down. Bonding layer **1232** facing down is bonded with bonding layer **1222** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1237**. That is, silicon substrate **1224** and components formed thereon can be bonded with silicon substrate **1202** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1232** are in contact with the bonding contacts in bonding layer **1222** at bonding interface **1237**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **12E**, it is understood that in some examples, silicon substrate **1202** and components formed thereon (e.g., transistors **1204**, **1206**, **1214**, and **1216**) can be flipped upside down, and bonding layer **1222** facing down can be bonded with bonding layer **1232** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1237** as well.

(174) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1237** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1232** and the bonding contacts in bonding layer **1222** are aligned and in contact with one another, such that memory stack **1226** and NAND memory strings **1228** formed therethrough can be coupled to transistors **1214**, **1216**, **1204**, and **1206** through the bonded bonding contacts across bonding interface **1237**, according to some implementations.

(175) Method **1400** proceeds to operation **1412**, as illustrated in FIG. **14**, in which the first substrate or the second substrate is thinned. As illustrated in FIG. **12F**, silicon substrate **1224** (shown in FIG. **12E**) is thinned to become a semiconductor layer **1234** having single crystalline silicon. Silicon substrate **1224** can be thinned by processes including, but not limited to, wafer

grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. It is understood that although not shown in FIG. 12F, in some examples, silicon substrate **1202** may be thinned to become a semiconductor layer having single crystalline silicon.

(176) Method **1400** proceeds to operation **1414**, as illustrated in FIG. 14, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned second substrate or above the array of NAND memory strings. As illustrated in FIG. 12F, a pad-out interconnect layer **1236** is formed on semiconductor layer **1234** (the thinned silicon substrate **1224**) above NAND memory strings **1228**. Pad-out interconnect layer **1236** can include interconnects, such as contact pads **1238**, formed in one or more ILD layers. Contact pads **1238** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1235** are formed, extending vertically through semiconductor layer **1234**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1235** can couple contact pads **1238** in pad-out interconnect layer **1236** to the interconnects in interconnect layer **1230**. It is understood that in some examples, contacts **1235** may be formed in silicon substrate **1224** before thinning (the formation of semiconductor layer **1234**) and be exposed from the backside of silicon substrate **1224** (where the thinning occurs) after the thinning. It is further understood that although not shown in FIG. 12F, in some examples, a pad-out interconnect layer may be formed on the thinned silicon substrate **1202**, and contacts may be formed through the thinned silicon substrate **1202** to couple the pad-out interconnect layer and interconnect layer **1208** across the thinned silicon substrate **1202**.

(177) In some implementations, a semiconductor layer having polysilicon is formed. To form the semiconductor layer, the first substrate is removed and replaced with the semiconductor layer. As illustrated in FIG. 12G, silicon substrate **1224** (shown in FIG. 12F) is removed, for example, using wafer grinding, dry etch, wet etch, CMP, any other suitable processes, to expose the channel structures (e.g., bottom open channel structure **812C** in FIG. 8C) of NAND memory strings **1228** from the source end. As illustrated in FIG. 12H, a semiconductor layer **1240** having polysilicon is formed to be in contact with the sources of NAND memory strings **1228**. Semiconductor layer **1240** can be formed by depositing polysilicon using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Similarly, pad-out interconnect layer **1236** including contact pads **1238** can be formed on semiconductor layer **1240**. Contacts **1242** can be formed through semiconductor layer **1240** having polysilicon after the formation of semiconductor layer **1240**.

(178) FIGS. 13A-13H illustrate another fabrication process for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure. FIG. 15 illustrates a flowchart of another method **1500** for forming the 3D memory devices in FIGS. 10A and 10B, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. 13A-13H and 15 include 3D memory devices **1100**, **1101**, and **1103** depicted in FIGS. 11A-11C. FIGS. 13A-13H and 15 will be described together. It is understood that the operations shown in method **1500** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. 15. For example, operation **1502**, **1504**, and **1506** may be performed in parallel.

(179) Referring to FIG. 15, method **1500** starts at operation **1502**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(180) As illustrated in FIG. 13A, a stack structure, such as a memory stack **1304** including

interleaved conductive layers and dielectric layers is formed on a silicon substrate **1302**. To form memory stack **1304**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **1302**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1304** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1304** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1304** and silicon substrate **1302**. (181) As illustrated in FIG. **13A**, NAND memory strings **1306** are formed above silicon substrate **1302**, each of which extends vertically through memory stack **1304** to be in contact with silicon substrate **1302**. In some implementations, fabrication processes to form NAND memory string **1306** include forming a channel hole through memory stack **1304** (or the dielectric stack) and into silicon substrate **1302** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1306** may vary depending on the types of channel structures of NAND memory strings **1306** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(182) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **13A**, an interconnect layer **1308** is formed above memory stack **1304** and NAND memory strings **1306**. Interconnect layer **1308** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1306**. In some implementations, interconnect layer **1308** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1308** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **13A** can be collectively referred to as interconnect layer **1308**.

(183) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **13A**, a bonding layer **1310** is formed above interconnect layer **1308**. Bonding layer **1310** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1308** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1308** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling

the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(184) Method **1500** proceeds to operation **1504**, as illustrated in FIG. **15**, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **13B**, a plurality of transistors **1314** and **1316** are formed on a silicon substrate **1312**. Transistors **1314** and **1316** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1312** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1314** and **1316**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1312** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1314** is different from the thickness of gate dielectric of transistor **1316**, for example, by depositing a thicker silicon oxide film in the region of transistor **1314** than the region of transistor **1316**, or by etching back part of the silicon oxide film deposited in the region of transistor **1316**. It is understood that the details of fabricating transistors **1314** and **1316** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(185) In some implementations, an interconnect layer **1318** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **13B**, an interconnect layer **1318** can be formed above transistors **1314** and **1316**. Interconnect layer **1318** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1314** and **1316**. In some implementations, interconnect layer **1318** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1318** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **13B** can be collectively referred to as interconnect layer **1318**.

(186) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **13B**, a bonding layer **1320** is formed above interconnect layer **1318**. Bonding layer **1320** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1318** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1318** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(187) Method **1500** proceeds to operation **1506**, as illustrated in FIG. **15**, in which a second transistor is formed on a third substrate. The third substrate can be a silicon substrate having single crystalline silicon. In some implementations, any two or all of operations **1502**, **1504**, and **1506** are performed in parallel to reduce process time.

(188) As illustrated in FIG. **13C**, a plurality of transistors **1324** and **1326** are formed on a silicon substrate **1322**. Transistors **1324** and **1326** can be formed by a plurality of processes including, but

not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1322** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1324** and **1326**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1322** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1324** is different from the thickness of gate dielectric of transistor **1326**, for example, by depositing a thicker silicon oxide film in the region of transistor **1324** than the region of transistor **1326**, or by etching back part of the silicon oxide film deposited in the region of transistor **1326**. It is understood that the details of fabricating transistors **1324** and **1326** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(189) In some implementations, an interconnect layer **1328** is formed above the transistor on the third substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **13C**, an interconnect layer **1328** can be formed above transistors **1324** and **1326**. Interconnect layer **1328** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1324** and **1326**. In some implementations, interconnect layer **1328** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1328** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **13C** can be collectively referred to as interconnect layer **1328**.

(190) In some implementations, a third bonding layer is formed above interconnect layer. The third bonding layer can include a plurality of third bonding contacts. As illustrated in FIG. **13C**, a bonding layer **1330** is formed above interconnect layer **1328**. Bonding layer **1330** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1328** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1328** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(191) Method **1500** proceeds to operation **1508**, as illustrated in FIG. **15**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a first bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(192) As illustrated in FIG. **13D**, silicon substrate **1302** and components formed thereon (e.g., memory stack **1304** and NAND memory strings **1306** formed therethrough) are flipped upside down. Bonding layer **1310** facing down is bonded with bonding layer **1320** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1332**. That is, silicon substrate **1302** and components formed thereon can be bonded with silicon substrate **1312** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1310** are in contact with the bonding contacts in bonding layer **1320** at bonding interface **1332**. In some



implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 13D, it is understood that in some examples, silicon substrate **1312** and components formed thereon (e.g., transistors **1314** and **1316**) can be flipped upside down, and bonding layer **1320** facing down can be bonded with bonding layer **1310** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1332** as well.

(193) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1332** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1310** and the bonding contacts in bonding layer **1320** are aligned and in contact with one another, such that memory stack **1304** and NAND memory strings **1306** formed therethrough can be coupled to transistors **1314** and **1316** through the bonded bonding contacts across bonding interface **1332**, according to some implementations.

(194) In some implementations, the second substrate is thinned, and a contact through the thinned second substrate is formed. As illustrated in FIG. 13E, silicon substrate **1312** (shown in FIG. 13D) is thinned to become a semiconductor layer **1334** having single crystalline silicon. Silicon substrate **1312** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(195) As illustrated in FIG. 13E, one or more contacts **1336** each extending vertically through semiconductor layer **1334** is formed. Contacts **1336** can be coupled to the interconnects in interconnect layer **1318**. Contacts **1336** can be formed by first patterning contact holes through semiconductor layer **1334** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **1336** may be formed in silicon substrate **1312** before thinning (the formation of semiconductor layer **1334**, e.g., in FIG. 13B) and be exposed from the backside of silicon substrate **1312** (where the thinning occurs) after the thinning.

(196) In some implementations, a bonding layer is on the thinned second substrate. The bonding layer can include a plurality of bonding contacts. As illustrated in FIG. 13E, a bonding layer **1338** is formed on semiconductor layer **1334**, i.e., the backside of silicon substrate **1312** (where the thinning occurs) after the thinning. Bonding layer **1338** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the surface of semiconductor layer **1334** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **1336** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor. It is understood that in some examples, bonding layer **1338** may be a dielectric layer (e.g., a silicon oxide layer) without bonding contacts for fusion bonding, instead of hybrid bonding. It is further understood that in some examples, bonding layer **1338** may be omitted to expose the silicon surface of semiconductor layer **1334** for anodic bonding or fusion bonding, instead of hybrid bonding.

(197) Method **1500** proceeds to operation **1510**, as illustrated in FIG. 15, in which the third substrate and the second substrate are bonded in a face-to-back manner. The third bonding contact in the third bonding layer can be in contact with the fourth bonding contact in the fourth bonding layer at a second bonding interface after bonding the third and second substrates. The bonding can include hybrid bonding.

(198) As illustrated in FIG. 13F, silicon substrate **1302** and components formed thereon after bonding with silicon substrate **1312** (e.g., memory stack **1304**, NAND memory strings **1306**, and

transistors **1314** and **1316**) are flipped upside down. Bonding layer **1338** facing down is bonded with bonding layer **1330** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1340**. That is, silicon substrate **1302** and components formed thereon can be bonded with silicon substrate **1322** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1338** are in contact with the bonding contacts in bonding layer **1330** at bonding interface **1340**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **13F**, it is understood that in some examples, silicon substrate **1322** and components formed thereon (e.g., transistors **1324** and **1326**) can be flipped upside down, and bonding layer **1330** facing down can be bonded with bonding layer **1338** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1340** as well.

(199) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1340** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1338** and the bonding contacts in bonding layer **1330** are aligned and in contact with one another, such that memory stack **1304**, NAND memory strings **1306**, and transistors **1314** and **1316** can be coupled to transistors **1324** and **1326** through contacts **1336** through semiconductor layer **1334** and the bonded bonding contacts across bonding interface **1340**, according to some implementations. It is understood that in some examples, anodic bonding or fusion bonding, instead of hybrid bonding, may be performed to bond silicon substrates **1302** and **1322** (and components formed thereon) at bonding interface **1340** without bonding contacts in bonding layer **1338**.

(200) Method **1500** proceeds to operation **1512**, as illustrated in FIG. **15**, in which the first substrate or the third substrate is thinned. As illustrated in FIG. **13G**, silicon substrate **1322** (shown in FIG. **13F**) is thinned to become a semiconductor layer **1342** having single crystalline silicon. Silicon substrate **1322** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(201) Method **1500** proceeds to operation **1514**, as illustrated in FIG. **15**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned third substrate or above the array of NAND memory strings. As illustrated in FIG. **13G**, a pad-out interconnect layer **1346** is formed on semiconductor layer **1342** (the thinned silicon substrate **1322**). Pad-out interconnect layer **1346** can include interconnects, such as contact pads **1348**, formed in one or more ILD layers. Contact pads **1348** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1344** are formed extending vertically through semiconductor layer **1342**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1344** can couple contact pads **1348** in pad-out interconnect layer **1346** to the interconnects in interconnect layer **1328**. It is understood that in some examples, contacts **1344** may be formed in silicon substrate **1322** before thinning (the formation of semiconductor layer **1342**, e.g., in FIG. **13C**) and be exposed from the backside of silicon substrate **1322** (where the thinning occurs) after the thinning.

(202) In some implementations, the first substrate is thinned, and the pad-out interconnect layer is formed on the thinned first substrate. As illustrated in FIG. **13H**, silicon substrate **1302** (shown in FIG. **13F**) is thinned to become a semiconductor layer **1303** having single crystalline silicon. Silicon substrate **1302** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. As illustrated in FIG. **13H**, pad-out interconnect layer **1346** is formed on semiconductor layer **1303** (the thinned silicon substrate **1302**). Pad-out interconnect layer **1346** can include interconnects, such as contact pads **1348**, formed in one or more ILD layers. In some implementations, after the bonding and

thinning, contacts **1335** are formed extending vertically through semiconductor layer **1303**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1335** can couple contact pads **1348** in pad-out interconnect layer **1346** to the interconnects in interconnect layer **1308**. It is understood that in some examples, contacts **1335** may be formed in silicon substrate **1302** before thinning (i.e., before the formation of semiconductor layer **1303**, e.g., in FIG. **13A**) without fully penetrating through silicon substrate **1302** and be exposed from the backside of silicon substrate **1302** (where the thinning occurs) after the thinning. It is also understood that in some examples, the first substrate (e.g., silicon substrate **1302** or semiconductor layer **1303** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**.

(203) FIGS. **16A** and **16B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **9A** and **9B**, according to various aspects of the present disclosure. 3D memory devices **1600** and **1601** may be examples of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **16A**, 3D memory device **1600** can include stacked first, second, and third semiconductor structures **102**, **104**, and **106**. In some implementations, first semiconductor structure **102** on one side of 3D memory device **1600** includes a semiconductor layer **1002**, a bonding layer **1008**, and a memory cell array vertically between semiconductor layer **1002** and bonding layer **1008**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer **1002** (e.g., as shown in FIGS. **8A-8C**). Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**). Bonding layer **1008** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(204) In some implementations, second semiconductor structure **104** in the intermediate of 3D memory device **1600** includes a semiconductor layer **1004**, a bonding layer **1010**, and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1004** and bonding layer **1010**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Similar to bonding layer **1008** in first semiconductor structure **102**, bonding layer **1010** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. That is, bonding layers **1008** and **1010** can be disposed on opposite sides of bonding interface **103**, and the bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(205) In some implementations, third semiconductor structure **106** on another side of 3D memory device **1600** includes a semiconductor layer **1006** and some of the peripheral circuits of the memory cell array, such that semiconductor layer **1006** is disposed vertically between the peripheral circuits and bonding interface **105**. The transistors (e.g., planar transistors **500** and 3D

transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1006**. Semiconductor layer **1006** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1006** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. It is understood that different from bonding interface **103** between first and second semiconductor structures **102** and **104**, which is between bonding layers **1008** and **1010** and results from hybrid bonding, bonding interface **105** between second and third semiconductor structures **104** and **106** may result from transfer bonding, as described below in detail, and thus, may not be formed between two bonding layers. That is, third semiconductor structure **106** of 3D memory device **1600** in FIG. **16A** does not include a bonding layer with bonding contacts, according to some implementations. As a result, instead of bonding contacts, through contacts (e.g., ILVs/TSVs) across bonding interface **105** and through semiconductor layers **1004** and **1006** vertically between second and third semiconductor structures **104** and **106** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **104** and **106**.

(206) It is understood that in some examples, second and third semiconductor structures **104** and **106** may also include bonding layers **1012** and **1014**, respectively, disposed on opposite sides of bonding interface **105**, as shown in FIG. **16B**. In FIG. **16B**, second semiconductor structure **104** of a 3D memory device **1601** can include two bonding layers **1010** and **1012** on two sides thereof, and bonding layer **1012** can be disposed vertically between semiconductor layer **1004** and bonding interface **105**. Third semiconductor structure **106** of 3D memory device **1601** can include bonding layer **1014** disposed vertically between bonding interface **105** and semiconductor layer **1006**. Each bonding layer **1012** or **1014** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. The bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, bonding contacts across bonding interface **105** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layers **1004** and **1006** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **104** and **106**.

(207) As shown in FIGS. **16A** and **16B**, since third and second semiconductor structures **106** and **104** are bonded in a back-to-back manner (e.g., semiconductor layer **1006** being disposed on the bottom side of third semiconductor structure **106**, while semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104** in FIGS. **16A** and **16B**), the transistors in third and second semiconductor structures **106** and **104** are disposed back-to-back, according to some implementations. In some implementations, semiconductor layer **1006** is disposed vertically between the transistors of the peripheral circuits in third semiconductor structure **106** and bonding interface **105**, and the transistors of the peripheral circuits in second semiconductor structure **104** are disposed vertically between bonding interface **103** and semiconductor layer **1004**. Moreover, since first and second semiconductor structures **102** and **104** are bonded in a face-to-face manner (e.g., semiconductor layer **1002** being disposed on the bottom side of first semiconductor structure **102**, while semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104** in FIGS. **16A** and **16B**), the transistors of peripheral circuits in second semiconductor structure **104** and the memory cell array in first semiconductor structure **102** are disposed face to face, facing each other, according to some implementations. It is understood that pad-out interconnect layer **902** in FIG. **9A** or **9B** is omitted from 3D memory devices **1600** and **1601** in FIGS. **16A** and **16B** for case of illustration and may be included in 3D memory devices **1600** and **1601** as described above with respect to FIGS. **9A** and **9B**.

(208) As described above, second and third semiconductor structures **104** and **106** can have peripheral circuits having transistors with different applied voltages. For example, third semiconductor structure **106** may be one example of semiconductor structure **408** including LLV

circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and second semiconductor structure **104** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1006** and **1004** in third and second semiconductor structures **106** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and third semiconductor structure **106** may include LLV circuits **402**, and the thickness of semiconductor layer **1006** in third semiconductor structure **106** may be smaller than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in third and second semiconductor structures **106** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and third semiconductor structure **106** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in second semiconductor structure **104** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in third semiconductor structure **106**.

(209) FIGS. **17A-17C** illustrate side views of various examples of 3D memory devices **1600** and **1601** in FIGS. **16A** and **16B**, according to various aspects of the present disclosure. As shown in FIG. **17A**, as one example of 3D memory devices **1600** and **1601** in FIGS. **16A** and **16B**, 3D memory device **1700** is a bonded chip including first semiconductor structure **102**, second semiconductor structure **104**, and third semiconductor structure **106**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **17A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, and second and third semiconductor structures **104** and **106** are bonded at bonding interface **105** therebetween, according to some implementations.

(210) As shown in FIG. **17A**, third semiconductor structure **106** can include semiconductor layer **1006** having semiconductor materials. In some implementations, semiconductor layer **1006** is a silicon substrate having single crystalline silicon. In some implementations, semiconductor layer **1006** is a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the backside of second semiconductor structure **104** by transfer bonding. Third semiconductor structure **106** can also include a device layer **1702** above and in contact with semiconductor layer **1006**. In some implementations, device layer **1702** includes a first peripheral circuit **1704** and a second peripheral circuit **1706**. First peripheral circuit **1704** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and second peripheral circuit **1706** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **1704** includes a plurality of transistors **1708** in contact with semiconductor layer **1006**, and second peripheral circuit **1706** includes a plurality of transistors **1710** in contact with semiconductor layer **1006**. Transistors **1708** and **1710** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1708** or **1710** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1708** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **1710** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **1708** than transistor **1710**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1708** and **1710**) can be formed on or in semiconductor layer **1006** as well.

(211) In some implementations, third semiconductor structure **106** further includes an interconnect layer **1712** above device layer **1702** to transfer electrical signals to and from peripheral circuits **1706** and **1704**. As shown in FIG. **17A**, device layer **1702** (including transistors **1708** and **1710** of peripheral circuits **1704** and **1706**) can be disposed vertically between bonding interface **105** and interconnect layer **1712**. Interconnect layer **1712** can include a plurality of interconnects. The

interconnects in interconnect layer **1712** can be coupled to transistors **1708** and **1710** of peripheral circuits **1704** and **1706** in device layer **1702**. Interconnect layer **1712** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **1712** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1702** are coupled to one another through the interconnects in interconnect layer **1712**. For example, peripheral circuit **1704** may be coupled to peripheral circuit **1706** through interconnect layer **1712**. The interconnects in interconnect layer **1712** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1712** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(212) In some implementations, the interconnects in interconnect layer **1712** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **1712** can occur after the high-temperature processes in forming device layers **1714** and **1702** in second and third semiconductor structures **104** and **106**, as well as being separated from the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **1712** having Cu can become feasible.

(213) As shown in FIG. **17A**, second semiconductor structure **104** can further include one or more contacts **1723** extending vertically through semiconductor layer **1006**. In some implementations, contacts **1723** are coupled to the interconnects in interconnect layer **1712**. Contact **1723** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1723** includes W. In some implementations, contact **1723** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1006**. Depending on the thickness of semiconductor layer **1006**, contact **1723** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(214) Second semiconductor structure **104** can be bonded with third semiconductor structure **106** in a back-to-back manner at bonding interface **105**. Second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, bonding interface **105** is disposed vertically between interconnect layer **1112** and semiconductor layer **1004** as a result of transfer bonding, which transfers semiconductor layer **1004** from another substrate and bonds semiconductor layer **1004** onto third semiconductor structure **106** as described below in detail. In some implementations, bonding interface **105** is the place at which interconnect layer **1112** and semiconductor layer **1004** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of interconnect layer **1112** of third semiconductor structure **106** and the bottom surface of semiconductor layer **1004** of second semiconductor structure **104**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1004** and/or between bonding interface **105** and interconnect layer **1112** to facilitate the transfer bonding of semiconductor layer **1004** onto interconnect layer **1112**. Thus, it is understood that bonding interface **105** may include the surfaces of the dielectric layer(s) in some examples.

(215) Second semiconductor structure **104** can include a device layer **1714** below and in contact with semiconductor layer **1004**. In some implementations, device layer **1714** includes a third peripheral circuit **1716** and a fourth peripheral circuit **1718**. Third peripheral circuit **1716** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and fourth peripheral circuit **1718** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral

circuit **1716** includes a plurality of transistors **1720**, and fourth peripheral circuit **1718** includes a plurality of transistors **1722** as well. Transistors **1720** and **1722** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1720** or **1722** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1720** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1722** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **1720** than transistor **1722**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1720** and **1722**) can be formed on or in semiconductor layer **1004** as well.

(216) Moreover, the different voltages applied to different transistors **1720**, **1722**, **1708**, and **1710** in second and third semiconductor structures **104** and **106** can lead to differences of device dimensions between second and third semiconductor structures **104** and **106**. In some implementations, the thickness of the gate dielectric of transistor **1720** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1708** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **1720** than transistor **1708**. In some implementations, the thickness of the gate dielectric of transistor **1722** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **1710** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **1722** and transistor **1710**. In some implementations, the thickness of semiconductor layer **1006** in which transistor **1708** (e.g., in LLV circuit **402**) is formed is smaller than the thickness of semiconductor layer **1004** in which transistor **1720** (e.g., in HV circuit **406**) is formed due to the lower voltage applied to transistor **1708** than transistor **1720**.

(217) As shown in FIG. **17A**, second semiconductor structure **104** can further include an interconnect layer **1726** below device layer **1714** to transfer electrical signals to and from peripheral circuits **1716** and **1718**. As shown in FIG. **17A**, interconnect layer **1726** can be vertically between bonding interface **103** and device layer **1714** (including transistors **1720** and **1722** of peripheral circuits **1716** and **1718**). Interconnect layer **1726** can include a plurality of interconnects coupled to transistors **1720** and **1722** of peripheral circuits **1716** and **1718** in device layer **1714**. Interconnect layer **1726** can further include one or more ILD layers in which the interconnects can form. That is, interconnect layer **1126** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1714** are coupled to one another through the interconnects in interconnect layer **1726**. For example, peripheral circuit **1716** may be coupled to peripheral circuit **1718** through interconnect layer **1726**. The interconnects in interconnect layer **1726** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1726** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1726** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(218) As shown in FIG. **17A**, second semiconductor structure **104** can further include one or more contacts **1724** extending vertically through semiconductor layer **1004**. In some implementations, contacts **1724** are coupled to the interconnects in interconnect layer **1726**. In some implementations, contact **1724** is in contact with contact **1723**, such that contacts **1723** and **1724** couple the interconnects in interconnect layer **1726** to the interconnects in interconnect layer **1712** to make an electrical connection across bonding interface **105** between second and third semiconductor structures **104** and **106** and through semiconductor layers **1004** and **1006**. Contact **1724** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1724** includes W. In some implementations, contact **1724** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **1724** can be an ILV having a depth in the submicron-level (e.g.,

between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(219) As shown in FIG. 17A, second semiconductor structure **104** can further include a bonding layer **1010** at bonding interface **103** and above and in contact with interconnect layer **1726**. Bonding layer **1010** can include a plurality of bonding contacts **1011** and dielectrics electrically isolating bonding contacts **1011**. Bonding contacts **1011** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, bonding contacts **1011** of bonding layer **1010** include Cu. The remaining area of bonding layer **1010** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1011** and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously.

(220) As shown in FIG. 17A, first semiconductor structure **102** can further include a bonding layer **1008** at bonding interface **103**, e.g., on the opposite side of bonding interface **103** with respect to bonding layer **1010** in second semiconductor structure **104**. Bonding layer **1008** can include a plurality of bonding contacts **1009** and dielectrics electrically isolating bonding contacts **1009**. Bonding contacts **1009** can include conductive materials, such as Cu. The remaining area of bonding layer **1008** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1009** and surrounding dielectrics in bonding layer **1008** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1010** of second semiconductor structure **104** and the bottom surface of bonding layer **1008** of first semiconductor structure **102**.

(221) As shown in FIG. 17A, first semiconductor structure **102** can further include an interconnect layer **1728** below and in contact with bonding layer **1008** to transfer electrical signals. Interconnect layer **1728** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1728** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **1728** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **1728** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1728** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(222) As shown in FIG. 17A, first semiconductor structure **102** can include a memory cell array, such as an array of NAND memory strings **208** below and in contact with interconnect layer **1728**. In some implementations, interconnect layer **1728** is vertically between NAND memory strings **208** and bonding interface **103**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **1727**. Memory stack **1727** may be an example of memory stack **804** in FIGS. 8A-8C, and the conductive layer and dielectric layer in memory stack **1727** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **1727** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **1727**.



(223) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(224) As shown in FIG. **17A**, first semiconductor structure **102** can further include semiconductor layer **1002** disposed below memory stack **1727** and in contact with the sources of NAND memory strings **208**. In some implementations, NAND memory strings **208** are disposed vertically between bonding interface **103** and semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **1727** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(225) As shown in FIG. **17A**, third semiconductor structure **106** can further include a pad-out interconnect layer **902** above and in contact with interconnect layer **1712**. In some implementations, device layer **1702** having transistors **1708** and **1710** is disposed vertically between pad-out interconnect layer **902** and semiconductor layer **1006**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **1732**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **1712** can be formed on the same side of semiconductor layer **1006**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **1700** and external devices, e.g., for pad-out purposes.

(226) As a result, peripheral circuits **1704**, **1706**, **1716**, and **1718** in third and second semiconductor structures **106** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **1712**, **1726**, and **1728**, bonding layers **1008** and **1010**, as well as contacts **1723** and **1724**. Moreover, peripheral circuits **1704**, **1706**, **1716**, and **1718** and NAND memory strings **208** in 3D memory device **1700** can be further coupled to external devices through pad-out interconnect layer **902**.

(227) It is understood that in some examples, similar to bonding interface **103**, bonding interface **105** may result from hybrid bonding and thus, be disposed vertically between two bonding layers each including bonding contacts in second and third semiconductor structures **104** and **106**, respectively. For example, as shown in FIG. **17B**, a 3D memory device **1701** may include bonding layers **1012** and **1014** in second and third semiconductor structures **104** and **106**, respectively, at bonding interface **105**, i.e., on opposite sides of bonding interface **105**. Bonding layer **1012** or **1014** can include a plurality of bonding contacts **1013** or **1015** and dielectrics electrically isolating bonding contacts **1013** or **1015**. Bonding contacts **1013** and **1015** can include conductive materials, such as Cu. The remaining area of bonding layer **1012** or **1014** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1013** or **1015** and surrounding dielectrics in bonding layer **1012** or **1014** can be used for hybrid bonding. In some implementations, bonding interface **105** is the place at which bonding layers **1012** and **1014** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of bonding layer **1014** of third semiconductor structure **106** and the bottom surface of bonding layer **1012** of second semiconductor structure **104**. Contact **1723** can be coupled to contact **1724** through bonding contacts **1013** and **1015** of bonding layers **1012** and **1014** across bonding interface **105**.

(228) It is also understood that the pad-out of 3D memory devices is not limited to from third semiconductor structure **106** having transistors **1708** and **1710** as shown in FIG. **17A** (corresponding to FIG. **9A**) and may be from first semiconductor structure **102** having NAND memory strings **208** (corresponding to FIG. **9B**). For example, as shown in FIG. **17B**, 3D memory

device **1701** may include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1002** of first semiconductor structure **102** on which NAND memory strings **208** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **1730** extending vertically through semiconductor layer **1002**. In some implementations, contact **1730** couples the interconnects in interconnect layer **1728** in first semiconductor structure **102** to contact pads **1732** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **1730** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1730** includes W. In some implementations, contact **1730** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **1730** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu$ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu$ m and 100  $\mu$ m). In some implementations, in FIG. **17B**, third semiconductor structure **106** of 3D memory device **1701** further includes a passivation layer **1734**, replacing pad-out interconnect layer **902** in FIG. **17B**. Passivation layer **1734** can include dielectric materials, such as silicon nitride and/or silicon oxide. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1700** and **1701** are not repeated for case of description.

(229) It is further understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIGS. **17A** and **17B** and may be any other suitable semiconductor materials. For example, as shown in FIG. **17C**, a 3D memory device **1703** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **1703** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **1703** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1700** and **1703** are not repeated for case of description.

(230) FIGS. **18A-18F** illustrate a fabrication process for forming the 3D memory devices in FIGS. **16A** and **16B**, according to some aspects of the present disclosure. FIG. **20** illustrates a flowchart of a method **2000** for forming the 3D memory devices in FIGS. **16A** and **16B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **18A-18F** and **20** include 3D memory devices **1700**, **1701**, and **1703** depicted in FIGS. **17A-17C**. FIGS. **18A-18F** and **20** will be described together. It is understood that the operations shown in method **2000** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **20**. In one example, operation **2002** may be performed after operation **2008** or in parallel with operations **2004-2008**. In another example, operation **2010** may be performed before operations **2006** and **2008**.

(231) Referring to FIG. **20**, method **2000** starts at operation **2002**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(232) As illustrated in FIG. **18D**, a stack structure, such as a memory stack **1826** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **1824**. To form memory stack **1826**, in some implementations, a dielectric stack (not shown) including interleaved

sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **1824**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1826** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1826** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1826** and silicon substrate **1824**. (233) As illustrated in FIG. **18D**, NAND memory strings **1828** are formed above silicon substrate **1824**, each of which extends vertically through memory stack **1826** to be in contact with silicon substrate **1824**. In some implementations, fabrication processes to form NAND memory string **1828** include forming a channel hole through memory stack **1826** (or the dielectric stack) and into silicon substrate **1824** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1828** may vary depending on the types of channel structures of NAND memory strings **1828** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(234) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18D**, an interconnect layer **1830** is formed above memory stack **1826** and NAND memory strings **1828**. Interconnect layer **1830** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1828**. In some implementations, interconnect layer **1830** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1830** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, chemical mechanical polishing (CMP), wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **18D** can be collectively referred to as interconnect layer **1830**.

(235) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **18D**, a bonding layer **1832** is formed above interconnect layer **1830**. Bonding layer **1832** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1830** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1830** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer

before depositing the conductor.

(236) Method **2000** proceeds to operation **2004**, as illustrated in FIG. **20**, in which a first transistor is formed on a first side (e.g., a first surface) of a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. The first side can be the front side on which devices are formed on the second substrate.

(237) As illustrated in FIG. **18A**, a plurality of transistors **1804** and **1806** are formed on the front side of a silicon substrate **1802**. Transistors **1804** and **1806** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1802** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1804** and **1806**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1802** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1804** is different from the thickness of gate dielectric of transistor **1806**, for example, by depositing a thicker silicon oxide film in the region of transistor **1804** than the region of transistor **1806**, or by etching back part of the silicon oxide film deposited in the region of transistor **1806**. It is understood that the details of fabricating transistors **1804** and **1806** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(238) In some implementations, an interconnect layer **1808** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18A**, an interconnect layer **1808** can be formed above transistors **1804** and **1806**. Interconnect layer **1808** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1804** and **1806**. In some implementations, interconnect layer **1808** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1808** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **18A** can be collectively referred to as interconnect layer **1808**. In some implementations, the interconnects in interconnect layer **1808** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(239) In some implementations, a second bonding layer is formed above the interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **18A**, a bonding layer **1822** is formed above interconnect layer **1808**. Bonding layer **1822** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1808** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1808** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(240) Method **2000** proceeds to operation **2006**, as illustrated in FIG. **20**, in which a semiconductor layer is formed on a second side (e.g., a second surface) of the second substrate opposite to the first side. The semiconductor layer can include single crystalline silicon. The second side can be the

backside of the second substrate. In some implementations, to form the semiconductor layer, another substrate and the second substrate are bonded in a face-to-back manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(241) In some implementations, the second substrate is thinned prior to forming the semiconductor layer, such that the semiconductor layer is formed on the second side of the thinned second substrate. As illustrated in FIG. **18B**, silicon substrate **1802** (shown in FIG. **18A**) is thinned to become a semiconductor layer **1809** having single crystalline silicon. Silicon substrate **1802** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, as shown in FIG. **18BD**, a handle substrate **1801** (a.k.a., carrier wafer) is attached to bonding layer **1822**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **1802**, such as thinning, contact formation, and bonding.

(242) In some implementations, a first contact through the thinned second substrate is formed. As illustrated in FIG. **18B**, one or more contacts **1817** each extending vertically through semiconductor layer **1809** (i.e., the thinned silicon substrate **1802**) are formed. Contacts **1817** can be coupled to the interconnects in interconnect layer **1808**. Contacts **1817** can be formed by first patterning contact holes through semiconductor layer **1809** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **1817** may be formed in silicon substrate **1802** before thinning (the formation of semiconductor layer **1809**, e.g., in FIG. **18A**) and be exposed from the backside of silicon substrate **1802** (where the thinning occurs) after the thinning.

(243) As illustrated in FIG. **18B**, a semiconductor layer **1810**, such as a single crystalline silicon layer, is formed on the backside (the side where the thinning occurs) of semiconductor layer **1809** (i.e., the thinned silicon substrate **1802**). Semiconductor layer **1810** can be attached to the backside of semiconductor layer **1810** to form a bonding interface **1812** vertically between semiconductor layer **1810** and semiconductor layer **1809**. In some implementations, to form semiconductor layer **1810**, another silicon substrate (not shown in FIG. **18B**) and semiconductor layer **1809** (i.e., the thinned silicon substrate **1802**) are bonded in a face-to-back manner (flipping thinned silicon substrate **1802** upside down and having the components formed on silicon substrate **1802**, such as transistors **1804** and **1806**, facing away from the other silicon substrate) using transfer bonding, thereby forming bonding interface **1812**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **1810** attached to the backside of semiconductor layer **1809** (i.e., the thinned silicon substrate **1802**). The details of various transfer bonding processes are described above with respect to FIGS. **48A-48D** and FIGS. **49A-49D** and thus, are not repeated for case of description.

(244) Referring to FIG. **20**, method **2000** proceeds to operation **2008**, in which a second transistor is formed on the semiconductor layer. As illustrated in FIG. **18C**, a plurality of transistors **1814** and **1816** are formed on semiconductor layer **1810** having single crystalline silicon. Transistors **1814** and **1816** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **1810** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1814** and **1816**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **1810** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1814** is different from the thickness of gate dielectric of transistor **1816**, for example, by depositing a thicker silicon oxide film in the region of transistor **1814** than the region of transistor **1816**, or by etching back part of the silicon

oxide film deposited in the region of transistor **1816**. It is understood that the details of fabricating transistors **1814** and **1816** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(245) In some implementations, an interconnect layer **1820** is formed above the transistor on the semiconductor layer. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18C**, an interconnect layer **1820** can be formed above transistors **1814** and **1816**. Interconnect layer **1820** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1814** and **1816**. In some implementations, interconnect layer **1820** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1820** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **18C** can be collectively referred to as interconnect layer **1820**. Different from interconnect layer **1808**, in some implementations, the interconnects in interconnect layer **1820** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **1820** may become feasible since there is no more high temperature processes after the fabrication of interconnect layer **1820**.

(246) In some implementations, a second contact through the semiconductor layer and coupled to the first contact is formed. As illustrated in FIG. **18C**, one or more contacts **1818** each extending vertically through semiconductor layer **1810** are formed. Contact **1818** can be aligned to be in contact with contact **1817** at bonding interface **1812**. Contacts **1818** and **1817** can couple the interconnects in interconnect layers **1820** and **1808** across bonding interface **1812** and through semiconductor layers **1810** and **1809**. Contacts **1818** can be formed by first patterning contact holes through semiconductor layer **1810** and aligned with contacts **1817** at bonding interface **1812** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(247) Method **2000** proceeds to operation **2010**, as illustrated in FIG. **20**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(248) As illustrated in FIG. **18E**, after removing handle substrate **1801** (e.g., shown in FIG. **18C**) to expose bonding layer **1822**, thinned silicon substrate **1802** (i.e., semiconductor layer **1809**) and components formed thereon (e.g., transistors **1804** and **1806**) are flipped upside down. Bonding layer **1822** facing down is bonded with bonding layer **1832** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1834**. That is, thinned silicon substrate **1802** and components formed thereon can be bonded with silicon substrate **1824** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1822** are in contact with the bonding contacts in bonding layer **1832** at bonding interface **1834**. Transistors **1806** and **1804** and NAND memory strings **1828** can face toward each other after the bonding. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **18E**, it is

understood that in some examples, silicon substrate **1824** and components formed thereon (e.g., memory stack **1826** and NAND memory strings **1828**) can be flipped upside down, and bonding layer **1832** facing down can be bonded with bonding layer **1822** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1834** as well.

(249) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1834** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1832** and the bonding contacts in bonding layer **1822** are aligned and in contact with one another, such that memory stack **1826** and NAND memory strings **1828** formed therethrough can be coupled to transistors **1814**, **1816**, **1804**, and **1806** through the bonded bonding contacts across bonding interface **1834**, according to some implementations. It is understood that in some examples, a bonding layer may be formed above interconnect layer **1820**, instead of interconnect layer **1808**, and thinned silicon substrate **1802** and components formed thereon can be bonded with silicon substrate **1824** and components formed thereon in a back-to-face manner, such that transistors **1816** and **1814** and NAND memory strings **1828** may face toward each other after the bonding.

(250) It is understood that in some examples, operation **2010** may be performed before operations **2006** and **2008**. That is, after the formation of the array of NAND memory strings on the first substrate at operation **2002** and the formation of the first transistor on the first side of the second substrate at operation **2004** (operations **2002** and **2004** may be performed in parallel), method **2000** may proceed to operation **2010** to bond the first and second substrates in a face-to-face matter. Method **2000** then may proceed to operation **2006** to form the semiconductor layer on the second side of the second substrate and operation **2008** to form the second transistor on the semiconductor layer. Accordingly, since the bonded first substrate (e.g., silicon substrate **1824** in FIG. **18D**) can serve as the base substrate when performing operations **2006** and **2008**, the attachment of the handle substrate (e.g., handle substrate **1801** in FIG. **18B**) may not be needed to simplify the process.

(251) Method **2000** skips optional operation **2012** and proceeds to operation **2014**, as illustrated in FIG. **20**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **18F**, a pad-out interconnect layer **1836** is formed above interconnect layer **1820** and transistors **1814** and **1816** on semiconductor layer **1810**. Pad-out interconnect layer **1836** can include interconnects, such as contact pads **1838**, formed in one or more ILD layers. Contact pads **1838** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(252) In some implementations, to form a pad-out interconnect layer on the first substrate, after operation **2010**, method **2000** proceeds to optional operation **2012**, as illustrated in FIG. **20**, in which the first substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **1824** (shown in FIG. **18E**) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **1824**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **1824** before thinning and be exposed from the backside of silicon substrate **1824** (where the thinning occurs) after the thinning.

(253) Method **2000** proceeds to operation **2014**, as illustrated in FIG. **20**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **1824**.

(254) FIGS. **19A-19F** illustrate another fabrication process for forming the 3D memory devices in FIGS. **16A** and **16B**, according to some aspects of the present disclosure. FIG. **21** illustrates a flowchart of another method **2100** for forming the 3D memory devices in FIGS. **16A** and **16B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **19A-19F** and **21** include 3D memory devices **1700**, **1701**, and **1703** depicted in FIGS. **17A-17C**. FIGS. **19A-19F** and **21** will be described together. It is understood that the operations shown in method **2100** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **21**. In one example, operation **2102**, **2104**, and **2106** may be performed in parallel. In another example, operation **2110** may be performed before operation **2108**.

(255) Referring to FIG. **21**, method **2100** starts at operation **2102**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(256) As illustrated in FIG. **19A**, a stack structure, such as a memory stack **1904** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **1902**. To form memory stack **1904**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **1902**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1904** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1904** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1904** and silicon substrate **1902**.

(257) As illustrated in FIG. **19A**, NAND memory strings **1906** are formed above silicon substrate **1902**, each of which extends vertically through memory stack **1904** to be in contact with silicon substrate **1902**. In some implementations, fabrication processes to form NAND memory string **1906** include forming a channel hole through memory stack **1904** (or the dielectric stack) and into silicon substrate **1902** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1906** may vary depending on the types of channel structures of NAND memory strings **1906** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(258) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **19A**, an interconnect layer **1908** is formed above memory stack **1904** and NAND memory strings **1906**. Interconnect layer **1908** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1906**. In some implementations, interconnect layer **1908** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1908** can include conductive materials deposited by one or more



thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19A** can be collectively referred to as interconnect layer **1908**.

(259) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **19A**, a bonding layer **1910** is formed above interconnect layer **1308**. Bonding layer **1910** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1908** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1908** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(260) Method **2100** proceeds to operation **2104**, as illustrated in FIG. **21**, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **19B**, a plurality of transistors **1914** and **1916** are formed on a silicon substrate **1912**. Transistors **1914** and **1916** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1912** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1914** and **1916**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1912** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1914** is different from the thickness of gate dielectric of transistor **1916**, for example, by depositing a thicker silicon oxide film in the region of transistor **1914** than the region of transistor **1916**, or by etching back part of the silicon oxide film deposited in the region of transistor **1916**. It is understood that the details of fabricating transistors **1914** and **1916** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(261) In some implementations, an interconnect layer **1918** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **19B**, an interconnect layer **1918** can be formed above transistors **1914** and **1916**. Interconnect layer **1918** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1914** and **1916**. In some implementations, interconnect layer **1918** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1918** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19B** can be collectively referred to as interconnect layer **1918**.

(262) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG.

**19B**, a bonding layer **1920** is formed above interconnect layer **1918**. Bonding layer **1920** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1918** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1918** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(263) Method **2100** proceeds to operation **2106**, as illustrated in FIG. **21**, in which a second transistor is formed on a third substrate. The third substrate can be a silicon substrate having single crystalline silicon. In some implementations, any two or all of operations **2102**, **2104**, and **2106** are performed in parallel to reduce process time.

(264) As illustrated in FIG. **19C**, a plurality of transistors **1924** and **1926** are formed on a silicon substrate **1922**. Transistors **1924** and **1926** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1922** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1924** and **1926**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1922** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1924** is different from the thickness of gate dielectric of transistor **1926**, for example, by depositing a thicker silicon oxide film in the region of transistor **1924** than the region of transistor **1926**, or by etching back part of the silicon oxide film deposited in the region of transistor **1926**. It is understood that the details of fabricating transistors **1924** and **1926** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(265) In some implementations, an interconnect layer **1928** is formed above the transistor on the third substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **19C**, an interconnect layer **1928** can be formed above transistors **1924** and **1926**. Interconnect layer **1928** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1924** and **1926**. In some implementations, interconnect layer **1928** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1928** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **19C** can be collectively referred to as interconnect layer **1928**.

(266) In some implementations, at least one of the second substrate or the third substrate is thinned. As illustrated in FIG. **19D**, silicon substrate **1912** (shown in FIG. **19B**) is thinned to become a semiconductor layer **1935** having single crystalline silicon. Similarly, silicon substrate **1922** (shown in FIG. **19C**) is thinned to become a semiconductor layer **1923** having single crystalline silicon. Silicon substrate **1912** or **1922** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, as shown in FIG. **19D**, a handle substrate **1901** is attached to bonding layer **1920**, and a handle substrate **1903** is attached to interconnect layer **1928**, for example, using

adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrates **1912** and **1922**, such as thinning, contact formation, and bonding.

(267) In some implementations, a first contact through the thinned second substrate is formed. In some implementations, a second contact through the thinned third substrate is formed, such that the second contact is coupled to the first contact after bonding the thinned third and second substrates. As illustrated in FIG. **19D**, one or more contacts **1936** each extending vertically through semiconductor layer **1935** (i.e., the thinned silicon substrate **1912**) are formed. Contacts **1936** can be coupled to the interconnects in interconnect layer **1918**. Similarly, one or more contacts **1937** each extending vertically through semiconductor layer **1923** (i.e., the thinned silicon substrate **1922**) are formed. Contacts **1937** can be coupled to the interconnects in interconnect layer **1928**. Contact **1937** or **1936** can be formed by first patterning contact holes through semiconductor layer **1923** or **1935** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **1936** may be formed in silicon substrate **1912** before thinning (the formation of semiconductor layer **1935**, e.g., in FIG. **19B**) and be exposed from the backside of silicon substrate **1912** (where the thinning occurs) after the thinning. Similarly, contacts **1937** may be formed in silicon substrate **1922** before thinning (the formation of semiconductor layer **1923**, e.g., in FIG. **19C**) and be exposed from the backside of silicon substrate **1922** (where the thinning occurs) after the thinning.

(268) In some implementations, a third bonding layer is formed on a second side of the thinned second substrate opposite to a first side on which the transistor is formed, and a fourth bonding layer is formed on a second side of the thinned third substrate opposite to a first side on which the transistor is formed. The third bonding layer can include a plurality of third bonding contacts, and the fourth bonding layer can include a plurality of fourth bonding contacts. As illustrated in FIG. **19D**, a bonding layer **1939** is formed on the backside of semiconductor layer **1935** (i.e., the thinned silicon substrate **1912**), and a bonding layer **1941** is formed on the backside of semiconductor layer **1923** (i.e., the thinned silicon substrate **1922**). Bonding layer **1939** or **1941** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the surface of semiconductor layer **1935** or **1923** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **1936** and **1937** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(269) Method **2100** proceeds to operation **2108**, as illustrated in FIG. **21**, in which the third substrate and the second substrate are bonded in a back-to-back manner. The third bonding contact in the third bonding layer can be in contact with the fourth bonding contact in the fourth bonding layer at a first bonding interface after bonding the third and second substrates. The bonding can include hybrid bonding.

(270) As illustrated in FIG. **19D**, thinned silicon substrate **1922** (i.e., semiconductor layer **1923**) and components formed thereon (e.g., transistors **1924** and **1926**) are flipped upside down. Bonding layer **1941** on the backside of thinned silicon substrate **1922** facing up is bonded with bonding layer **1939** on the backside of thinned silicon substrate **1912** facing down, i.e., in a back-to-back manner, thereby forming a bonding interface **1940**. That is, thinned silicon substrate **1922** and components formed thereon can be bonded with thinned silicon substrate **1912** and components formed thereon in a back-to-back manner, such that the bonding contacts in bonding layer **1941** are in contact with the bonding contacts in bonding layer **1939** at bonding interface **1940**. In some

implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 13D, it is understood that in some examples, thinned silicon substrate **1912** and components formed thereon (e.g., transistors **1914** and **1916**) can be flipped upside down, and bonding layer **1939** facing up can be bonded with bonding layer **1941** facing down, i.e., in a back-to-back manner, thereby forming bonding interface **1940** as well.

(271) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1940** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1939** and the bonding contacts in bonding layer **1941** are aligned and in contact with one another, such that contacts **1936** can be coupled to contacts **1937**, and transistors **1924** and **1926** can be coupled to transistors **1914** and **1916** through the bonded bonding contacts across bonding interface **1940** and contacts **1936** and **1937**, according to some implementations. It is understood that in some examples, anodic bonding or fusion bonding, instead of hybrid bonding, may be performed to bond thinned silicon substrates **1912** and **1922** (and components formed thereon) at bonding interface **1940** in a back-to-back manner without bonding contacts in bonding layer **1939** and/or bonding layer **1941**.

(272) Method **2100** proceeds to operation **2110**, as illustrated in FIG. 21, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a first bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(273) As illustrated in FIG. 19E, handle substrate **1901** (shown in FIG. 19D) attached to bonding layer **1920** is removed and expose bonding layer **1920**, and silicon substrate **1902** and components formed thereon (e.g., memory stack **1904** and NAND memory strings **1906** formed therethrough) are flipped upside down. Bonding layer **1910** facing down is bonded with bonding layer **1920** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1932**. That is, silicon substrate **1902** and components formed thereon can be bonded with thinned silicon substrate **1912** (i.e., semiconductor layer **1935**) and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1910** are in contact with the bonding contacts in bonding layer **1920** at bonding interface **1932**. Transistors **1914** and **1916** and NAND memory strings **1906** can face toward each other after the bonding. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 19E, it is understood that in some examples, thinned silicon substrate **1912** and components formed thereon (e.g., transistors **1914** and **1916**) can be flipped upside down, and bonding layer **1920** facing down can be bonded with bonding layer **1910** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1932** as well.

(274) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1932** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1910** and the bonding contacts in bonding layer **1920** are aligned and in contact with one another, such that memory stack **1904** and NAND memory strings **1906** formed therethrough can be coupled to transistors **1914** and **1916** through the bonded bonding contacts across bonding interface **1932**, according to some implementations. It is understood that in some examples, a bonding layer may be formed above interconnect layer **1928**, instead of interconnect layer **1918**, and thinned silicon substrate **1922** (i.e., semiconductor layer **1923**) and components formed thereon can be bonded with silicon substrate **1902** and components formed thereon in a face-to-face manner, such that transistors **1926** and **1924** and NAND memory strings **1906** may face toward each other after the bonding.

(275) It is understood that in some examples, operation **2110** may be performed before operation **2108**. That is, after the formation of the array of NAND memory strings on the first substrate at operation **2102**, the formation of the first transistor on the second substrate at operation **2104**, and

the formation of the second transistor on the third substrate at operation **2106** (operations **2102**, **2104**, and **2106** may be performed in parallel), method **2100** may perform operation **2110** to bond the first and second substrates in a face-to-face matter. Method **2100** then may proceed to operation **2108** to bond the third and second substrates in a back-to-back manner. Accordingly, since the bonded first substrate (e.g., silicon substrate **1902** in FIG. **19A**) can serve as the base substrate when performing operation **2108**, the attachment of the carrier substrate (e.g., carrier substrate **1901** in FIG. **19D**) can be skipped to simplify the process.

(276) Method **2100** proceeds to optional operation **2112**, as illustrated in FIG. **21**, in which the first substrate is thinned. As illustrated in FIG. **19F**, silicon substrate **1902** (shown in FIG. **19E**) is thinned to become a semiconductor layer **1934** having single crystalline silicon. Silicon substrate **1902** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(277) Method **2100** proceeds to operation **2114**, as illustrated in FIG. **21**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. As illustrated in FIG. **19F**, a pad-out interconnect layer **1948** is formed on semiconductor layer **1934** (the thinned silicon substrate **1902**). Pad-out interconnect layer **1948** can include interconnects, such as contact pads **1938**, formed in one or more ILD layers. Contact pads **1938** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1944** are formed extending vertically through semiconductor layer **1934**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1944** can couple contact pads **1938** in pad-out interconnect layer **1948** to the interconnects in interconnect layer **1908**. In some implementations, handle substrate **1903** (e.g., shown in FIG. **19E**) attached to interconnect layer **1928** is removed to expose interconnect layer **1928**, and a passivation layer **1942** is then formed on interconnect layer **1928** by depositing dielectric materials, such as silicon nitride, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. It is understood that in some examples, contacts **1944** may be formed in silicon substrate **1902** before thinning (the formation of semiconductor layer **1934**, e.g., in FIG. **19A**) and be exposed from the backside of silicon substrate **1902** (where the thinning occurs) after the thinning.

(278) In some implementations, after operation **2110**, optional operation **2112** is skipped, and method **2100** proceeds to operation **2114**, as illustrated in FIG. **21**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. Although not shown in FIG. **19F**, it is understood that in some examples, a pad-out interconnect layer having contact pads may be formed above interconnect layer **1908** and transistors **1926** and **1924** after removing handle substrate **1903**. It is further understood that in some examples, the first substrate (e.g., silicon substrate **1902** or semiconductor layer **1934** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**.

(279) FIGS. **22A** and **22B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **9A** and **9B**, according to various aspects of the present disclosure. 3D memory devices **2200** and **2201** may be examples of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **22A**, 3D memory device **2200** can include stacked first, second, and third semiconductor structures **102**, **104**, and **106**. In some implementations, first semiconductor structure **102** on one side of 3D memory device **2200** includes semiconductor layer **1002** and a memory cell array vertically between semiconductor layer **1002** and bonding interface **103**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer

**1002** (e.g., as shown in FIGS. **8A-8C**). Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**).

(280) In some implementations, second semiconductor structure **104** in the intermediate of 3D memory device **2200** includes a semiconductor layer **1004**, a bonding layer **1012**, and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1004** and bonding layer **1012**. In some implementations, semiconductor layer **1004** is disposed vertically between bonding interface **103** and the peripheral circuits of second semiconductor structure **104**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Bonding interface **103** between first and second semiconductor structures **102** and **104** may result from transfer bonding. Through contacts (e.g., ILVs/TSVs) across bonding interface **103** and through semiconductor layer **1004** vertically between first and second semiconductor structures **102** and **104** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **102** and **104**. Bonding layer **1012** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding.

(281) In some implementations, third semiconductor structure **106** on another side of 3D memory device **2200** includes a semiconductor layer **1006**, a bonding layer **1014**, and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1006** and bonding interface **105**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1006**. Semiconductor layer **1006** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1006** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Similar to bonding layer **1012**, bonding layer **1014** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding. Bonding interface **105** is vertically between and in contact with bonding layers **1012** and **1014**, respectively, according to some implementations. That is, bonding layers **1012** and **1014** can be disposed on opposite sides of bonding interface **105**, and the bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **105** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(282) It is understood that in some examples, first and second semiconductor structures **102** and **104** may also include bonding layers **1008** and **1010**, respectively, disposed on opposite sides of bonding interface **103**, as shown in FIG. **22B**. In FIG. **22B**, second semiconductor structure **104** of a 3D memory device **2201** can include two bonding layers **1010** and **1012** on two sides thereof, and bonding layer **1010** can be disposed vertically between semiconductor layer **1004** and bonding interface **103**. First semiconductor structure **102** of 3D memory device **2201** can include bonding layer **1008** disposed vertically between bonding interface **103** and semiconductor layer **1002**. Each

bonding layer **1008** or **1010** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. The bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, bonding contacts across bonding interface **103** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1004** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(283) As shown in FIGS. **22A** and **22B**, since third and second semiconductor structures **106** and **104** are bonded in a face-to-face manner (e.g., semiconductor layer **1006** being disposed on the bottom side of third semiconductor structure **106**, while semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104** in FIGS. **22A** and **22B**), the transistors in third semiconductor structure **106** and the transistors in second semiconductor structure **104** face toward each other, according to some implementations. In some implementations, semiconductor layer **1004** is disposed vertically between the transistors of the peripheral circuits in second semiconductor structure **104** and bonding interface **103**, and the transistors of the peripheral circuits in third semiconductor structure **106** are disposed vertically between bonding interface **105** and semiconductor layer **1006**. Moreover, since first and second semiconductor structures **102** and **104** are bonded in a face-to-back manner (e.g., semiconductor layers **1002** and **1004** being disposed on the top sides of first and second semiconductor structures **102** and **104**, respectively, in FIGS. **22A** and **22B**), the transistors of peripheral circuits in second semiconductor structure **104** and the memory cell array in first semiconductor structure **102** face toward the same direction (e.g., the negative y-direction in FIGS. **22A** and **22B**), according to some implementations. It is understood that pad-out interconnect layer **902** in FIG. **9A** or **9B** is omitted from 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B** for ease of illustration and may be included in 3D memory devices **2200** and **2201** as described above with respect to FIGS. **9A** and **9B**.

(284) As described above, second and third semiconductor structures **104** and **106** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and third semiconductor structure **106** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1006** and **1004** in third and second semiconductor structures **106** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1006** in third semiconductor structure **106** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in third and second semiconductor structures **106** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in third semiconductor structure **106** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**.

(285) FIGS. **23A-23C** illustrate side views of various examples of 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B**, according to various aspects of the present disclosure. As shown in FIG. **23A**, as one example of 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B**, 3D memory device **2300** is a bonded chip including first semiconductor structure **102**, second semiconductor structure **104**, and third semiconductor structure **106**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **23A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, and second and third semiconductor structures **104** and **106**

are bonded at bonding interface **105** therebetween, according to some implementations.

(286) As shown in FIG. **23A**, third semiconductor structure **106** can include semiconductor layer **1006** having semiconductor materials. In some implementations, semiconductor layer **1006** is a silicon substrate having single crystalline silicon. Third semiconductor structure **106** can also include a device layer **2302** above and in contact with semiconductor layer **1006**. In some implementations, device layer **2302** includes a first peripheral circuit **2304** and a second peripheral circuit **1106**. First peripheral circuit **2304** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **2306** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **2304** includes a plurality of transistors **2308** in contact with semiconductor layer **1006**, and second peripheral circuit **2306** includes a plurality of transistors **2310** in contact with semiconductor layer **1006**. Transistors **2308** and **2310** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2308** or **2310** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2308** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2310** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **2308** than transistor **2310**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **2308** and **2310**) can be formed on or in semiconductor layer **1006** as well.

(287) In some implementations, third semiconductor structure **106** further includes an interconnect layer **2312** above device layer **2302** to transfer electrical signals to and from peripheral circuits **2306** and **2304**. As shown in FIG. **23A**, interconnect layer **2312** can be vertically between bonding interface **105** and device layer **2302** (including transistors **2308** and **2310** of peripheral circuits **2304** and **2306**). Interconnect layer **2312** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnects in interconnect layer **2312** can be coupled to transistors **2308** and **2310** of peripheral circuits **2304** and **2306** in device layer **2302**. Interconnect layer **2312** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **2312** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **2302** are coupled to one another through the interconnects in interconnect layer **2312**. For example, peripheral circuit **2304** may be coupled to peripheral circuit **2306** through interconnect layer **2312**. The interconnects in interconnect layer **2312** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2312** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(288) In some implementations, the interconnects in interconnect layer **2312** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **2312** can be separated from the high-temperature processes in forming first and second semiconductor structures **102** and **104**, the interconnects of interconnect layer **2312** having Cu can become feasible.

(289) As shown in FIG. **23A**, third semiconductor structure **106** can further include a bonding layer **1014** at bonding interface **105** and above and in contact with interconnect layer **2312**. Bonding layer **1014** can include a plurality of bonding contacts **1015** and dielectrics electrically isolating bonding contacts **1015**. Bonding contacts **1015** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, bonding contacts **1015** of bonding layer **1014** include Cu. The remaining area of bonding layer **1014** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon



oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1015** and surrounding dielectrics in bonding layer **1014** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously. (290) As shown in FIG. 23A, second semiconductor structure **104** can also include a bonding layer **1012** at bonding interface **105**, e.g., on the opposite side of bonding interface **105** with respect to bonding layer **1014** in third semiconductor structure **106**. Bonding layer **1012** can include a plurality of bonding contacts **1013** and dielectrics electrically isolating bonding contacts **1013**. Bonding contacts **1013** can include conductive materials, such as Cu. The remaining area of bonding layer **1012** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1013** and surrounding dielectrics in bonding layer **1012** can be used for hybrid bonding. In some implementations, bonding interface **105** is the place at which bonding layers **1014** and **1012** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of bonding layer **1014** of third semiconductor structure **106** and the bottom surface of bonding layer **1012** of second semiconductor structure **104**.

(291) As shown in FIG. 23A, second semiconductor structure **104** further includes an interconnect layer **2326** above and in contact with bonding layer **1012** to transfer electrical signals. Interconnect layer **2326** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. Interconnect layer **2326** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2326** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2326** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(292) In some implementations, the interconnects in interconnect layer **2326** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **2326** can occur after the high-temperature processes in forming components (e.g., NAND memory strings **208**) in first semiconductor structure **102** and components in a device layer **2314** in second semiconductor structure **104**, as well as being separated from the high-temperature processes in forming third semiconductor structure **106**, the interconnects of interconnect layer **2326** having Cu can become feasible.

(293) As shown in FIG. 23A, second semiconductor structure **104** can further include device layer **2314** above and in contact with interconnect layer **2326**. In some implementations, device layer **2314** includes a third peripheral circuit **2316** and a fourth peripheral circuit **2318**. In some implementations, the devices in device layer **2314** are coupled to one another through the interconnects in interconnect layer **2326**. For example, peripheral circuit **2316** may be coupled to peripheral circuit **2318** through interconnect layer **2326**. Third peripheral circuit **2316** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **2318** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **2316** includes a plurality of transistors **2320**, and fourth peripheral circuit **2318** includes a plurality of transistors **2322** as well. Transistors **2320** and **2322** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2320** or **2322** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2320** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **2322** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **2320** than transistor **2322**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of

transistors **2320** and **2322**) can be formed on or in semiconductor layer **1004** as well.

(294) Moreover, the different voltages applied to different transistors **2320**, **2322**, **2308**, and **2310** in second and third semiconductor structures **104** and **106** can lead to differences of device dimensions between second and third semiconductor structures **104** and **106**. In some implementations, the thickness of the gate dielectric of transistor **2308** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2320** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **2308** than transistor **2320**. In some implementations, the thickness of the gate dielectric of transistor **2322** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **2310** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **2322** and transistor **2310**. In some implementations, the thickness of semiconductor layer **1006** in which transistor **2308** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **2320** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **2308** than transistor **2320**.

(295) First semiconductor structure **102** can be bonded on top of second semiconductor structure **104** in a face-to-back manner at bonding interface **103**. As shown in FIG. 23A, second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of first semiconductor structure **102** by transfer bonding. In some implementations, bonding interface **103** is disposed vertically between an interconnect layer **2328** of first semiconductor structure **102** and semiconductor layer **1004** as a result of transfer bonding, which transfers semiconductor layer **1004** from another substrate and bonds semiconductor layer **1004** onto first semiconductor structure **102** as described below in detail. In some implementations, bonding interface **103** is the place at which interconnect layer **2328** and semiconductor layer **1004** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the bottom surface of interconnect layer **2328** of first semiconductor structure **102** and the top surface of semiconductor layer **1004** of second semiconductor structure **104**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **103** and semiconductor layer **1004** and/or between bonding interface **103** and interconnect layer **2328** to facilitate the transfer bonding of semiconductor layer **1004** onto interconnect layer **2328**. Thus, it is understood that bonding interface **103** may include the surfaces of the dielectric layer(s) in some examples.

(296) As shown in FIG. 23A, second semiconductor structure **104** can further include one or more contacts **2324** extending vertically through semiconductor layer **1004**. Contact **2324** can extend vertically further through bonding interface **103** to be in contact with the interconnects in interconnect layer **2328**. In some implementations, contact **2324** is coupled to the interconnects in interconnect layer **2326**. Contact **2324** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2324** includes W. In some implementations, contact **2324** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **2324** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(297) As shown in FIG. 23A, first semiconductor structure **102** can further include interconnect layer **2328** on the opposite side of bonding interface **103** with respect to semiconductor layer **1004** to transfer electrical signals. Interconnect layer **2328** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **2328** also include local interconnects, such as bit line contacts and word line contacts. Contacts **2324** through semiconductor layer **1004** can couple the interconnects in interconnect layer **2328** to the interconnects in interconnect layer **2326**. Interconnect layer **2328** can

further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2328** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2328** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(298) As shown in FIG. **23A**, first semiconductor structure **102** can include a memory cell array, such as an array of NAND memory strings **208** above and in contact with interconnect layer **2328**. In some implementations, interconnect layer **2328** is vertically between NAND memory strings **208** and bonding interface **103**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **2327**. Memory stack **2327** may be an example of memory stack **804** in FIGS. **8A-8C**, and the conductive layer and dielectric layer in memory stack **2327** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **2327** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **2327**.

(299) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(300) As shown in FIG. **23A**, first semiconductor structure **102** can further include semiconductor layer **1002** disposed above memory stack **2327** and in contact with the sources of NAND memory strings **208**. In some implementations, NAND memory strings **208** are disposed vertically between bonding interface **103** and semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **2327** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(301) As shown in FIG. **23A**, first semiconductor structure **102** can further include a pad-out interconnect layer **902** above and in contact with semiconductor layer **1002**. In some implementations, semiconductor layer **1002** is disposed vertically between pad-out interconnect layer **902** and NAND memory strings **208**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **2332**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **2328** can be formed on opposite sides of semiconductor layer **1002**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **2300** and external devices, e.g., for pad-out purposes.

(302) As shown in FIG. **11A**, first semiconductor structure **102** can further include one or more contacts **2330** extending vertically through semiconductor layer **1002**. In some implementations, contact **2330** couples the interconnects in interconnect layer **2328** to contact pads **2332** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **2330** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1130** includes W. In some implementations, contact **2330** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of

semiconductor layer **1002**, contact **2330** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(303) As a result, peripheral circuits **2304**, **2306**, **2316**, and **2318** in third and second semiconductor structures **106** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **2312**, **2326**, and **2328**, bonding layers **1014** and **1012**, as well as contacts **2324**. Moreover, peripheral circuits **2304**, **2306**, **2316**, and **2318** and NAND memory strings **208** in 3D memory device **2300** can be further coupled to external devices through contacts **2330** and pad-out interconnect layer **902**.

(304) It is understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIG. 23A and may be any other suitable semiconductor materials. For example, as shown in FIG. 23B, a 3D memory device **2301** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **2301** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **2301** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **2300** and **2301** are not repeated for ease of description.

(305) It is also understood that the pad-out of 3D memory devices is not limited to from first semiconductor structure **102** having NAND memory strings **208** as shown in FIGS. 23A and 23B (corresponding to FIG. 9B) and may be from third semiconductor structure **106** having peripheral circuit **2304** (corresponding to FIG. 9A). For example, as shown in FIG. 23C, a 3D memory device **2303** may include pad-out interconnect layer **902** in third semiconductor structure **106**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1006** of third semiconductor structure **106** on which transistors **2308** of peripheral circuit **2304** are formed. In some implementations, third semiconductor structure **106** further includes one or more contacts **2334** extending vertically through semiconductor layer **1006**. In some implementations, contact **2334** couples the interconnects in interconnect layer **2312** in third semiconductor structure **106** to contact pads **2332** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1006**. Contact **2334** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2334** includes W. In some implementations, contact **2334** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1006**. Depending on the thickness of semiconductor layer **1006**, contact **2334** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **2300** and **2303** are not repeated for ease of description.

(306) It is further understood that in some examples, similar to bonding interface **105**, bonding interface **103** may result from hybrid bonding and thus, be disposed vertically between two bonding layers each including bonding contacts in second and third semiconductor structures **104** and **106**, respectively. For example, as shown in FIG. 23C, 3D memory device **2303** may include bonding layers **1008** and **1010** in first and second semiconductor structures **102** and **104**, respectively, at bonding interface **103**, i.e., on opposite sides of bonding interface **103**. Bonding layer **1008** or **1010** can include a plurality of bonding contacts **1009** or **1011** and dielectrics electrically isolating bonding contacts **1009** or **1011**. Bonding contacts **1009** and **1011** can include conductive materials, such as Cu. The remaining area of bonding layer **1008** or **1010** can be formed

with dielectric materials, such as silicon oxide. Bonding contacts **1009** and **1011** and surrounding dielectrics in bonding layer **1008** or **1010** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1010** of second semiconductor structure **104** and the bottom surface of bonding layer **1008** of first semiconductor structure **102**. Contact **2324** can be coupled to bonding contacts **1011**, and interconnect layer **2328** can be coupled to bonding contacts **1009**.

(307) FIGS. **24A-24F** illustrate a fabrication process for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. FIG. **26** illustrates a flowchart of a method **2600** for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **24A-24F** and **26** include 3D memory devices **2300**, **2301**, and **2303** depicted in FIGS. **23A-23C**. FIGS. **24A-24F** and **26** will be described together. It is understood that the operations shown in method **2600** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **26**. For example, operation **2602** may be performed after operation **2608** or in parallel with operations **2604-2608**.

(308) Referring to FIG. **26**, method **2600** starts at operation **2602**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(309) As illustrated in FIG. **24A**, a stack structure, such as a memory stack **2426** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **2424**. To form memory stack **2426**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **2424**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **2426** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **2426** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **2426** and silicon substrate **2424**.

(310) As illustrated in FIG. **24A**, NAND memory strings **2428** are formed above silicon substrate **2424**, each of which extends vertically through memory stack **2426** to be in contact with silicon substrate **2424**. In some implementations, fabrication processes to form NAND memory string **2428** include forming a channel hole through memory stack **2426** (or the dielectric stack) and into silicon substrate **2424** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **2428** may vary depending on the types of channel structures of NAND memory strings **2428** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(311) In some implementations, an interconnect layer is formed above the array of NAND memory

strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. 24A, an interconnect layer **2430** is formed above memory stack **2426** and NAND memory strings **2428**. Interconnect layer **2430** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **2428**. In some implementations, interconnect layer **2430** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2430** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 24A can be collectively referred to as interconnect layer **2430**.

(312) Method **2600** proceeds to operation **2604**, as illustrated in FIG. 26, in which a semiconductor layer is formed above the array of NAND memory strings. The semiconductor layer can include single crystalline silicon. In some implementations, to form the semiconductor layer, another substrate and the second substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(313) As illustrated in FIG. 24B, a semiconductor layer **2410**, such as a single crystalline silicon layer, is formed above interconnect layer **2430** and NAND memory strings **2428**. Semiconductor layer **2410** can be attached above interconnect layer **2430** to form a bonding interface **2412** vertically between semiconductor layer **2410** and interconnect layer **2430**. In some implementations, to form semiconductor layer **2410**, another silicon substrate (not shown in FIG. 24B) and silicon substrate **2424** are bonded in a face-to-face manner (having the components formed on silicon substrate **2424**, such as NAND memory strings **2428**, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface **2412**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **2410** attached above interconnect layer **2430**. The details of various transfer bonding processes are described above with respect to FIGS. 48A-48D and FIGS. 49A-49D and thus, are not repeated for case of description.

(314) Referring to FIG. 26, method **2600** proceeds to operation **2606**, in which a first transistor is formed on the semiconductor layer. As illustrated in FIG. 24C, a plurality of transistors **2414** and **2416** are formed on semiconductor layer **2410** having single crystalline silicon. Transistors **2414** and **2416** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **2410** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2414** and **2416**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **2410** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2414** is different from the thickness of gate dielectric of transistor **2416**, for example, by depositing a thicker silicon oxide film in the region of transistor **2414** than the region of transistor **2416**, or by etching back part of the silicon oxide film deposited in the region of transistor **2416**. It is understood that the details of fabricating transistors **2414** and **2416** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(315) In some implementations, an interconnect layer **2420** is formed above the transistor on the semiconductor layer. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 24C, an interconnect layer **2420** can be formed above transistors

**2414** and **2416**. Interconnect layer **2420** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2414** and **2416**. In some implementations, interconnect layer **2420** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2420** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **24C** can be collectively referred to as interconnect layer **2420**. In some implementations, the interconnects in interconnect layer **2420** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **2420** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **2420**.

(316) In some implementations, a contact through the semiconductor layer is formed. As illustrated in FIG. **24C**, one or more contacts **2418** each extending vertically through semiconductor layer **2410** is formed. Contact **2418** can extend vertically further through bonding interface **2412** to be in contact with the interconnects in interconnect layer **2430**. Contacts **2418** can couple the interconnects in interconnect layers **2420** and **2430**. Contacts **2418** can be formed by first patterning contact holes through semiconductor layer **2410** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(317) In some implementations, a first bonding layer is formed above the interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **24C**, a bonding layer **2422** is formed above interconnect layer **2420**. Bonding layer **2422** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1220** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2420** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(318) Method **2600** proceeds to operation **2608**, as illustrated in FIG. **26**, in which a second transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **24D**, a plurality of transistors **2404** and **2406** are formed on a silicon substrate **2402**. Transistors **2404** and **2406** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2402** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2404** and **2406**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2402** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2404** is different from the thickness of gate dielectric of transistor **2406**, for example, by depositing a thicker silicon oxide film in the region of transistor **2404** than the region of transistor **2406**, or by etching back part of the silicon oxide film deposited in the region of transistor **2406**. It

is understood that the details of fabricating transistors **2404** and **2406** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(319) In some implementations, an interconnect layer **2408** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 24D, an interconnect layer **2408** can be formed above transistors **2404** and **2406**. Interconnect layer **2408** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2404** and **2406**. In some implementations, interconnect layer **2408** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2408** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 24D can be collectively referred to as interconnect layer **2408**. In some implementations, the interconnects in interconnect layer **2408** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **2408** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **2408**.

(320) In some implementations, a second bonding layer is formed above the interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. 24D, a bonding layer **2432** is formed above interconnect layer **2408**. Bonding layer **2432** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **2408** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2408** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(321) Method **2600** proceeds to operation **2610**, as illustrated in FIG. 26, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(322) As illustrated in FIG. 24E, silicon substrate **2424** and components formed thereon (e.g., memory stack **2426**, NAND memory strings **2428**, and transistors **2416** and **2414**) are flipped upside down. Bonding layer **2422** facing down is bonded with bonding layer **2432** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **2412**. That is, silicon substrate **2424** and components formed thereon can be bonded with silicon substrate **2402** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **2422** are in contact with the bonding contacts in bonding layer **2432** at bonding interface **2412**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 24E, it is understood that in some examples, silicon substrate **2402** and components formed thereon (e.g., transistors **2404** and **2406**) can be flipped upside down, and bonding layer **2432** facing down can be



bonded with bonding layer **2422** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **2412** as well.

(323) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **2412** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **2422** and the bonding contacts in bonding layer **2432** are aligned and in contact with one another, such that memory stack **2426** and NAND memory strings **2428** formed therethrough and transistors **2416** and **2414** can be coupled to transistors **2404** and **2406** through the bonded bonding contacts across bonding interface **1237**, according to some implementations.

(324) Method **2600** proceeds to operation **2612**, as illustrated in FIG. **26**, in which the first substrate or the second substrate is thinned. As illustrated in FIG. **24F**, silicon substrate **2424** (shown in FIG. **24E**) is thinned to become a semiconductor layer **2434** having single crystalline silicon. Silicon substrate **2424** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. It is understood that although not shown in FIG. **24F**, in some examples, silicon substrate **2402** may be thinned to become a semiconductor layer having single crystalline silicon.

(325) Method **2600** proceeds to operation **2614**, as illustrated in FIG. **26**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned second substrate or above the array of NAND memory strings. As illustrated in FIG. **24F**, a pad-out interconnect layer **2436** is formed on semiconductor layer **2434** (the thinned silicon substrate **2424**) above NAND memory strings **2428**. Pad-out interconnect layer **2436** can include interconnects, such as contact pads **2438**, formed in one or more ILD layers. Contact pads **2438** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **2435** are formed, extending vertically through semiconductor layer **2434**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **2435** can couple contact pads **2438** in pad-out interconnect layer **2436** to the interconnects in interconnect layer **2430**. It is understood that in some examples, contacts **2435** may be formed in silicon substrate **2424** before thinning (the formation of semiconductor layer **2434**) and be exposed from the backside of silicon substrate **2424** (where the thinning occurs) after the thinning. It is also understood that although not shown in FIG. **24F**, in some examples, a pad-out interconnect layer may be formed on the thinned silicon substrate **2402**, and contacts may be formed through the thinned silicon substrate **2402** to couple the pad-out interconnect layer and interconnect layer **2408** across the thinned silicon substrate **2402**. It is further understood that in some examples, the first substrate (e.g., silicon substrate **2424** or semiconductor layer **2434** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**.

(326) FIGS. **25A-25F** illustrate another fabrication process for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. FIG. **27** illustrates a flowchart of another method **2700** for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **25A-25F** and **27** include 3D memory devices **2300**, **2301**, and **2303** depicted in FIGS. **23A-23C**. FIGS. **25A-25F** and **27** will be described together. It is understood that the operations shown in method **2700** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **27**. For example, operation **2702**, **2704**, and **2706** may be performed in parallel.

(327) Referring to FIG. **27**, method **2700** starts at operation **2702**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having

single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(328) As illustrated in FIG. 25A, a stack structure, such as a memory stack **2504** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **2502**. To form memory stack **2504**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **2502**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **2504** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **2504** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **2504** and silicon substrate **2502**.

(329) As illustrated in FIG. 25A, NAND memory strings **2506** are formed above silicon substrate **2502**, each of which extends vertically through memory stack **2504** to be in contact with silicon substrate **2502**. In some implementations, fabrication processes to form NAND memory string **2506** include forming a channel hole through memory stack **2504** (or the dielectric stack) and into silicon substrate **2502** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **2506** may vary depending on the types of channel structures of NAND memory strings **2506** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. 8A-8C) and thus, are not elaborated for case of description.

(330) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. 25A, an interconnect layer **2508** is formed above memory stack **2504** and NAND memory strings **2506**. Interconnect layer **2508** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **2506**. In some implementations, interconnect layer **2508** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2508** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 25A can be collectively referred to as interconnect layer **2508**.

(331) In some implementations, a first bonding layer is formed above the array of NAND memory strings. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. 25A, a bonding layer **2510** is formed above interconnect layer **2508**. Bonding layer **2510** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **2508** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer by first patterning contact

holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor. It is understood that in some examples, bonding layer **2510** may be a dielectric layer (e.g., a silicon oxide layer) without bonding contacts for fusion bonding, instead of hybrid bonding.

(332) Method **2700** proceeds to operation **2704**, as illustrated in FIG. 27, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 25B, a plurality of transistors **2514** and **2516** are formed on a silicon substrate **2512**. Transistors **2514** and **2516** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2512** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2514** and **2516**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2512** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2514** is different from the thickness of gate dielectric of transistor **2516**, for example, by depositing a thicker silicon oxide film in the region of transistor **2514** than the region of transistor **2516**, or by etching back part of the silicon oxide film deposited in the region of transistor **2516**. It is understood that the details of fabricating transistors **2514** and **2516** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(333) In some implementations, an interconnect layer **2518** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 25B, an interconnect layer **2518** can be formed above transistors **2514** and **2516**. Interconnect layer **2518** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2514** and **2516**. In some implementations, interconnect layer **2518** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2518** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 25B can be collectively referred to as interconnect layer **2518**.

(334) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. 25B, a bonding layer **2520** is formed above interconnect layer **2518**. Bonding layer **2520** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **2518** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2518** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(335) Method **2700** proceeds to operation **2706**, as illustrated in FIG. 27, in which a second transistor is formed on a third substrate. The third substrate can be a silicon substrate having single

crystalline silicon. In some implementations, any two or all of operations **2702**, **2704**, and **2706** are performed in parallel to reduce process time.

(336) As illustrated in FIG. **25C**, a plurality of transistors **2524** and **2526** are formed on a silicon substrate **2522**. Transistors **2524** and **2526** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2522** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2524** and **2526**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2522** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2524** is different from the thickness of gate dielectric of transistor **2526**, for example, by depositing a thicker silicon oxide film in the region of transistor **2524** than the region of transistor **2526**, or by etching back part of the silicon oxide film deposited in the region of transistor **2526**. It is understood that the details of fabricating transistors **2524** and **2526** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(337) In some implementations, an interconnect layer **2528** is formed above the transistor on the third substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **25C**, an interconnect layer **2528** can be formed above transistors **2524** and **2526**. Interconnect layer **2528** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2524** and **2526**. In some implementations, interconnect layer **2528** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2528** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **25C** can be collectively referred to as interconnect layer **2528**.

(338) In some implementations, a third bonding layer is formed above interconnect layer. The third bonding layer can include a plurality of third bonding contacts. As illustrated in FIG. **25C**, a bonding layer **2530** is formed above interconnect layer **2528**. Bonding layer **2530** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **2528** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2528** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(339) Method **2700** proceeds to operation **2708**, as illustrated in FIG. **27**, in which the second substrate and the third substrate are bonded in a face-to-face manner. The second bonding contact in the second bonding layer can be in contact with the third bonding contact in the third bonding layer at a first bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(340) As illustrated in FIG. **25D**, silicon substrate **2512** and components formed thereon (e.g., transistors **2514** and **2516**) are flipped upside down. Bonding layer **2520** facing down is bonded with bonding layer **2530** facing up, i.e., in a face-to-face manner, thereby forming a bonding

interface **2540**. That is, silicon substrate **2512** and components formed thereon can be bonded with silicon substrate **2522** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **2530** are in contact with the bonding contacts in bonding layer **2520** at bonding interface **2540**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **25D**, it is understood that in some examples, silicon substrate **2522** and components formed thereon (e.g., transistors **2524** and **2526**) can be flipped upside down, and bonding layer **2530** facing down can be bonded with bonding layer **2520** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **2540** as well.

(341) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **2540** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **2520** and the bonding contacts in bonding layer **2530** are aligned and in contact with one another, such that transistors **2524** and **2526** can be coupled to transistors **2514** and **2516** through the bonded bonding contacts across bonding interface **2540**, according to some implementations.

(342) In some implementations, the second substrate is thinned, and a contact through the thinned second substrate is formed. As illustrated in FIG. **25E**, silicon substrate **2512** (shown in FIG. **25D**) is thinned to become a semiconductor layer **2534** having single crystalline silicon. Silicon substrate **2512** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(343) As illustrated in FIG. **25E**, one or more contacts **2536** each extending vertically through semiconductor layer **2534** is formed. Contacts **2536** can be coupled to the interconnects in interconnect layer **2518**. Contacts **2536** can be formed by first patterning contact holes through semiconductor layer **2534** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **2536** may be formed in silicon substrate **2512** before thinning (the formation of semiconductor layer **2534**, e.g., in FIG. **25B**) and be exposed from the backside of silicon substrate **2512** (where the thinning occurs) after the thinning.

(344) In some implementations, a fourth bonding layer is formed on the thinned second substrate. The fourth bonding layer can include a plurality of fourth bonding contacts. As shown in FIG. **25E**, a bonding layer **2511** is formed on semiconductor layer **2534**, i.e., the backside of silicon substrate **2512** (where the thinning occurs) after the thinning. Bonding layer **2511** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the surface of semiconductor layer **2534** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **2536** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (gluc) layer, a barrier layer, and/or a seed layer before depositing the conductor. It is understood that in some examples, bonding layer **2511** may be a dielectric layer (e.g., a silicon oxide layer) without bonding contacts for fusion bonding, instead of hybrid bonding. It is further understood that in some examples, the bonding layer may be omitted to expose the silicon surface of semiconductor layer **2534** for anodic bonding or fusion bonding, instead of hybrid bonding.

(345) Method **2700** proceeds to operation **2710**, as illustrated in FIG. **15**, in which the first substrate and the second substrate are bonded in a face-to-back manner. The first bonding contact in the first bonding layer can be in contact with the fourth bonding contact in the fourth bonding layer at a second bonding interface after bonding the third and second substrates. The bonding can include hybrid bonding.

(346) As illustrated in FIG. 25E, silicon substrate **2502** and components formed thereon (e.g., memory stack **2504** and NAND memory strings **2506**) are flipped upside down. Bonding layer **2510** on interconnect layer **2508** facing down is bonded with bonding layer **2511** on semiconductor layer **2534** facing up, i.e., in a face-to-back manner, thereby forming a bonding interface **2532**. That is, silicon substrate **2502** and components formed thereon can be bonded with thinned silicon substrate **2512** (i.e., semiconductor layer **2534**) and components formed thereon after bonding with silicon **2522** in a face-to-back manner at bonding interface **2532**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 25E, it is understood that in some examples, silicon substrate **2512** and components formed thereon (e.g., transistors **2516**, **2514**, **2524**, and **2526**) can be flipped upside down, and the bonding layer on semiconductor layer **2534** facing down can be bonded with the bonding layer on interconnect layer **2508** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **2532** as well.

(347) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **2532** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **2510** on interconnect layer **2508** and the bonding contacts in bonding layer **2511** on semiconductor layer **2534** are aligned and in contact with one another, such that memory stack **2504** and NAND memory strings **2506** can be coupled to transistors **2514**, **2516**, **2524**, and **2526** through contacts **2536** through semiconductor layer **2534** and the bonded bonding contacts across bonding interface **2540**, according to some implementations. It is understood that in some examples, anodic bonding or fusion bonding, instead of hybrid bonding, may be performed to bond silicon substrate **2502** and thinned silicon substrate **2512** (and components formed thereon) at bonding interface **2532** without bonding contacts in the bonding layers. It is further understood that in some examples, silicon substrate **2522**, instead of silicon substrate **2512**, may be thinned and bonded with silicon substrate **2502** in a similar face-to-back manner as described above.

(348) Method **2700** proceeds to operation **2712**, as illustrated in FIG. 27, in which the first substrate or the third substrate is thinned. As illustrated in FIG. 25F, silicon substrate **2522** (shown in FIG. 25E) is thinned to become a semiconductor layer **2542** having single crystalline silicon. Silicon substrate **2522** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. It is understood that although not shown in FIG. 25F, in some examples, silicon substrate **2502** may be thinned to become a semiconductor layer having single crystalline silicon.

(349) Method **2700** proceeds to operation **2714**, as illustrated in FIG. 27, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned third substrate or above the array of NAND memory strings. As illustrated in FIG. 25F, a pad-out interconnect layer **2546** is formed on semiconductor layer **2542** (the thinned silicon substrate **2522**). Pad-out interconnect layer **2546** can include interconnects, such as contact pads **2548**, formed in one or more ILD layers. Contact pads **2548** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **2544** are formed, extending vertically through semiconductor layer **2542**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **2544** can couple contact pads **2548** in pad-out interconnect layer **2546** to the interconnects in interconnect layer **2528**. It is understood that in some examples, contacts **2544** may be formed in silicon substrate **2522** before thinning (the formation of semiconductor layer **2542**, e.g., in FIG. 25C) and be exposed from the backside of silicon substrate **2522** (where the thinning occurs) after the thinning. It is further understood that although not shown in FIG. 25F, in some examples, a pad-out interconnect layer may be formed on the thinned silicon substrate **2502** above NAND memory strings **2506**, and contacts may be formed

through the thinned silicon substrate **2502** to couple the pad-out interconnect layer and interconnect layer **2508** across the thinned silicon substrate **2502**.

(350) FIGS. **28A** and **28B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **9A** and **9B**, according to various aspects of the present disclosure. 3D memory devices **2800** and **2801** may be examples of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **28A**, 3D memory device **2800** can include stacked first, second, and third semiconductor structures **102**, **104**, and **106**. In some implementations, first semiconductor structure **102** on one side of 3D memory device **2800** includes semiconductor layer **1002** and a memory cell array vertically between semiconductor layer **1002** and bonding interface **103**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer **1002** (e.g., as shown in FIGS. **8A-8C**). Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**).

(351) In some implementations, second semiconductor structure **104** in the intermediate of 3D memory device **2800** includes a semiconductor layer **1004** and some of the peripheral circuits of the memory cell array. In some implementations, bonding interface **103** is disposed vertically between semiconductor layer **1004** and the peripheral circuits of second semiconductor structure **104**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Bonding interface **103** between first and second semiconductor structures **102** and **104** may result from transfer bonding. Through contacts (e.g., ILVs/TSVs) across bonding interface **103** and through semiconductor layer **1004** vertically between first and second semiconductor structures **102** and **104** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(352) In some implementations, third semiconductor structure **106** on another side of 3D memory device **2800** includes a semiconductor layer **1006** and some of the peripheral circuits of the memory cell array. In some implementations, bonding interface **105** is disposed vertically between semiconductor layer **1006** and the peripheral circuits of third semiconductor structure **106**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1006**. Semiconductor layer **1006** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1006** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Bonding interface **105** between third and second semiconductor structures **106** and **104** may result from transfer bonding. Through contacts (e.g., ILVs/TSVs) across bonding interface **105** and through semiconductor layer **1006** vertically between third and second semiconductor structures **106** and **104** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **106** and **104**.

(353) It is understood that in some examples, first and second semiconductor structures **102** and **104** may also include bonding layers **1008** and **1010**, respectively, disposed on opposite sides of

bonding interface **103**, and third and second semiconductor structures **106** and **104** may also include bonding layers **1014** and **1012**, respectively, disposed on opposite sides of bonding interface **105**, as shown in FIG. **28B**. In FIG. **28B**, second semiconductor structure **104** of a 3D memory device **2801** can include two bonding layers **1010** and **1012** on two sides thereof. Bonding layer **1010** can be disposed vertically between semiconductor layer **1004** and bonding interface **103**, and bonding layer **1012** can be disposed vertically between the peripheral circuits of second semiconductor structure **104** and bonding interface **105**. First semiconductor structure **102** of 3D memory device **2801** can include bonding layer **1008** disposed vertically between bonding interface **103** and semiconductor layer **1002**. Third semiconductor structure **106** of 3D memory device **2801** can include bonding layer **1014** disposed vertically between bonding interface **105** and semiconductor layer **1006**. Each bonding layer **1008**, **1010**, **1012**, or **1014** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. The bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, bonding contacts across bonding interface **103** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1004** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**. Similarly, the bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, bonding contacts across bonding interface **105** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1006** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **106** and **104**.

(354) As shown in FIGS. **28A** and **28B**, since third and second semiconductor structures **106** and **104** are bonded in a back-to-face manner (e.g., semiconductor layers **1006** and **1004** being disposed on the bottom sides of third and second semiconductor structures **106** and **104**, respectively, in FIGS. **28A** and **28B**), the transistors in third semiconductor structure **106** and the transistors in second semiconductor structure **104** face toward the same direction (e.g., the positive y-direction in FIGS. **28A** and **28B**), according to some implementations. In some implementations, semiconductor layer **1004** is disposed vertically between the transistors of the peripheral circuits in second semiconductor structure **104** and bonding interface **103**, and semiconductor layer **1006** is disposed vertically between the transistors of the peripheral circuits in third semiconductor structure **106** and bonding interface **105**. Moreover, since first and second semiconductor structures **102** and **104** are bonded in a face-to-back manner (e.g., semiconductor layers **1002** and **1004** being disposed on the bottom sides of first and second semiconductor structures **102** and **104**, respectively, in FIGS. **28A** and **228**), the transistors of peripheral circuits in second and third semiconductor structures **104** and **106** and the memory cell array in first semiconductor structure **102** face toward the same direction (e.g., the positive y-direction in FIGS. **28A** and **28B**), according to some implementations. It is understood that pad-out interconnect layer **902** in FIG. **9A** or **9B** is omitted from 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B** for case of illustration and may be included in 3D memory devices **2800** and **2801** as described above with respect to FIGS. **9A** and **9B**.

(355) As described above, second and third semiconductor structures **104** and **106** can have peripheral circuits having transistors with different applied voltages. For example, third semiconductor structure **106** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and second semiconductor structure **104** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1006** and **1004** in third and second semiconductor structures **106** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and third semiconductor structure **106** may include LLV circuits **402**, and the thickness of semiconductor



layer **1004** in second semiconductor structure **104** may be larger than the thickness of semiconductor layer **1006** in third semiconductor structure **106**. Moreover, in some implementations, the gate dielectrics of the transistors in third and second semiconductor structures **106** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and third semiconductor structure **106** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in second semiconductor structure **104** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in third semiconductor structure **106**.

(356) FIGS. **29A** and **29B** illustrate side views of various examples of 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B**, according to various aspects of the present disclosure. As shown in FIG. **29A**, as one example of 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B**, 3D memory device **2900** is a bonded chip including first semiconductor structure **102**, second semiconductor structure **104**, and third semiconductor structure **106**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **29A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, and second and third semiconductor structures **104** and **106** are bonded at bonding interface **105** therebetween, according to some implementations.

(357) As shown in FIG. **29A**, first semiconductor structure **102** can include semiconductor layer **1002** having semiconductor materials. In some implementations, semiconductor layer **1006** is a silicon substrate having single crystalline silicon. First semiconductor structure **102** can include a memory cell array, such as an array of NAND memory strings **208** on semiconductor layer **1002**. The sources of NAND memory strings **208** can be in contact with semiconductor layer **1002**. In some implementations, NAND memory strings **208** are disposed vertically between bonding interface **103** and semiconductor layer **1002**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **2927**. Memory stack **2927** may be an example of memory stack **804** in FIGS. **8A-8C**, and the conductive layer and dielectric layer in memory stack **2927** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **2927** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **2927**. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(358) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(359) As shown in FIG. **29A**, first semiconductor structure **102** can further include an interconnect layer **2928** above and in contact with NAND memory strings **208** to transfer electrical signals to and from NAND memory strings **208**. Interconnect layer **2928** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **2928** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **2928** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2928** can include conductive materials including, but not limited to **2928 W**, Co, Cu, Al, silicides, or any combination

thereof. The ILD layers in interconnect layer **1128** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(360) Second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a back-to-face manner at bonding interface **103**. Second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a layer of single crystalline silicon transferred from a silicon substrate or a SOI substrate and attached to the top surface of first semiconductor structure **102** by transfer bonding. In some implementations, bonding interface **103** is disposed vertically between interconnect layer **2928** and semiconductor layer **1004** as a result of transfer bonding, which transfers semiconductor layer **1004** from another substrate and bonds semiconductor layer **1004** onto first semiconductor structure **102** as described below in detail. In some implementations, bonding interface **103** is the place at which interconnect layer **2928** and semiconductor layer **1004** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of interconnect layer **2928** of first semiconductor structure **102** and the bottom surface of semiconductor layer **1004** of second semiconductor structure **104**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1004** and/or between bonding interface **105** and interconnect layer **2928** to facilitate the transfer bonding of semiconductor layer **1004** onto interconnect layer **1112**. Thus, it is understood that bonding interface **103** may include the surfaces of the dielectric layer(s) in some examples.

(361) As shown in FIG. **29A**, second semiconductor structure **104** can also include a device layer **2914** above and in contact with semiconductor layer **1006**. In some implementations, device layer **2914** includes a first peripheral circuit **2916** and a second peripheral circuit **2918**. First peripheral circuit **2916** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **2918** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **2916** includes a plurality of transistors **2920** in contact with semiconductor layer **1004**, and second peripheral circuit **2918** includes a plurality of transistors **2922** in contact with semiconductor layer **1006**. Transistors **2920** and **2922** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2920** or **2922** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2920** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2922** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **2920** than transistor **2922**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **2920** and **2922**) can be formed on or in semiconductor layer **1004** as well.

(362) In some implementations, second semiconductor structure **104** further includes an interconnect layer **2926** above device layer **2914** to transfer electrical signals to and from peripheral circuits **2916** and **2918**. As shown in FIG. **29A**, interconnect layer **2926** can be vertically between bonding interface **105** and device layer **2914** (including transistors **2920** and **2922** of peripheral circuits **2916** and **2918**). Interconnect layer **2926** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnects in interconnect layer **2926** can be coupled to transistors **2920** and **2922** of peripheral circuits **2916** and **2918** in device layer **2914**. Interconnect layer **2926** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **2926** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **2914** are coupled to one another through the interconnects in interconnect layer **2926**. For example, peripheral circuit **2916** may be coupled to peripheral circuit **2918** through interconnect layer **2926**. The interconnects in

interconnect layer **2926** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2926** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **2926** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(363) As shown in FIG. **29A**, second semiconductor structure **104** can further include one or more contacts **2924** extending vertically through semiconductor layer **1004**. In some implementations, contact **2924** couples the interconnects in interconnect layer **2926** to the interconnects in interconnect layer **2928** to make an electrical connection across bonding interface **103** between second and first semiconductor structures **104** and **102**. Contact **2924** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2924** includes W. In some implementations, contact **2924** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **2924** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(364) Third semiconductor structure **106** can be bonded on top of second semiconductor structure **104** in a back-to-face manner at bonding interface **105**. Third semiconductor structure **106** can include semiconductor layer **1006** having semiconductor materials. In some implementations, semiconductor layer **1006** is a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of second semiconductor structure **104** by transfer bonding. In some implementations, bonding interface **105** is disposed vertically between interconnect layer **2926** and semiconductor layer **1006** as a result of transfer bonding, which transfers semiconductor layer **1006** from another substrate and bonds semiconductor layer **1006** onto second semiconductor structure **104** as described below in detail. In some implementations, bonding interface **105** is the place at which interconnect layer **2926** and semiconductor layer **1006** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of interconnect layer **2926** of second semiconductor structure **104** and the bottom surface of semiconductor layer **1006** of third semiconductor structure **106**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1006** and/or between bonding interface **105** and interconnect layer **2926** to facilitate the transfer bonding of semiconductor layer **1006** onto interconnect layer **2926**. Thus, it is understood that bonding interface **105** may include the surfaces of the dielectric layer(s) in some examples.

(365) Third semiconductor structure **106** can include a device layer **2902** above and in contact with semiconductor layer **1006**. In some implementations, device layer **2902** includes a third peripheral circuit **2904** and a fourth peripheral circuit **2906**. Third peripheral circuit **2904** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **2906** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **2904** includes a plurality of transistors **2908**, and fourth peripheral circuit **2906** includes a plurality of transistors **2910** as well. Transistors **2908** and **2910** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2908** or **2910** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2908** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **2910** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **2908** than transistor **2910**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **2908** and

**2910**) can be formed on or in semiconductor layer **1006** as well.

(366) Moreover, the different voltages applied to different transistors **2920**, **2922**, **2908**, and **2910** in second and third semiconductor structures **104** and **106** can lead to differences of device dimensions between second and third semiconductor structures **104** and **106**. In some implementations, the thickness of the gate dielectric of transistor **2920** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2908** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **2920** than transistor **2908**. In some implementations, the thickness of the gate dielectric of transistor **2922** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **2910** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **2922** and transistor **2910**. In some implementations, the thickness of semiconductor layer **1004** in which transistor **2920** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1006** in which transistor **2908** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **2920** than transistor **2908**.

(367) As shown in FIG. **29A**, third semiconductor structure **106** can further include an interconnect layer **2912** above device layer **2902** to transfer electrical signals to and from peripheral circuits **2904** and **2906**. As shown in FIG. **29A**, device layer **1114** (including transistors **1120** and **1122** of peripheral circuits **1116** and **1118**) can be vertically between bonding interface **105** and interconnect layer **2912**. Interconnect layer **2912** can include a plurality of interconnects coupled to transistors **2908** and **2910** of peripheral circuits **2904** and **2906** in device layer **2902**. Interconnect layer **2912** can further include one or more ILD layers in which the interconnects can form. That is, interconnect layer **2912** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **2902** are coupled to one another through the interconnects in interconnect layer **2912**. For example, peripheral circuit **2904** may be coupled to peripheral circuit **2906** through interconnect layer **2912**. The interconnects in interconnect layer **2912** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1126** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(368) In some implementations, the interconnects in interconnect layer **2912** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **2912** can occur after the high-temperature processes in forming device layers **1114** and **1102** in second and third semiconductor structures **104** and **106**, as well as after the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **2912** having Cu can become feasible. In some implementations, the interconnects in interconnect layer **2912** includes Cu as the conductive metal material, but not other conductive metal materials, such as W.

(369) As shown in FIG. **29A**, third semiconductor structure **106** can further include one or more contacts **2925** extending vertically through semiconductor layer **1006**. In some implementations, contact **2925** couples the interconnects in interconnect layer **2912** to the interconnects in interconnect layer **2926** to make an electrical connection across bonding interface **105** between second and third semiconductor structures **104** and **106**. Contact **2925** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2925** includes Cu. For example, contact **2925** may include Cu as the conductive metal material, but not other conductive metal materials, such as W. In some implementations, contact **2925** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1006**. Depending on the thickness of semiconductor layer **1006**, contact **2925** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu$ m), or a TSV having a depth in the micron- or tens micron-level (e.g.,

between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(370) As shown in FIG. 29A, third semiconductor structure **106** can further include a pad-out interconnect layer **902** above and in contact with interconnect layer **2912**. In some implementations, interconnect layer **2912** is disposed vertically between pad-out interconnect layer **902** and device layer **2902** including transistors **2908** and **2910**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **2932**, in one or more ILD layers. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **2900** and external devices, e.g., for pad-out purposes.

(371) As a result, peripheral circuits **2904**, **2906**, **2916**, and **2918** in third and second semiconductor structures **106** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **2912**, **2926**, and **2928**, as well as contacts **2925** and **2924**. Moreover, peripheral circuits **2904**, **2906**, **2916**, and **2918** and NAND memory strings **208** in 3D memory device **2900** can be further coupled to external devices through pad-out interconnect layer **902**.

(372) It is understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIG. 29A and may be any other suitable semiconductor materials. For example, as shown in FIG. 29B, a 3D memory device **2901** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **2901** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **2901** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof.

(373) It is also understood that the pad-out of 3D memory devices is not limited to from third semiconductor structure **106** having peripheral circuits **2904** and **2906** as shown in FIG. 29A (corresponding to FIG. 9A) and may be from first semiconductor structure **102** having NAND memory strings **208** (corresponding to FIG. 9B). For example, as shown in FIG. 29B, 3D memory device **2901** may include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1002** of first semiconductor structure **102** on which NAND memory strings **208** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **2934** extending vertically through semiconductor layer **1002**. In some implementations, contact **2934** couples the interconnects in interconnect layer **2928** in first semiconductor structure **102** to contact pads **2932** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **2934** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2934** includes W. In some implementations, contact **2934** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **2934** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **2900** and **2901** are not repeated for case of description.

(374) Although not shown in FIGS. 29A and 29B, it is understood that in some examples, bonding interface **105** may result from hybrid bonding and thus, be disposed vertically between two bonding layers each including bonding contacts in second and third semiconductor structures **104** and **106**, respectively, as described above in detail. Similarly, in some examples, bonding interface **103** may result from hybrid bonding and thus, be disposed vertically between two bonding layers each including bonding contacts in second and first semiconductor structures **104** and **102**,

respectively, as described above in detail.

(375) FIGS. **30A-30F** illustrate a fabrication process for forming the 3D memory devices in FIGS. **28A** and **28B**, according to some aspects of the present disclosure. FIG. **32** illustrates a flowchart of a method **3200** for forming the 3D memory devices in FIGS. **28A** and **28B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **30A-30F** and **32** include 3D memory devices **2900** and **2901** depicted in FIGS. **29A** and **29B**. FIGS. **30A-30F** and **32** will be described together. It is understood that the operations shown in method **3200** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **32**.

(376) Referring to FIG. **32**, method **3200** starts at operation **3202**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(377) As illustrated in FIG. **30A**, a stack structure, such as a memory stack **3026** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **3024**. To form memory stack **3026**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **3024**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **3026** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **3026** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **3026** and silicon substrate **3024**.

(378) As illustrated in FIG. **30A**, NAND memory strings **3028** are formed above silicon substrate **3024**, each of which extends vertically through memory stack **3026** to be in contact with silicon substrate **3024**. In some implementations, fabrication processes to form NAND memory string **3028** include forming a channel hole through memory stack **3026** (or the dielectric stack) and into silicon substrate **3024** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **3028** may vary depending on the types of channel structures of NAND memory strings **3028** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(379) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **30A**, an interconnect layer **3030** is formed above memory stack **3026** and NAND memory strings **3028**. Interconnect layer **3030** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **3028**. In some implementations, interconnect layer **3030** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3030** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating,

electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 30A can be collectively referred to as interconnect layer 3030.

(380) Method 3200 proceeds to operation 3204, as illustrated in FIG. 32, in which a first semiconductor layer is formed above the array of NAND memory strings. The first semiconductor layer can include single crystalline silicon. In some implementations, to form the first semiconductor layer, another substrate and the first substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the first semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(381) As illustrated in FIG. 30B, a semiconductor layer 3010, such as a single crystalline silicon layer, is formed above interconnect layer 3030 and NAND memory strings 3028. Semiconductor layer 3010 can be attached above interconnect layer 3030 to form a bonding interface 3012 vertically between semiconductor layer 3010 and interconnect layer 3030. In some implementations, to form semiconductor layer 3010, another silicon substrate (not shown in FIG. 30B) and silicon substrate 3024 are bonded in a face-to-face manner (having the components formed on silicon substrate 3024, such as NAND memory strings 3028, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface 3012. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer 3010 attached above interconnect layer 3030. The details of various transfer bonding processes are described above with respect to FIGS. 48A-48D and FIGS. 49A-49D and thus, are not repeated for ease of description.

(382) Referring to FIG. 32, method 3200 proceeds to operation 3206 in which a first transistor is formed on the first semiconductor layer. As illustrated in FIG. 30C, a plurality of transistors 3014 and 3016 are formed on semiconductor layer 3010 having single crystalline silicon. Transistors 3014 and 3016 can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer 3010 by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors 3014 and 3016. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer 3010 by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor 3014 is different from the thickness of gate dielectric of transistor 3016, for example, by depositing a thicker silicon oxide film in the region of transistor 3014 than the region of transistor 3016, or by etching back part of the silicon oxide film deposited in the region of transistor 3016. It is understood that the details of fabricating transistors 3014 and 3016 may vary depending on the types of the transistors (e.g., planar transistors 500 or 3D transistors 600 in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(383) In some implementations, an interconnect layer 3020 is formed above the transistor on the semiconductor layer. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 30C, an interconnect layer 3020 can be formed above transistors 3014 and 3016. Interconnect layer 3020 can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors 3014 and 3016. In some implementations, interconnect layer 3020 includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer 3020 can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or

more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 30C can be collectively referred to as interconnect layer **3020**.

(384) In some implementations, a contact through the semiconductor layer is formed. As illustrated in FIG. 30C, one or more contacts **3018** each extending vertically through semiconductor layer **3010** is formed. Contacts **3018** can couple the interconnects in interconnect layers **3020** and **3030**. Contacts **3018** can be formed by first patterning contact holes through semiconductor layer **3010** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(385) Method **3200** proceeds to operation **3208**, as illustrated in FIG. 32, in which a second semiconductor layer is formed above the first transistor. The second semiconductor layer can include single crystalline silicon. In some implementations, to form the second semiconductor layer, another substrate and the first substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the second semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(386) As illustrated in FIG. 30D, a semiconductor layer **3002**, such as a single crystalline silicon layer, is formed above interconnect layer **3020** and transistors **3014** and **3016**. Semiconductor layer **3002** can be attached above interconnect layer **3020** to form a bonding interface **3034** vertically between semiconductor layer **3002** and interconnect layer **3020**. In some implementations, to form semiconductor layer **3002**, another silicon substrate (not shown in FIG. 30D) and silicon substrate **3024** are bonded in a face-to-face manner (having the components formed on silicon substrate **3024**, such as NAND memory strings **3028** and transistors **3014** and **3016**, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface **3034**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **3002** attached above interconnect layer **3020**. The details of various transfer bonding processes are described above with respect to FIGS. 48A-48D and FIGS. 49A-49D and thus, are not repeated for ease of description.

(387) Referring to FIG. 32, method **3200** proceeds to operation **3206** in which a second transistor is formed on the second semiconductor layer. As illustrated in FIG. 30E, a plurality of transistors **3004** and **3006** are formed on semiconductor layer **3002** having single crystalline silicon.

Transistors **3004** and **3006** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **3002** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3004** and **3006**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **3002** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3004** is different from the thickness of gate dielectric of transistor **3006**, for example, by depositing a thicker silicon oxide film in the region of transistor **3004** than the region of transistor **3006**, or by etching back part of the silicon oxide film deposited in the region of transistor **3006**. It is understood that the details of fabricating transistors **3004** and **3006** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(388) In some implementations, an interconnect layer **3008** is formed above the transistor on the semiconductor layer. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 30E, an interconnect layer **3008** can be formed above transistors **3004** and **3006**. Interconnect layer **3008** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3004** and **3006**. In some



implementations, interconnect layer **3008** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3008** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **30C** can be collectively referred to as interconnect layer **3008**. Different from interconnect layer **3020**, in some implementations, the interconnects in interconnect layer **3008** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **3008** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **3008**.

(389) In some implementations, a contact through the semiconductor layer is formed. As illustrated in FIG. **30E**, one or more contacts **3019** each extending vertically through semiconductor layer **3002** is formed. Contacts **3019** can couple the interconnects in interconnect layers **3008** and **3020**. Contacts **3019** can be formed by first patterning contact holes through semiconductor layer **3002** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(390) Method **3200** skips optional operation **3212** and proceeds to operation **3214**, as illustrated in FIG. **32**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **30F**, a pad-out interconnect layer **3036** is formed above interconnect layer **3008** and transistors **3004** and **3006** on semiconductor layer **3002**. Pad-out interconnect layer **3036** can include interconnects, such as contact pads **3038**, formed in one or more ILD layers. Contact pads **3038** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(391) In some implementations, to form a pad-out interconnect layer on the first substrate, after operation **3210**, method **3200** proceeds to optional operation **3212**, as illustrated in FIG. **32**, in which the first substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **3024** (shown in FIG. **30E**) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **3024**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **3024** before thinning and be exposed from the backside of silicon substrate **3024** (where the thinning occurs) after the thinning.

(392) Method **3200** proceeds to operation **3214**, as illustrated in FIG. **32**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **3024**.

(393) FIGS. **31A-31F** illustrate another fabrication process for forming the 3D memory devices in FIGS. **28A** and **28B**, according to some aspects of the present disclosure. FIG. **33** illustrates a flowchart of another method **3300** for forming the 3D memory devices in FIGS. **28A** and **28B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted

in FIGS. **31A-31F** and **33** include 3D memory devices **2900** and **2901** depicted in FIGS. **29A** and **29B**. FIGS. **31A-31F** and **33** will be described together. It is understood that the operations shown in method **3300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **33**. For example, operation **3302**, **3304**, and **3306** may be performed in parallel.

(394) Referring to FIG. **33**, method **3300** starts at operation **3302**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(395) As illustrated in FIG. **31A**, a stack structure, such as a memory stack **3104** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **3102**. To form memory stack **3104**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **3102**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **3104** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **3104** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **3104** and silicon substrate **3102**.

(396) As illustrated in FIG. **31A**, NAND memory strings **3106** are formed above silicon substrate **3102**, each of which extends vertically through memory stack **3104** to be in contact with silicon substrate **3102**. In some implementations, fabrication processes to form NAND memory string **3106** include forming a channel hole through memory stack **3104** (or the dielectric stack) and into silicon substrate **3102** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **3106** may vary depending on the types of channel structures of NAND memory strings **3106** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(397) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **31A**, an interconnect layer **3108** is formed above memory stack **3104** and NAND memory strings **3106**. Interconnect layer **3108** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **3106**. In some implementations, interconnect layer **3108** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3108** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects

illustrated in FIG. 31A can be collectively referred to as interconnect layer **3108**.

(398) Method **3300** proceeds to operation **3304**, as illustrated in FIG. 33, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 31B, a plurality of transistors **3114** and **3116** are formed on a silicon substrate **3112**. Transistors **3114** and **3116** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3112** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3114** and **3116**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3112** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3114** is different from the thickness of gate dielectric of transistor **3116**, for example, by depositing a thicker silicon oxide film in the region of transistor **3114** than the region of transistor **3116**, or by etching back part of the silicon oxide film deposited in the region of transistor **3116**. It is understood that the details of fabricating transistors **3114** and **3116** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(399) In some implementations, an interconnect layer **3118** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 31B, an interconnect layer **3118** can be formed above transistors **3114** and **3116**. Interconnect layer **3118** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3114** and **3116**. In some implementations, interconnect layer **3118** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3118** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 31B can be collectively referred to as interconnect layer **3118**.

(400) Method **3300** proceeds to operation **3306**, as illustrated in FIG. 33, in which a second transistor is formed on a third substrate. The third substrate can be a silicon substrate having single crystalline silicon. In some implementations, any two or all of operations **3302**, **3304**, and **3306** are performed in parallel to reduce process time.

(401) As illustrated in FIG. 31C, a plurality of transistors **3124** and **3126** are formed on a silicon substrate **3122**. Transistors **3124** and **3126** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3122** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3124** and **3126**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3122** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3124** is different from the thickness of gate dielectric of transistor **3126**, for example, by depositing a thicker silicon oxide film in the region of transistor **3124** than the region of transistor **3126**, or by etching back part of the silicon oxide film deposited in the region of transistor **3126**. It is understood that the details of fabricating transistors **3124** and **3126** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(402) In some implementations, an interconnect layer **3128** is formed above the transistor on the

third substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 31C, an interconnect layer **3128** can be formed above transistors **3124** and **3126**. Interconnect layer **3128** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3124** and **3126**. In some implementations, interconnect layer **3128** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1928** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 31C can be collectively referred to as interconnect layer **3128**.

(403) In some implementations, at least one of the second substrate or the third substrate is thinned. As illustrated in FIG. 31D, silicon substrate **3112** (shown in FIG. 31B) is thinned to become a semiconductor layer **3135** having single crystalline silicon. Similarly, as illustrated in FIG. 31E, silicon substrate **3122** (shown in FIG. 31C) is thinned to become a semiconductor layer **3123** having single crystalline silicon. Silicon substrate **3112** or **3122** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, handle substrates (not shown) are attached to interconnect layers **3118** and **3128**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrates **3112** and **3122**, such as thinning, contact formation, and bonding.

(404) In some implementations, a first contact through the thinned second substrate and coupled to the interconnect layer is formed. In some implementations, a second contact through the thinned third substrate and coupled to the interconnect layer is formed. As illustrated in FIG. 31D, one or more contacts **3136** each extending vertically through semiconductor layer **3135** (i.e., the thinned silicon substrate **3112**) are formed. Contacts **3136** can be coupled to the interconnects in interconnect layer **3118**. Similarly, as illustrated in FIG. 31E, one or more contacts **3137** each extending vertically through semiconductor layer **3123** (i.e., the thinned silicon substrate **3122**) are formed. Contacts **3137** can be coupled to the interconnects in interconnect layer **3128**. Contact **3137** or **3136** can be formed by first patterning contact holes through semiconductor layer **3123** or **3135** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **3136** may be formed in silicon substrate **3112** before thinning (the formation of semiconductor layer **3135**, e.g., in FIG. 31B) and be exposed from the backside of silicon substrate **3112** (where the thinning occurs) after the thinning. Similarly, contacts **3137** may be formed in silicon substrate **3122** before thinning (the formation of semiconductor layer **3123**, e.g., in FIG. 31C) and be exposed from the backside of silicon substrate **3122** (where the thinning occurs) after the thinning.

(405) Method **3300** proceeds to operation **3308**, as illustrated in FIG. 33, in which the first substrate and the second substrate are bonded in a face-to-back manner. As illustrated in FIG. 31D, silicon substrate **3102** and components formed thereon (e.g., memory stack **3104** and NAND memory strings **3106**) is bonded to thinned silicon substrate **3112** (i.e., semiconductor layer **3135**) and components formed thereon (e.g., transistors **3114** and **3116**) in a face-to-back manner, i.e., the frontside of silicon substrate **3102** facing toward the backside of thinned silicon substrate **3112**, to form a bonding interface **3132**. The bonding can be performed using fusion bonding or anodic bonding depending on the materials at bonding interface **3132**, e.g., SiO.sub.2—Si or SiO.sub.2—SiO.sub.2. As a result of the bonding, contacts **3136** couple the interconnects in interconnect layer

**3118** to the interconnects in interconnect layer **3108**.

(406) Method **3300** proceeds to operation **3310**, as illustrated in FIG. **33**, in which the second substrate and the third substrate are bonded in a face-to-back manner. As illustrated in FIG. **31E**, thinned silicon substrate **3112** (i.e., semiconductor layer **3135**) and components formed thereon (e.g., transistors **3114** and **3116**) is bonded to thinned silicon substrate **3122** (i.e., semiconductor layer **3123**) and components formed thereon (e.g., transistors **3124** and **3126**) in a face-to-back manner, i.e., the frontside of thinned silicon substrate **3112** facing toward the backside of thinned silicon substrate **3122**, to form a bonding interface **3140**. The bonding can be performed using fusion bonding or anodic bonding depending on the materials at bonding interface **3140**, e.g., SiO.sub.2—Si or SiO.sub.2—SiO.sub.2. As a result of the bonding, contacts **3137** couple the interconnects in interconnect layer **3128** to the interconnects in interconnect layer **3118**. It is understood that the sequence of bonding silicon substrates **3102**, **3112**, and **3222** may switch to any suitable order in other examples.

(407) Method **3300** skips optional operation **3312** and proceeds to operation **3314**, as illustrated in FIG. **33**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **31F**, a pad-out interconnect layer **3146** is formed above interconnect layer **3128** and transistors **3124** and **3126** on semiconductor layer **3123**. Pad-out interconnect layer **3146** can include interconnects, such as contact pads **3148**, formed in one or more ILD layers. Contact pads **3148** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(408) In some implementations, to form a pad-out interconnect layer on the first substrate, after operation **3310**, method **3300** proceeds to optional operation **3312**, as illustrated in FIG. **33**, in which the first substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **3102** (shown in FIG. **31E**) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **3102**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **3102** before thinning and be exposed from the backside of silicon substrate **3102** (where the thinning occurs) after the thinning.

(409) Method **3300** proceeds to operation **3314**, as illustrated in FIG. **33**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **3102**.

(410) FIGS. **34A** and **34B** illustrate schematic views of cross-sections of 3D memory devices **3400** and **3401** having three stacked semiconductor structures, according to various aspects of the present disclosure. 3D memory devices **3400** and **3401** may be examples of 3D memory device **101** in FIG. **1B** in which first semiconductor structure **102** including the memory cell array is disposed vertically between second semiconductor structure **104** including some of the peripheral circuits and third semiconductor structure **106** including some of the peripheral circuits. In other words, as shown in FIGS. **34A** and **34B**, first semiconductor structure **102** including the memory cell array of 3D memory devices **900** and **901** is disposed in the intermediate of 3D memory devices **3400** and **3401**, second semiconductor structure **104** including some of the peripheral circuits is disposed on one side of 3D memory devices **3400** and **3401**, and third semiconductor structure **106** including some of the peripheral circuits is disposed on another side of 3D memory devices **3400** and **3401** in the vertical direction, according to some implementations. Second and third semiconductor structures **104** and **106** each including peripheral circuits can be separated by first semiconductor

structure **102** including the memory cell array in three stacked semiconductor structures **102**, **104**, and **106**.

(411) Moreover, as shown in FIGS. **34A** and **34B**, 3D memory device **3400** or **3401** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **34A**, third semiconductor structure **106** including some of the peripheral circuits on one side of 3D memory device **3400** may include pad-out interconnect layer **902**. In another example shown in FIG. **34B**, second semiconductor structure **104** including some of the peripheral circuits on one side of 3D memory device **3401** may include pad-out interconnect layer **902**. In either example, 3D memory device **3400** or **3401** may be pad-out from one peripheral circuit side to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory devices **3400** and **3401**.

(412) FIGS. **35A** and **35B** illustrate schematic views of cross-sections of 3D memory devices **3400** and **3401** in FIGS. **34A** and **34B**, according to some aspects of the present disclosure. 3D memory devices **3500** and **3501** may be examples of 3D memory devices **3400** and **3401** in FIGS. **34A** and **34B**. As shown in FIG. **35A**, 3D memory device **3500** can include stacked first, second, and third semiconductor structures **102**, **104**, and **106**. In some implementations, first semiconductor structure **102** in the intermediate of 3D memory device **3500** includes semiconductor layer **1002**, a bonding layer **3502**, and a memory cell array vertically between bonding layer **3502** and semiconductor layer **1002**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer **1002** (e.g., as shown in FIGS. **8A-8C**).

Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**). Through contacts (e.g., ILVs/TSVs) can make direct, short-distance (e.g., submicron-level) electrical connections through semiconductor layer **1002**. Bonding layer **3502** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding. In some implementations, bonding layer **3502** is disposed vertically between bonding interface **3503** and the memory cell array in first semiconductor structure **102**.

(413) In some implementations, second semiconductor structure **104** on one side of 3D memory device **3500** includes a semiconductor layer **1004** and some of the peripheral circuits of the memory cell array. In some implementations, semiconductor layer **1004** is disposed vertically between bonding interface **103** and the peripheral circuits of second semiconductor structure **104**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a layer transferred from a silicon substrate or an SOI substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Through contacts (e.g., ILVs/TSVs) can make direct, short-distance (e.g., submicron-level) electrical connections through semiconductor layer **1004**.

(414) Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. Through contacts (e.g., ILVs/TSVs) through semiconductor layers **1002** and **1004** and in contact with each other at bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent

semiconductor structures **102** and **104**.

(415) In some implementations, third semiconductor structure **106** on another side of 3D memory device **3500** includes a semiconductor layer **1006**, a bonding layer **1014**, and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1006** and bonding interface **3503**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1006**. Semiconductor layer **1006** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **1006** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Bonding layer **1014** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding.

(416) Bonding interface **3503** is vertically between and in contact with bonding layers **3502** and **1014**, respectively, according to some implementations. That is, bonding layers **3502** and **1014** can be disposed on opposite sides of bonding interface **3503**, and the bonding contacts of bonding layer **3502** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **3503**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **3503** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **106**.

(417) It is understood that in some examples, first and second semiconductor structures **102** and **104** may also include bonding layers **1008** and **1010**, respectively, disposed on opposite sides of bonding interface **103**, as shown in FIG. 35B. In FIG. 35B, first semiconductor structure **102** of a 3D memory device **3501** can include two bonding layers **1008** and **3502** on two sides thereof, and bonding layer **1008** can be disposed vertically between semiconductor layer **1002** and bonding interface **103**. Second semiconductor structure **104** of 3D memory device **3501** can include bonding layer **1010** disposed vertically between bonding interface **103** and semiconductor layer **1004**. Each bonding layer **1008** or **1010** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. The bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, bonding contacts across bonding interface **103** in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layers **1002** and **1004** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **104** and **102**.

(418) As shown in FIGS. 35A and 35B, since third and first semiconductor structures **106** and **102** are bonded in a face-to-face manner (e.g., semiconductor layer **1006** being disposed on the top side of third semiconductor structure **106**, while semiconductor layer **1002** being disposed on the bottom side of first semiconductor structure **102** in FIGS. 35A and 35B), the transistors in third semiconductor structure **106** and the memory cell array in first semiconductor structure **102** face toward each other, according to some implementations. In some implementations, semiconductor layer **1004** is disposed vertically between the transistors of the peripheral circuits in second semiconductor structure **104** and bonding interface **103**, and the transistors of the peripheral circuits in third semiconductor structure **106** are disposed vertically between bonding interface **105** and semiconductor layer **1006**. Moreover, since first and second semiconductor structures **102** and **104** are bonded in a back-to-back manner (e.g., semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104**, while semiconductor layer **1002** being disposed on the bottom side of first semiconductor structure **102** in FIGS. 35A and 35B), the transistors of peripheral circuits in second semiconductor structure **104** and the memory cell array in first semiconductor structure **102** face away from each other, according to some implementations. It is understood that pad-out interconnect layer **902** in FIGS. 9A and 9B is omitted from 3D memory device **3500** in FIG. 35 for case of illustration and may be included in 3D memory device **3500** as

described above with respect to FIGS. 9A and 9B.

(419) As described above, second and third semiconductor structures **104** and **106** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. 4B, and third semiconductor structure **106** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. 4B, or vice versa. Thus, in some implementations, semiconductor layers **1006** and **1004** in third and second semiconductor structures **106** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1006** in third semiconductor structure **106** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in third and second semiconductor structures **106** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, third semiconductor structure **106** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in third semiconductor structure **106** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**.

(420) FIGS. 36A and 36B illustrate side views of various examples of 3D memory devices **3500** and **3501** in FIGS. 35A and 35B, according to various aspects of the present disclosure. As shown in FIG. 36A, as one example of 3D memory devices **3500** and **3501** in FIGS. 35A and 35B, 3D memory device **3600** is a bonded chip including first semiconductor structure **102**, second semiconductor structure **104**, and third semiconductor structure **106**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. 36A), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, and first and third semiconductor structures **102** and **106** are bonded at bonding interface **3503** therebetween, according to some implementations.

(421) As shown in FIG. 36A, third semiconductor structure **106** can include semiconductor layer **1006** having semiconductor materials. In some implementations, semiconductor layer **1006** is a silicon substrate having single crystalline silicon. Third semiconductor structure **106** can also include a device layer **3602** above and in contact with semiconductor layer **1006**. In some implementations, device layer **3602** includes a first peripheral circuit **3604** and a second peripheral circuit **3606**. First peripheral circuit **3604** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **3606** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **3604** includes a plurality of transistors **3608** in contact with semiconductor layer **1006**, and second peripheral circuit **3606** includes a plurality of transistors **3610** in contact with semiconductor layer **1006**. Transistors **3608** and **3610** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **3608** or **3610** includes a gate dielectric, and the thickness of the gate dielectric of transistor **3608** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **3610** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **3608** than transistor **3610**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **3608** and **3610**) can be formed on or in semiconductor layer **1006** as well.

(422) In some implementations, third semiconductor structure **106** further includes an interconnect layer **3612** above device layer **3602** to transfer electrical signals to and from peripheral circuits **3606** and **3604**. As shown in FIG. 36A, interconnect layer **3612** can be disposed vertically between



bonding interface **3503** and device layer **3602** (including transistors **3608** and **3610** of peripheral circuits **3604** and **3606**). Interconnect layer **3612** can include a plurality of interconnects. The interconnects in interconnect layer **3612** can be coupled to transistors **3608** and **3610** of peripheral circuits **3604** and **3606** in device layer **3602**. Interconnect layer **3612** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **3612** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **3602** are coupled to one another through the interconnects in interconnect layer **3612**. For example, peripheral circuit **3604** may be coupled to peripheral circuit **3606** through interconnect layer **3612**. The interconnects in interconnect layer **3612** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **3612** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(423) As shown in FIG. **36A**, third semiconductor structure **106** can further include a bonding layer **1014** at bonding interface **3503** and above and in contact with interconnect layer **3612**. Bonding layer **1014** can include a plurality of bonding contacts **1015** and dielectrics electrically isolating bonding contacts **1015**. Bonding contacts **1015** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, the bonding contacts of bonding layer **1014** include Cu. The remaining area of bonding layer **1014** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1015** and surrounding dielectrics in bonding layer **1014** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously.

(424) As shown in FIG. **36A**, first semiconductor structure **102** can also include a bonding layer **3502** at bonding interface **3503**, e.g., on the opposite side of bonding interface **3503** with respect to bonding layer **1014** in third semiconductor structure **106**. Bonding layer **3502** can include a plurality of bonding contacts **3505** and dielectrics electrically isolating bonding contacts **3505**. Bonding contacts **3505** can include conductive materials, such as Cu. The remaining area of bonding layer **3502** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **3505** and surrounding dielectrics in bonding layer **3502** can be used for hybrid bonding. In some implementations, bonding interface **3503** is the place at which bonding layers **3502** and **1014** are met and bonded. In practice, bonding interface **3503** can be a layer with a certain thickness that includes the top surface of bonding layer **1014** of third semiconductor structure **106** and the bottom surface of bonding layer **3502** of first semiconductor structure **102**.

(425) As shown in FIG. **36A**, first semiconductor structure **102** can further include an interconnect layer **3628** above and in contact with bonding layer **3502** to transfer electrical signals. Interconnect layer **3628** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **3628** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **3628** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **3628** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **3628** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(426) As shown in FIG. **36A**, first semiconductor structure **102** can further include a memory cell array, such as an array of NAND memory strings **208** above and in contact with interconnect layer **3628**. In some implementations, interconnect layer **3628** is vertically between NAND memory strings **208** and bonding interface **3503**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some

implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **3627**. Memory stack **3627** may be an example of memory stack **804** in FIGS. **8A-8C**, and the conductive layer and dielectric layer in memory stack **3627** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **3627** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **3627**.

(427) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(428) As shown in FIG. **36A**, first semiconductor structure **102** can further include semiconductor layer **1002** disposed above memory stack **3627** and in contact with the sources of NAND memory strings **208**. In some implementations, semiconductor layer **1002** is disposed vertically between bonding interface **103** and NAND memory strings **208**. Semiconductor layer **1002** can include semiconductor materials. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **3627** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(429) As shown in FIG. **36A**, first semiconductor structure **102** can further include one or more contacts **3625** extending vertically through semiconductor layer **1002**. In some implementations, contacts **3625** are coupled to the interconnects in interconnect layer **3628**. Contact **3625** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **3625** includes W. In some implementations, contact **3625** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **3625** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(430) Second semiconductor structure **104** can be bonded with first semiconductor structure **102** in a back-to-back manner at bonding interface **103**. Second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, bonding interface **103** is disposed vertically between semiconductor layer **1002** and semiconductor layer **1004** as a result of anodic bonding or fusion bonding as described below in detail. In some implementations, bonding interface **103** is the place at which semiconductor layer **1002** and semiconductor layer **1004** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of semiconductor layer **1002** of first semiconductor structure **102** and the bottom surface of semiconductor layer **1004** of second semiconductor structure **104**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **103** and semiconductor layer **1004** and/or between bonding interface **103** and semiconductor layer **1002** to facilitate the fusion bonding or anodic bonding of semiconductor layers **1002** and **1004**. Thus, it is understood that bonding interface **103** may include the surfaces of the dielectric layer(s) in some examples. It is further understood that in some examples, bonding layers having bonding contacts (e.g., Cu contacts) may be formed vertically between bonding interface **103** and semiconductor layer **1004** and between bonding

interface **103** and semiconductor layer **1002** to achieve hybrid bonding of semiconductor layers **1002** and **1004**. In other words, a dielectric layer (e.g., silicon oxide layer) may be disposed vertically between semiconductor layer **1004** and semiconductor layer **1002** in some examples, which can serve as a shielding layer between the components formed on semiconductor layer **1002** and the components formed on semiconductor layer **1004**, for example, for reducing the impact across bonding interface **103** on the threshold voltage of transistors **3620** and **3622** caused by memory stack **3627** and NAND memory strings **208**.

(431) Second semiconductor structure **104** can include a device layer **3614** above and in contact with semiconductor layer **1004**. In some implementations, device layer **3614** includes a third peripheral circuit **3616** and a fourth peripheral circuit **3618**. Third peripheral circuit **3616** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **3618** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **3616** includes a plurality of transistors **3620**, and fourth peripheral circuit **3618** includes a plurality of transistors **3622** as well. Transistors **3620** and **3622** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **3620** or **3622** includes a gate dielectric, and the thickness of the gate dielectric of transistor **3620** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **3622** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **3620** than transistor **3622**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **3620** and **3622**) can be formed on or in semiconductor layer **1004** as well.

(432) Moreover, the different voltages applied to different transistors **3620**, **3622**, **3608**, and **3610** in second and third semiconductor structures **104** and **106** can lead to differences of device dimensions between second and third semiconductor structures **104** and **106**. In some implementations, the thickness of the gate dielectric of transistor **3608** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **3620** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **3608** than transistor **3620**. In some implementations, the thickness of the gate dielectric of transistor **3622** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **3610** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **3622** and transistor **3610**. In some implementations, the thickness of semiconductor layer **1006** in which transistor **3608** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **3620** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **3608** than transistor **3620**.

(433) As shown in FIG. **36A**, second semiconductor structure **104** can further include an interconnect layer **3626** above and in contact with device layer **3614** to transfer electrical signals to and from peripheral circuits **3616** and **3618**. As shown in FIG. **36A**, device layer **1714** (including transistors **1720** and **1722** of peripheral circuits **1716** and **1718**) can be vertically between bonding interface **103** and interconnect layer **3626**. Interconnect layer **3626** can include a plurality of interconnects coupled to transistors **3620** and **3622** of peripheral circuits **3616** and **3618** in device layer **3614**. Interconnect layer **3626** can further include one or more ILD layers in which the interconnects can form. That is, interconnect layer **3626** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **3614** are coupled to one another through the interconnects in interconnect layer **3626**. For example, peripheral circuit **3616** may be coupled to peripheral circuit **3618** through interconnect layer **3626**. The interconnects in interconnect layer **3626** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **3626** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(434) In some implementations, the interconnects in interconnect layer **3626** include Cu, which has

a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **3626** can occur after the high-temperature processes in forming device layer **3614** in second semiconductor structure **104** and devices in first semiconductor structure **102**, as well as being separated from the high temperature processes in forming third semiconductor structure **106**, the interconnects of interconnect layer **3626** having Cu can become feasible.

(435) As shown in FIG. **36A**, second semiconductor structure **104** can further include one or more contacts **3624** extending vertically through semiconductor layer **1004**. In some implementations, contacts **3624** are coupled to the interconnects in interconnect layer **3626**. In some implementations, contact **3624** is in contact with contact **3625**, such that contacts **3624** and **3625** couple the interconnects in interconnect layer **3626** to the interconnects in interconnect layer **3628** to make an electrical connection across bonding interface **103** between second and first semiconductor structures **104** and **102** and through semiconductor layers **1004** and **1002**. Contact **3624** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **3624** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **3624** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ).

(436) As shown in FIG. **36A**, second semiconductor structure **104** can further include a pad-out interconnect layer **902** above and in contact with interconnect layer **3626**. In some implementations, device layer **3614** having transistors **3620** and **3622** is disposed vertically between pad-out interconnect layer **902** and semiconductor layer **1004**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **3632**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **3626** can be formed on the same side of semiconductor layer **1004**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **3600** and external devices, e.g., for pad-out purposes.

(437) As a result, peripheral circuits **3604**, **3606**, **3616**, and **3618** in third and second semiconductor structures **106** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **3612**, **3626**, and **3628** and contacts **3624** and **3625**. Moreover, peripheral circuits **3604**, **3606**, **3616**, and **3618** and NAND memory strings **208** in 3D memory device **3600** can be further coupled to external devices through pad-out interconnect layer **902**.

(438) It is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having transistors **3620** and **3622** as shown in FIG. **36A** (corresponding to FIG. **34B**) and may be from third semiconductor structure **106** having transistors **3608** and **3610** (corresponding to FIG. **34A**). For example, as shown in FIG. **36B**, 3D memory device **3601** may include pad-out interconnect layer **902** in third semiconductor structure **106**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1006** of third semiconductor layer **1006** on which transistors **3608** and **3610** are formed. In some implementations, third semiconductor structure **106** further includes one or more contacts **3634** extending vertically through semiconductor layer **1006**. In some implementations, contact **3634** couples the interconnects in interconnect layer **3612** in third semiconductor structure **106** to contact pads **3632** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1006**. Contact **3634** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **3634** includes W. In some implementations, contact **3634** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1006**. Depending on the thickness

of semiconductor layer **1006**, contact **3634** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu$ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu$ m and 100  $\mu$ m).

(439) It is further understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIG. **36A** and may be any other suitable semiconductor materials. For example, as shown in FIG. **36B**, 3D memory device **3601** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **3601** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **3601** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **3600** and **3601** are not repeated for case of description.

(440) FIGS. **37A-37G** illustrate a fabrication process for forming the 3D memory device in FIGS. **35A** and **35B**, according to some aspects of the present disclosure. FIG. **38** illustrates a flowchart of another method **3800** for forming the 3D memory devices in FIGS. **35A** and **35B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **37A-37G** and **38** include 3D memory devices **3600** and **3601** depicted in FIGS. **36A** and **36B**. FIGS. **37A-37G** and **38** will be described together. It is understood that the operations shown in method **3800** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **38**. For example, operation **3802**, **3804**, and **3806** may be performed in parallel.

(441) Referring to FIG. **38**, method **3800** starts at operation **3802**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(442) As illustrated in FIG. **37A**, a stack structure, such as a memory stack **3704** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **3702**. To form memory stack **3704**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **3702**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **3704** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **3704** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **3704** and silicon substrate **3702**.

(443) As illustrated in FIG. **37A**, NAND memory strings **3706** are formed above silicon substrate **3702**, each of which extends vertically through memory stack **3704** to be in contact with silicon substrate **3702**. In some implementations, fabrication processes to form NAND memory string **3706** include forming a channel hole through memory stack **3704** (or the dielectric stack) and into silicon substrate **3702** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a

tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **3706** may vary depending on the types of channel structures of NAND memory strings **3706** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for ease of description.

(444) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **37A**, an interconnect layer **3708** is formed above memory stack **3704** and NAND memory strings **3706**. Interconnect layer **3708** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **3706**. In some implementations, interconnect layer **3708** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3708** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **37A** can be collectively referred to as interconnect layer **3708**.

(445) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **37A**, a bonding layer **3710** is formed above interconnect layer **3708**. Bonding layer **3710** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **3708** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **3708** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(446) Method **3800** proceeds to operation **3804**, as illustrated in FIG. **38**, in which a first transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **37B**, a plurality of transistors **3714** and **3716** are formed on a silicon substrate **3712**. Transistors **3714** and **3716** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3712** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3714** and **3716**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3712** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3714** is different from the thickness of gate dielectric of transistor **3716**, for example, by depositing a thicker silicon oxide film in the region of transistor **3714** than the region of transistor **3716**, or by etching back part of the silicon oxide film deposited in the region of transistor **3716**. It is understood that the details of fabricating transistors **3714** and **3716** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(447) In some implementations, an interconnect layer **3718** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD

layers. As illustrated in FIG. 37B, an interconnect layer **3718** can be formed above transistors **3714** and **3716**. Interconnect layer **3718** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3714** and **3716**. In some implementations, interconnect layer **3718** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3718** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 37B can be collectively referred to as interconnect layer **3718**.

(448) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. 37B, a bonding layer **3720** is formed above interconnect layer **3718**. Bonding layer **3720** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **3718** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **3718** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(449) Method **3800** proceeds to operation **3806**, as illustrated in FIG. 38, in which a second transistor is formed on a third substrate. The third substrate can be a silicon substrate having single crystalline silicon. In some implementations, any two or all of operations **3802**, **3804**, and **3806** are performed in parallel to reduce process time.

(450) As illustrated in FIG. 37C, a plurality of transistors **3724** and **3726** are formed on a silicon substrate **3722**. Transistors **3724** and **3726** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3722** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3724** and **3726**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3722** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3724** is different from the thickness of gate dielectric of transistor **3726**, for example, by depositing a thicker silicon oxide film in the region of transistor **3724** than the region of transistor **3726**, or by etching back part of the silicon oxide film deposited in the region of transistor **3726**. It is understood that the details of fabricating transistors **3724** and **3726** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description.

(451) In some implementations, an interconnect layer is formed above the transistor on the third substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 37C, an interconnect layer **3742** can be formed above transistors **3724** and **3726**. Interconnect layer **3742** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3724** and **3726**. In some implementations, interconnect layer **3742** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3742** can include conductive materials deposited by one or more thin film deposition processes including,

but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 37C can be collectively referred to as interconnect layer **3742**. In some implementations, the interconnects in interconnect layer **3742** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **3742** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **3742**.

(452) Method **3800** proceeds to operation **3808**, as illustrated in FIG. **38**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a first bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(453) As illustrated in FIG. **37D**, silicon substrate **3702** and components formed thereon (e.g., memory stack **3704** and NAND memory strings **3706** formed therethrough) are flipped upside down. Bonding layer **3710** facing down is bonded with bonding layer **3720** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **3732**. That is, silicon substrate **3702** and components formed thereon can be bonded with silicon substrate **3712** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **3710** are in contact with the bonding contacts in bonding layer **3720** at bonding interface **3732**. Transistors **3714** and **3716** and NAND memory strings **3706** can face toward each other after the bonding. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **37D**, it is understood that in some examples, silicon substrate **3712** and components formed thereon (e.g., transistors **3714** and **3716**) can be flipped upside down, and bonding layer **3720** facing down can be bonded with bonding layer **3710** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **3732** as well.

(454) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **3732** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **3710** and the bonding contacts in bonding layer **3720** are aligned and in contact with one another, such that memory stack **3704** and NAND memory strings **3706** formed therethrough can be coupled to transistors **3714** and **3716** through the bonded bonding contacts across bonding interface **3732**, according to some implementations.

(455) In some implementations, the first substrate is thinned. As illustrated in FIG. **37E**, silicon substrate **3702** (shown in FIG. **37D**) is thinned to become a semiconductor layer **3734** having single crystalline silicon. Silicon substrate **3702** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(456) In some implementations, a first contact through the thinned first substrate is formed. As illustrated in FIG. **37E**, one or more contacts **3736** each extending vertically through semiconductor layer **3734** (i.e., the thinned silicon substrate **3702**) are formed. Contacts **3736** can be coupled to the interconnects in interconnect layer **3708**. Contact **3736** can be formed by first patterning contact holes through semiconductor layer **3734** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **3736** may be formed in silicon substrate **3702** before thinning (the formation of



semiconductor layer **3734**, e.g., in FIG. **37A**) and be exposed from the backside of silicon substrate **3702** (where the thinning occurs) after the thinning.

(457) In some implementations, the third substrate is thinned. As illustrated in FIG. **37F**, silicon substrate **3722** (shown in FIG. **37C**) is thinned to become a semiconductor layer **3728** having single crystalline silicon. Silicon substrate **3722** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, as shown in FIG. **37F**, a handle substrate **3743** is attached to interconnect layer **3742**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **3722**, such as thinning, contact formation, and bonding.

(458) In some implementations, a second contact through the thinned third substrate is formed. As illustrated in FIG. **37F**, one or more contacts **3737** each extending vertically through semiconductor layer **3728** (i.e., the thinned silicon substrate **3722**) are formed. Contacts **3737** can be coupled to the interconnects in interconnect layer **3708**. Contact **3736** can be formed by first patterning contact holes through semiconductor layer **3734** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **3737** may be formed in silicon substrate **3722** before thinning (the formation of semiconductor layer **3728**, e.g., in FIG. **37C**) and be exposed from the backside of silicon substrate **3722** (where the thinning occurs) after the thinning.

(459) Method **3800** proceeds to operation **3810**, as illustrated in FIG. **38**, in which the first substrate and the third substrate are bonded in a back-to-back manner. As illustrated in FIG. **37F**, thinned silicon substrate **3702** (i.e., semiconductor layer **3734**) and components formed thereon (e.g., memory stack **3704** and NAND memory strings **3706**) is bonded to thinned silicon substrate **3722** (i.e., semiconductor layer **3728**) and components formed thereon (e.g., transistors **3724** and **3726**) in a face-to-back manner, i.e., the backside of thinned silicon substrate **3702** facing toward the backside of thinned silicon substrate **3722**, to form a bonding interface **3740**. The bonding can be performed using fusion bonding or anodic bonding depending on the materials at bonding interface **3132**, e.g.,  $\text{SiO}_2\text{—Si}$  or  $\text{SiO}_2\text{—SiO}_2$ . As a result of the bonding, contact **3736** is aligned and in contact with contact **3736** at bonding interface **3740**, and bonded contacts **3736** and **3737** couple the interconnects in interconnect layer **3742** to the interconnects in interconnect layer **3708**, according to some implementations.

(460) In some implementations, a third bonding layer is formed on a second side of the thinned first substrate opposite to a first side on which the array of NAND memory strings is formed, and a fourth bonding layer is formed on a second side of the thinned third substrate opposite to a first side on which the transistor is formed. The third bonding layer can include a plurality of third bonding contacts, and the fourth bonding layer can include a plurality of fourth bonding contacts. Although not shown in FIG. **37F**, it is understood that the first substrate and the third substrate may be bonded in a back-to-back manner using hybrid bonding, such that the third bonding contacts in the third bonding layer are aligned and in contact with the fourth bonding contacts in the fourth bonding layer at bonding interface **3740** as described above in detail. Although not shown, in some implementations, semiconductor layer **3734** having single crystalline silicon (i.e., thinned silicon substrate **3702**) is replaced with a semiconductor layer having a different material (e.g., a polysilicon layer) before forming the third bonding layer, such that the third bonding layer is formed on the replaced semiconductor layer (e.g., the polysilicon layer). As a result, the third and fourth bonding layers can be in contact with semiconductor layers with different materials, such as polysilicon and single crystalline silicon, respectively.

(461) Method **3800** skips optional operation **3812** and proceeds to operation **3814**, as illustrated in FIG. **38**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **37G**, handle substrate **3743** (shown in

FIG. 37F) is removed, and a pad-out interconnect layer **3746** is formed above interconnect layer **3742** and transistors **3724** and **3726** on semiconductor layer **3728**. Pad-out interconnect layer **3746** can include interconnects, such as contact pads **3748**, formed in one or more ILD layers. Contact pads **3748** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(462) In some implementations, to form a pad-out interconnect layer on the second substrate, after operation **3810**, method **3800** proceeds to optional operation **3812**, as illustrated in FIG. **38**, in which the second substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **3702** (shown in FIG. 37F) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **3712**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **3712** before thinning and be exposed from the backside of silicon substrate **3712** (where the thinning occurs) after the thinning.

(463) Method **3800** proceeds to operation **3814**, as illustrated in FIG. **38**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned second substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **3712**. It is further understood that in some examples, the first substrate (e.g., silicon substrate **3702** or semiconductor layer **3734** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**.

(464) FIGS. **39A** and **39B** illustrate schematic views of cross-sections of 3D memory devices **3900** and **3901** having two stacked semiconductor structures, according to various aspects of the present disclosure. 3D memory devices **3900** and **3901** may be examples of 3D memory device **120** in FIG. **1C** in which first semiconductor structure **102** including the memory cell array is bonded to fourth semiconductor structure **108** including at least two separate portions of the peripheral circuits of the memory cell array disposed in different planes. In other words, as shown in FIGS. **39A** and **39B**, first semiconductor structure **102** including the memory cell array of 3D memory devices **3900** and **3901** is disposed on one side of 3D memory devices **3900** and **3901** in the vertical direction, according to some implementations.

(465) In some implementations, first semiconductor structure **102** includes a semiconductor layer **1002**, a bonding layer **1008**, and a memory cell array vertically between semiconductor layer **1002** and bonding layer **1008**. The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with semiconductor layer **1002** (e.g., as shown in FIGS. **8A-8C**).

Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate) or polysilicon (e.g., a deposited layer), for example, depending on the types of channel structures of the NAND memory strings (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**). Bonding layer **1008** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(466) In some implementations, fourth semiconductor structure **108** includes a semiconductor layer **3904**, a bonding layer **1010**, a first portion of the peripheral circuits of the memory cell array vertically between bonding layer **1010** and a first side of semiconductor layer **3904**, and a second portion of the peripheral circuits of the memory cell array in contact with a second side of

semiconductor layer **3904** opposite to the first side. That is, the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the first portion of the peripheral circuits and the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the second portion of the peripheral circuits can be in contact with opposite sides of semiconductor layer **3904**. Thus, the transistors of the two separate portions of the peripheral circuits are stacked over each other in different planes across semiconductor layer **3904**, according to some implementations. It is understood that in some examples, different from semiconductor layer **1002** in first semiconductor structure **102**, semiconductor layer **3904** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Through contacts (e.g., ILVs/TSVs) through semiconductor layer **3904** can make direct, short-distance (e.g., submicron-level) electrical connections between the two portions of the peripheral circuits on opposite sides of semiconductor layer **3904**.

(467) Similar to bonding layer **1008** in first semiconductor structure **102**, bonding layer **1010** in fourth semiconductor structure **108** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. That is, bonding layers **1008** and **1010** can be disposed on opposite sides of bonding interface **103**, and the bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **108**.

(468) Moreover, as shown in FIGS. **39A** and **39B**, 3D memory device **3900** or **3901** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **39A**, fourth semiconductor structure **108** including peripheral circuits may include pad-out interconnect layer **902**. In this example, 3D memory device **3900** may be pad-out from the peripheral circuit side to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **3900**. In another example shown in FIG. **39B**, first semiconductor structure **102** including memory cell array may include pad-out interconnect layer **902**.

(469) As shown in FIGS. **39A** and **39B**, 3D memory device **3900** or **3901** can include the memory cell array, a first peripheral circuit including a first transistor, a second peripheral circuit include a second transistor, a first semiconductor layer **3904** including a first side and a second side, and a second semiconductor layer **1002** including a third side and a fourth side. The memory cell array, the first transistor, and the second transistor can be in contact with three of the first, second, third, and fourth sides. The second and third sides can be disposed between the first and fourth sides, and the first transistor and the memory cell array can be in contact with the second and third sides, respectively. For example, as shown in FIGS. **39A** and **39B**, the memory cell array is in contact with the third side of second semiconductor layer **1002**, the first transistor is in contact with the second side of first semiconductor layer **3904**, and the second transistor is in contact with the first side of first semiconductor layer **3904**.

(470) Moreover, as described below in detail, semiconductor layer **3904** can be a single silicon substrate (e.g., a thinned double side silicon substrate), and the peripheral circuits in fourth semiconductor structure **108** can be formed on both sides (e.g., the front side and the backside) of the single silicon substrate, thereby reducing the device cost comparing with the architecture of using two silicon substrates and having the peripheral circuits formed on the front side of each silicon substrate.

(471) FIGS. **40A** and **40B** illustrate side views of various examples of 3D memory devices **3900** and **3901** in FIGS. **39A** and **39B**, according to various aspects of the present disclosure. As shown

in FIG. 40A, as one example of 3D memory devices **3900** and **3901** in FIGS. 39A and 39B, 3D memory device **4000** is a bonded chip including first semiconductor structure **102** and fourth semiconductor structure **108**, which are stacked over each another in different planes in the vertical direction (e.g., the y-direction in FIG. 40A), according to some implementations. First and fourth semiconductor structures **102** and **108** are bonded at bonding interface **103** therebetween, and fourth semiconductor structure **108** includes two separate device layers **4002** and **4014** on opposite sides thereof in the vertical direction (e.g., the y-direction in FIG. 40A), according to some implementations.

(472) As shown in FIG. 40A, fourth semiconductor structure **108** can include semiconductor layer **3904** having semiconductor materials. In some implementations, semiconductor layer **3904** is a silicon substrate having single crystalline silicon. Devices, such as transistors, can be formed on both sides of semiconductor layer **3904**. In some implementations, the thickness of semiconductor layer **3904** is between 1  $\mu\text{m}$  and 10  $\mu\text{m}$ . Fourth semiconductor structure **108** can also include a device layer **4002** above and in contact with a first side (e.g., toward the negative y-direction in FIG. 40A) of semiconductor layer **3904**. In some implementations, device layer **4002** includes a first peripheral circuit **4004** and a second peripheral circuit **4006**. First peripheral circuit **4004** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and second peripheral circuit **4006** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **4004** includes a plurality of transistors **4008** in contact with the first side of semiconductor layer **3904**, and second peripheral circuit **4006** includes a plurality of transistors **4010** in contact with the first side of semiconductor layer **1006**. Transistors **4008** and **4010** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **4008** or **4010** includes a gate dielectric, and the thickness of the gate dielectric of transistor **4008** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **4010** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **4008** than transistor **4010**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **4008** and **4010**) can be formed on the first side of semiconductor layer **3904** as well.

(473) In some implementations, fourth semiconductor structure **108** further includes an interconnect layer **4012** above device layer **4002** to transfer electrical signals to and from peripheral circuits **4006** and **4004**. As shown in FIG. 40A, device layer **4002** (including transistors **4008** and **4010** of peripheral circuits **4004** and **4006**) can be disposed vertically between semiconductor layer **3904** and interconnect layer **4012**. Interconnect layer **4012** can include a plurality of interconnects. The interconnects in interconnect layer **4012** can be coupled to transistors **4008** and **4010** of peripheral circuits **4004** and **4006** in device layer **4002**. Interconnect layer **4012** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **4012** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **4002** are coupled to one another through the interconnects in interconnect layer **4012**. For example, peripheral circuit **4004** may be coupled to peripheral circuit **4006** through interconnect layer **4012**. The interconnects in interconnect layer **4012** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4012** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(474) In some implementations, the interconnects in interconnect layer **4012** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **4012** can occur after the high-temperature processes in forming device layers **4014** and **4002** in

fourth semiconductor structure **108**, as well as being separated from the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **4012** having Cu can become feasible.

(475) Fourth semiconductor structure **108** can also include another device layer **4014** below and in contact with a second side (e.g., toward the positive y-direction in FIG. **40A**) of semiconductor layer **3904** opposite to the first side. Device layers **4014** and **4002** can thus be disposed in different planes in the vertical direction, i.e., stacked over one another on opposite sides of semiconductor layer **3904** in fourth semiconductor structure **108**. In some implementations, device layer **4014** includes a third peripheral circuit **4016** and a fourth peripheral circuit **4018**. Third peripheral circuit **4016** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and fourth peripheral circuit **4018** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **4016** includes a plurality of transistors **4020**, and fourth peripheral circuit **4018** includes a plurality of transistors **4022** as well. Transistors **4020** and **4022** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **4020** or **4022** includes a gate dielectric, and the thickness of the gate dielectric of transistor **4020** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **4022** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **4020** than transistor **4022**. In some implementations, the thickness of the gate dielectric of transistor **4020** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **4008** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **4020** than transistor **4008**. In some implementations, the thickness of the gate dielectric of transistor **4022** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **4010** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **4022** and transistor **4010**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1720** and **1722**) can be formed on the second side of semiconductor layer **3904** as well.

(476) As shown in FIG. **40A**, fourth semiconductor structure **108** can further include an interconnect layer **4026** below device layer **4014** to transfer electrical signals to and from peripheral circuits **4016** and **4018**. As shown in FIG. **40A**, interconnect layer **4026** can be vertically between bonding interface **103** and device layer **4014** (including transistors **4020** and **4022** of peripheral circuits **4016** and **4018**). Interconnect layer **4026** can include a plurality of interconnects coupled to transistors **4020** and **4022** of peripheral circuits **4016** and **4018** in device layer **4014**. Interconnect layer **4026** can further include one or more ILD layers in which the interconnects can form. That is, interconnect layer **4026** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **4014** are coupled to one another through the interconnects in interconnect layer **4026**. For example, peripheral circuit **4016** may be coupled to peripheral circuit **4018** through interconnect layer **4026**. The interconnects in interconnect layer **4026** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4026** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **4026** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(477) As shown in FIG. **40A**, fourth semiconductor structure **108** can further include one or more contacts **4024** extending vertically through semiconductor layer **3904**. In some implementations, contacts **4024** couples the interconnects in interconnect layer **4026** to the interconnects in interconnect layer **4012** to make an electrical connection between opposite sides of semiconductor layer **3904**. Contact **4024** can include conductive materials including, but not limited to, W, Co, Cu,

Al, silicides, or any combination thereof. In some implementations, contact **4024** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **3904**. Depending on the thickness of semiconductor layer **3904**, contact **4024** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu$ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu$ m and 100  $\mu$ m).

(478) As shown in FIG. **40A**, fourth semiconductor structure **108** can further include a bonding layer **1010** at bonding interface **103** and below and in contact with interconnect layer **4026**.

Bonding layer **1010** can include a plurality of bonding contacts **1011** and dielectrics electrically isolating bonding contacts **1011**. Bonding contacts **1011** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, bonding contacts **1011** of bonding layer **1010** include Cu. The remaining area of bonding layer **1010** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1011** and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously.

(479) As shown in FIG. **40A**, first semiconductor structure **102** can also include a bonding layer **1008** at bonding interface **103**, e.g., on the opposite side of bonding interface **103** with respect to bonding layer **1010** in fourth semiconductor structure **108**. Bonding layer **1008** can include a plurality of bonding contacts **1009** and dielectrics electrically isolating bonding contacts **1009**. Bonding contacts **1009** can include conductive materials, such as Cu. The remaining area of bonding layer **1008** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **1009** and surrounding dielectrics in bonding layer **1008** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1010** of second semiconductor structure **104** and the bottom surface of bonding layer **1008** of first semiconductor structure **102**.

(480) As shown in FIG. **40A**, first semiconductor structure **102** can further include an interconnect layer **4028** above and in contact with bonding layer **1008** to transfer electrical signals. Interconnect layer **4028** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **4028** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **4028** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **4028** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4028** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(481) As shown in FIG. **40A**, first semiconductor structure **102** can further include a memory cell array, such as an array of NAND memory strings **208** below and in contact with interconnect layer **4028**. In some implementations, interconnect layer **4028** is vertically between NAND memory strings **208** and bonding interface **103**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **4027**. Memory stack **4027** may be an example of memory stack **804** in FIGS. **8A-8C**, and the conductive layer and dielectric layer in memory stack **4027** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **4027**. The interleaved conductive layers and dielectric layers in memory stack **4027** alternate in the vertical direction, according to some implementations. Each

conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **4027**.

(482) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(483) As shown in FIG. **40A**, first semiconductor structure **102** can further include semiconductor layer **1002** disposed below memory stack **4027** and in contact with the sources of NAND memory strings **208**. In some implementations, NAND memory strings **208** are disposed vertically between bonding interface **103** and semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **4027** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in semiconductor layer **1002** as well.

(484) As shown in FIG. **40A**, fourth semiconductor structure **108** can further include a pad-out interconnect layer **902** above and in contact with interconnect layer **4012**. In some implementations, device layer **4002** having transistors **4008** and **4010** is disposed vertically between pad-out interconnect layer **902** and semiconductor layer **3904**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **4032**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **4012** can be formed on the same side of semiconductor layer **3904**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **3900** and external devices, e.g., for pad-out purposes.

(485) As a result, peripheral circuits **4004**, **4006**, **4016**, and **4018** on different sides of fourth semiconductor structure **108** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **4012**, **4026**, and **4028**, bonding layers **1008** and **1010**, as well as contacts **4024**. Moreover, peripheral circuits **4004**, **4006**, **4016**, and **4018** and NAND memory strings **208** in 3D memory device **3900** can be further coupled to external devices through pad-out interconnect layer **902**.

(486) It is understood that the pad-out of 3D memory devices is not limited to from fourth semiconductor structure **108** having transistors **4008**, **4010**, **4020**, and **4022** as shown in FIG. **40A** (corresponding to FIG. **39A**) and may be from first semiconductor structure **102** having NAND memory strings **208** (corresponding to FIG. **39B**). For example, as shown in FIG. **40B**, 3D memory device **4001** may include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1002** of first semiconductor structure **102** on which NAND memory strings **208** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **4030** extending vertically through semiconductor layer **1002**. In some implementations, contact **4030** couples the interconnects in interconnect layer **4028** in first semiconductor structure **102** to contact pads **4032** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **4030** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **4030** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **4030** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ). In some

implementations, in FIG. 40B, fourth semiconductor structure **108** of 3D memory device **4001** further includes a passivation layer **4034**, replacing pad-out interconnect layer **902** in FIG. 40A. Passivation layer **4034** can include dielectric materials, such as silicon nitride and/or silicon oxide. (487) It is also understood that the material of semiconductor layer **1002** in first semiconductor structure **102** is not limited to single crystalline silicon as described above with respect to FIG. 40A and may be any other suitable semiconductor materials. For example, as shown in FIG. 40B, 3D memory device **4001** may include semiconductor layer **1002** having polysilicon in first semiconductor structure **102**. NAND memory strings **208** of 3D memory device **4001** in contact with semiconductor layer **1002** having polysilicon can include any suitable channel structures disclosed herein that are in contact with a polysilicon layer, such as bottom open channel structure **812C**. In some implementations, NAND memory strings **208** of 3D memory device **4001** are “floating gate” type of NAND memory strings, and semiconductor layer **1002** having polysilicon is in contact with the “floating gate” type of NAND memory strings as the source plate thereof. It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **4000** and **4001** are not repeated for case of description.

(488) FIGS. 41A-41E illustrate a fabrication process for forming the 3D memory devices in FIGS. 39A and 39B, according to some aspects of the present disclosure. FIGS. 42A-42I illustrate another fabrication process for forming the 3D memory devices in FIGS. 39A and 39B, according to some aspects of the present disclosure. FIG. 43 illustrates a flowchart of a method **4300** for forming the 3D memory devices in FIGS. 39A and 39B, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. 41A-41E, 42A-42F, and 43 include 3D memory devices **4000** and **4001** depicted in FIGS. 40A and 40B. FIGS. 41A-41E, 42A-42I, and 43 will be described together. It is understood that the operations shown in method **4300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. 43. For example, operation **4302** and **4304** may be performed in parallel.

(489) Referring to FIG. 43, method **4300** starts at operation **4302**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(490) As illustrated in FIGS. 41A and 42E, a stack structure, such as a memory stack **4104** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **4102**. To form memory stack **4104**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **4102**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **4104** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **4104** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **4104** and silicon substrate **4102**.

(491) As illustrated in FIGS. 41A and 42E, NAND memory strings **4106** are formed above silicon substrate **4102**, each of which extends vertically through memory stack **4104** to be in contact with silicon substrate **4102**. In some implementations, fabrication processes to form NAND memory string **4106** include forming a channel hole through memory stack **4104** (or the dielectric stack) and



into silicon substrate **4102** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **4106** may vary depending on the types of channel structures of NAND memory strings **4106** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for ease of description.

(492) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIGS. **41A** and **42E**, an interconnect layer **4108** is formed above memory stack **4104** and NAND memory strings **4106**. Interconnect layer **4108** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **4106**. In some implementations, interconnect layer **4108** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4108** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIGS. **41A** and **42E** can be collectively referred to as interconnect layer **4108**.

(493) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIGS. **41A** and **42E**, a bonding layer **4110** is formed above interconnect layer **4108**. Bonding layer **4110** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **4108** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **4108** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(494) Method **4300** proceeds to operation **4304**, as illustrated in FIG. **43**, in which a first transistor is formed on a first side of a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIGS. **41B** and **42A**, a plurality of transistors **4114** and **4116** are formed on one side of a silicon substrate **4112**. Transistors **4114** and **4116** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **4112** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **4114** and **4116**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **4112** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **4114** is different from the thickness of gate dielectric of transistor **4116**, for example, by depositing a thicker silicon oxide film in the region of transistor **4114** than the region of transistor **4116**, or by etching back part of the silicon oxide film deposited in the region of transistor **4116**. It is understood that the details of fabricating transistors **4114** and **4116** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(495) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIGS. **41B** and **42A**, an interconnect layer **4118** can be formed above transistors **4114** and **4116**. Interconnect layer **4118** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **4114** and **4116**. In some implementations, interconnect layer **4118** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4118** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIGS. **41B** and **42A** can be collectively referred to as interconnect layer **4118**. In some implementations, the interconnects in interconnect layer **4118** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(496) In some implementations, a second bonding layer is formed above the interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIGS. **41B** and **42A**, a bonding layer **4120** is formed above interconnect layer **4118**. Bonding layer **4120** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **4118** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **4118** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(497) Method **4300** proceeds to operation **4306**, as illustrated in FIG. **43**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(498) As illustrated in FIG. **41C**, silicon substrate **4102** and components formed thereon (e.g., memory stack **4104** and NAND memory strings **4106**) are flipped upside down. Bonding layer **4110** facing down is bonded with bonding layer **4120** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **4132**. That is, silicon substrate **4102** and components formed thereon can be bonded with silicon substrate **4112** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **4110** are in contact with the bonding contacts in bonding layer **4120** at bonding interface **4132**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **41C**, it is understood that in some examples, silicon substrate **4112** and components formed thereon (e.g., transistors **4114** and **4116**) can be flipped upside down, and bonding layer **4120** facing down can be bonded with bonding layer **4110** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **4132** as well.

(499) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **4132** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **4110** and the bonding contacts in bonding layer **4120** are aligned and in contact with one another, such that memory stack **4104** and NAND memory strings **4106** formed therethrough can be coupled to transistors **4114** and **4116** through the bonded bonding contacts across bonding

interface **4132**, according to some implementations.

(500) In some implementations, the second substrate is thinned after the bonding from the second side opposite to the first side. As illustrated in FIG. **41D**, silicon substrate **4112** (shown in FIG. **41C**) is thinned from another side opposite to the side on which transistors **4114** and **4116** are formed to become a semiconductor layer **4113** having single crystalline silicon. Silicon substrate **4112** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(501) Method **4300** proceeds to operation **4308**, as illustrated in FIG. **43**, in which a second transistor is formed on a second side of the second substrate opposite to the first side. As illustrated in FIG. **41D**, a plurality of transistors **4124** and **4126** are formed on the other side of thinned silicon substrate **4112** (i.e., semiconductor layer **4113**) opposite to the side on which transistors **4114** and **4116** are formed. Transistors **4124** and **4126** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed on the other side of semiconductor layer **4113** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **4124** and **4126**. In some implementations, isolation regions (e.g., STIs) are also formed on the other side of semiconductor layer **4113** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **4124** is different from the thickness of gate dielectric of transistor **4126**, for example, by depositing a thicker silicon oxide film in the region of transistor **4124** than the region of transistor **4126**, or by etching back part of the silicon oxide film deposited in the region of transistor **4126**. It is understood that the details of fabricating transistors **4124** and **4126** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(502) In some implementations, an interconnect layer **4128** is formed above the transistor. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **41D**, an interconnect layer **4128** can be formed above transistors **4124** and **4126**.

Interconnect layer **4128** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **4124** and **4126**. In some implementations, interconnect layer **4128** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4128** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **41D** can be collectively referred to as interconnect layer **4128**.

(503) Different from interconnect layer **4118**, in some implementations, the interconnects in interconnect layer **4128** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **4128** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **4128**.

(504) In some implementations, a contact through the thinned second substrate is formed. As illustrated in FIG. **41D**, one or more contacts **4136** each extending vertically through semiconductor layer **4113** (i.e., the thinned silicon substrate **4112**) are formed. Contacts **4136** can couple the interconnects in interconnect layer **4118** and the interconnects in interconnect layer **4128**. Contact **4136** can be formed by first patterning contact holes through semiconductor layer **4113** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the

dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(505) Method **4300** skips optional operation **4310** and proceeds to operation **4312**, as illustrated in FIG. **43**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **41E**, a pad-out interconnect layer **4140** is formed above interconnect layer **4128** and transistors **4126** and **4124** on semiconductor layer **4113**. Pad-out interconnect layer **4140** can include interconnects, such as contact pads **4138**, formed in one or more ILD layers. Contact pads **4138** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. It is understood that although not shown in FIG. **41E**, in some examples, silicon substrate **4102** may be thinned, and pad-out interconnect layer **4140** may be formed on thinned silicon substrate **4102**, instead of above transistors **4124** and **4126**.

(506) It is understood that in some examples, the sequence of operation **4306** and **4308** in method **4300** may be switched. In some implementations, after operation **4304**, method **4300** skips operation **4306** and proceeds to operation **4308**, as illustrated in FIG. **43**, in which a second transistor is formed on a second side of the second substrate opposite to the first side.

(507) In some implementations, the second substrate is thinned before the bonding from the second side opposite to the first side. As illustrated in FIG. **42C**, silicon substrate **4112** (shown in FIG. **42B**) is thinned from another side opposite to the side on which transistors **4114** and **4116** are formed to become semiconductor layer **4113** having single crystalline silicon. Silicon substrate **4112** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, as illustrated in FIG. **42B**, a handle substrate **4201** is attached to bonding layer **4120**, for example, using adhesive bonding, to allow the subsequent backside processes on silicon substrates **4112**, such as thinning, contact formation, and bonding.

(508) As illustrated in FIG. **42D**, transistors **4124** and **4126** are formed on the other side of thinned silicon substrate **4112** (i.e., semiconductor layer **4113**) opposite to the side on which transistors **4114** and **4116** are formed. Transistors **4124** and **4126** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed on the other side of semiconductor layer **4113** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **4124** and **4126**. In some implementations, isolation regions (e.g., STIs) are also formed on the other side of semiconductor layer **4113** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **4124** is different from the thickness of gate dielectric of transistor **4126**, for example, by depositing a thicker silicon oxide film in the region of transistor **4124** than the region of transistor **4126**, or by etching back part of the silicon oxide film deposited in the region of transistor **4126**. It is understood that the details of fabricating transistors **4124** and **4126** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(509) In some implementations, an interconnect layer **4128** is formed above the transistor. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **42D**, an interconnect layer **4128** can be formed above transistors **4124** and **4126**.

Interconnect layer **4128** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **4124** and **4126**. In some implementations, interconnect layer **4128** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4128** can include conductive

materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 42D can be collectively referred to as interconnect layer **4128**.

(510) Different from interconnect layer **4118**, in some implementations, the interconnects in interconnect layer **4128** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **4128** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **4128**.

(511) In some implementations, a contact through the thinned second substrate is formed. As illustrated in FIG. 42D, one or more contacts **4136** each extending vertically through semiconductor layer **4113** (i.e., the thinned silicon substrate **4112**) are formed after thinning silicon substrate **3112**. Contacts **4136** can couple the interconnects in interconnect layer **4118** and the interconnects in interconnect layer **4128**. Contact **4136** can be formed after thinning silicon substrate **3112** by first patterning contact holes through semiconductor layer **4113** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor. It is understood that in some examples, contacts **4136** may be formed in silicon substrate **4112** before thinning (i.e., before the formation of semiconductor layer **4113**, e.g., in FIG. 42B) without fully penetrating through silicon substrate **4112** and be exposed from the backside of silicon substrate **4112** (where the thinning occurs) after the thinning. In some examples, the contact hole and the spacer of contact **4136** may be sequentially formed in silicon substrate **4112** before thinning and may be thinned along with silicon substrate **4112** by the thinning process. The conductor of contact **4136** then may be formed through the thinned spacer after the thinning process.

(512) After operation **4308**, method **4300** returns to operation **4306**, as illustrated in FIG. 43, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(513) As illustrated in FIG. 42D, handle substrate **4201** (shown in FIG. 42C) is removed to expose bonding layer **4120**. In some implementations, another substrate (not shown) is attached to interconnect layer **4128** to provide support for the subsequent bonding process. As illustrated in FIG. 42E, silicon substrate **4102** and components formed thereon (e.g., memory stack **4104** and NAND memory strings **4106**) are flipped upside down. Bonding layer **4110** facing down is bonded with bonding layer **4120** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **4132**. That is, silicon substrate **4102** and components formed thereon can be bonded with the first side (on which transistors **4114** and **4116** are formed) of thinned silicon substrate **4112** (semiconductor layer **4113**) and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **4110** are in contact with the bonding contacts in bonding layer **4120** at bonding interface **4132**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. 42E, it is understood that in some examples, thinned silicon substrate **4112** and components formed thereon (e.g., transistors **4114**, **4116**, **4124**, and **4126**) can be flipped upside down, and bonding layer **4120** facing down can be bonded with bonding layer **4110** facing

up, i.e., in a face-to-face manner, thereby forming bonding interface **4132** as well.

(514) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **4132** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **4110** and the bonding contacts in bonding layer **4120** are aligned and in contact with one another, such that memory stack **4104** and NAND memory strings **4106** formed therethrough can be coupled to transistors **4114**, **4116**, **4124**, and **4126** through the bonded bonding contacts across bonding interface **4132**, according to some implementations. As illustrated in FIG. **42E**, in some implementations, after the bonding, a passivation layer **4242** is formed on interconnect layer **4128** by depositing dielectric materials, such as silicon nitride, using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof.

(515) Method **4300** proceeds to optional operation **4310**, as illustrated in FIG. **43**, in which the first substrate is thinned. As illustrated in FIG. **42F**, silicon substrate **4102** (shown in FIG. **42E**) is thinned to become a semiconductor layer **4235** having single crystalline silicon. Silicon substrate **4102** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(516) Method **4300** proceeds to operation **4312**, as illustrated in FIG. **43**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. As illustrated in FIG. **42F**, a pad-out interconnect layer **4208** is formed on semiconductor layer **4235** (the thinned silicon substrate **4102**). Pad-out interconnect layer **4208** can include interconnects, such as contact pads **4238**, formed in one or more ILD layers. Contact pads **4238** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **4244** are formed, extending vertically through semiconductor layer **4235**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **4244** can couple contact pads **4238** in pad-out interconnect layer **4208** to the interconnects in interconnect layer **4108**. It is understood that in some examples, contacts **4244** may be formed in silicon substrate **4102** before thinning (the formation of semiconductor layer **4235**, e.g., in FIG. **42E**) and be exposed from the backside of silicon substrate **4102** (where the thinning occurs) after the thinning.

(517) It is understood that in some examples, the first substrate (e.g., silicon substrate **4102** or semiconductor layer **4235** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**.

(518) After operation **4308**, as the first and second transistors are formed on both sides of the second substrate, respectively, the first substrate can be bonded with either the first side or the second side of the second substrate at operation **4306**. FIGS. **42D-42F** show a process in which the first substrate is bonded with the first side of the second substrate on which the first transistor is formed, e.g., bonding first substrate **4102** and components thereon (e.g., NAND memory strings **4106**) to one side of thinned second substrate **4112** (i.e., semiconductor layer **4113**) on which transistors **4114** and **4116** are formed. In some implementations, the first substrate is bonded with the second side of the second substrate on which the second transistor is formed.

(519) To bond the first substrate with the second side of the second substrate, in some implementations, the second bonding layer is formed above the interconnect layer above the second transistor, as opposed to the interconnect layer above the first transistor. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **42G**, bonding layer **4120** (e.g., shown in FIG. **42C**) is not formed above interconnect layer **4118**, and handle substrate **4201** is attached onto interconnect layer **4118**, as opposed to bonding layer **4120**. Instead, a bonding layer **4211** is formed above interconnect layer **4128**. Bonding layer **4211** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited

on the top surface of interconnect layer **4128** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **4128** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(520) As shown in FIGS. **42B**, **42C**, and **42G**, in some implementations, handle substrate **4201** is bonded to interconnect layer **4118** before thinning silicon substrate **4112** and forming transistors **4124** and **4126** and interconnect layer **4128** and bonding layer **4211** on the backside of thinned silicon substrate **4112**. That is, handle substrate **4201** can remain being bonded to interconnect layer **4118** without being removed and introducing another handle substrate **4201** on the opposite side of semiconductor layer **4113** (i.e., thinned silicon substrate **4112**), thereby simplifying the fabrication process and reducing the production cost.

(521) In some implementations, the thickness of the gate dielectric of transistor **4114** is larger than the thickness of the gate dielectric of transistor **4126**. For example, transistor **4114** may be one example of the transistors forming HV circuits **406**, and transistor **4126** may be one example of the transistors forming LLV circuits **402**. That is, transistors **4114** of HV circuits **406** may be formed on the front side of silicon substrate **4112** before the formation of transistors **4126** of LLV circuits **402** on the backside of silicon substrate **4112**, which may reduce the impact of the formation of transistor **4114** on transistor **4126** in a reversed order, thereby reducing the device defects of transistors **4126**.

(522) As illustrated in FIG. **42H**, silicon substrate **4102** and components formed thereon (e.g., memory stack **4104** and NAND memory strings **4106**) are flipped upside down. Bonding layer **4110** facing down is bonded with bonding layer **4211** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **4233**. That is, silicon substrate **4102** and components formed thereon can be bonded with the second side (on which transistors **4124** and **4126** are formed) of thinned silicon substrate **4112** (semiconductor layer **4113**) and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **4110** are in contact with the bonding contacts in bonding layer **4211** at bonding interface **4233**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **42H**, it is understood that in some examples, thinned silicon substrate **4112** and components formed thereon (e.g., transistors **4114**, **4116**, **4124**, and **4126**) can be flipped upside down, and bonding layer **4211** facing down can be bonded with bonding layer **4110** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **4233** as well.

(523) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **4233** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **4110** and the bonding contacts in bonding layer **4211** are aligned and in contact with one another, such that memory stack **4104** and NAND memory strings **4106** formed therethrough can be coupled to transistors **4114**, **4116**, **4124**, and **4126** through the bonded bonding contacts across bonding interface **4233**, according to some implementations.

(524) As illustrated in FIG. **42I**, silicon substrate **4102** (shown in FIG. **42H**) is thinned to become semiconductor layer **4235** having single crystalline silicon. Silicon substrate **4102** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(525) As illustrated in FIG. **42I**, pad-out interconnect layer **4208** is formed on semiconductor layer **4235** (the thinned silicon substrate **4102**). Pad-out interconnect layer **4208** can include interconnects, such as contact pads **4238**, formed in one or more ILD layers. In some

implementations, after the bonding and thinning, contacts **4244** are formed, extending vertically through semiconductor layer **4235**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **4244** can couple contact pads **4238** in pad-out interconnect layer **4208** to the interconnects in interconnect layer **4108**. It is understood that in some examples, contacts **4244** may be formed in silicon substrate **4102** before thinning (the formation of semiconductor layer **4235**, e.g., in FIG. **42E**) and be exposed from the backside of silicon substrate **4102** (where the thinning occurs) after the thinning. (526) It is understood that in some examples, the first substrate (e.g., silicon substrate **4102** or semiconductor layer **4235** after thinning) may be removed and replaced with a semiconductor layer having polysilicon in a similar manner as described above with respect to FIGS. **12G** and **12H**. (527) FIGS. **44A** and **44B** illustrate schematic views of cross-sections of 3D memory devices **4400** and **4401** having two stacked semiconductor structures **104** and **110**, according to some aspects of the present disclosure. 3D memory devices **4400** and **4401** may be examples of 3D memory device **121** in FIG. **1D** in which second semiconductor structure **104** including some of the peripheral circuits is bonded to a fifth semiconductor structure **110** including a memory cell array and some of the peripheral circuits of the memory cell array disposed in different planes. In other words, as shown in FIGS. **44A** and **44B**, the memory cell array in fifth semiconductor structure **110** is disposed in the intermediate of 3D memory devices **4400** and **4401** in the vertical direction, according to some implementations.

(528) In some implementations, second semiconductor structure **104** includes a semiconductor layer **1004**, a bonding layer **1010**, and some of the peripheral circuits vertically between semiconductor layer **1004** and bonding layer **1010**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits in second semiconductor structure **104** can be in contact with semiconductor layer **1004**. Semiconductor layer **1004** can include semiconductor materials. In some implementations, semiconductor layer **1004** on which the transistors are formed includes single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Bonding layer **1010** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(529) In some implementations, fifth semiconductor structure **110** includes a pad-out interconnect layer **902**, a semiconductor layer **4404**, a bonding layer **4406**, a memory cell array vertically between bonding layer **4406** and a first side of semiconductor layer **4404**, and some of the peripheral circuits of the memory cell array vertically between pad-out interconnect layer **902** and a second side of semiconductor layer **4404** opposite to the first side. That is, the transistors of some of the peripheral circuits and the memory cell array can be in contact with opposite sides of semiconductor layer **4404**. Thus, the transistors of the two separate portions of the peripheral circuits are stacked over each other in different planes and separated by the memory cell array in the vertical direction, according to some implementations.

(530) The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with the first side of semiconductor layer **4404** (e.g., as shown in FIGS. **8A-8C**). The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits in fifth semiconductor structure **110** can be in contact with the second side of semiconductor layer **1004**. Semiconductor layer **4404** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). In some implementations, semiconductor layer **4404** on which both the transistors and the memory cell array are formed includes single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance.

(531) Similar to bonding layer **1010** in second semiconductor structure **104**, bonding layer **4406** in fifth semiconductor structure **110** can also include conductive bonding contacts (not shown) and



dielectrics electrically isolating the bonding contacts. A bonding interface **4403** is vertically between and in contact with bonding layers **1010** and **4406**, respectively, according to some implementations. That is, bonding layers **1010** and **4406** can be disposed on opposite sides of bonding interface **4403**, and the bonding contacts of bonding layer **4406** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **4403**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **104** and **110**.

(532) As shown in FIGS. **44A** and **44B**, 3D memory devices **4400** and **4401** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **44A**, fifth semiconductor structure **110** including some of the peripheral circuits may include pad-out interconnect layer **902**. In another example shown in FIG. **44B**, second semiconductor structure **104** including some of the peripheral circuits may include pad-out interconnect layer **902**. In either example, 3D memory device **4400** or **4401** may be pad-out from one of the peripheral circuit sides to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **4400** or **4401**.

(533) As shown in FIGS. **44A** and **44B**, 3D memory device **4400** or **4401** can include the memory cell array, a first peripheral circuit including a first transistor, a second peripheral circuit include a second transistor, a first semiconductor layer **1004** including a first side and a second side, and a second semiconductor layer **4404** including a third side and a fourth side. The memory cell array, the first transistor, and the second transistor can be in contact with three of the first, second, third and fourth sides. The second and third sides can be disposed between the first and fourth sides, and the first transistor and the memory cell array can be in contact with the second and third sides, respectively. For example, as shown in FIGS. **44A** and **44B**, the memory cell array is in contact with the third side of second semiconductor layer **4404**, the first transistor is in contact with the second side of first semiconductor layer **1004**, and the second transistor is in contact with the fourth side of second semiconductor layer **4404**.

(534) FIGS. **45A** and **45B** illustrate side views of example of 3D memory devices **4400** and **4401** in FIGS. **44A** and **44B**, according to various aspects of the present disclosure. As shown in FIG. **45A**, as one example of 3D memory devices **4400** and **4401** in FIGS. **44A** and **44B**, 3D memory device **4500** is a bonded chip including second semiconductor structure **104** and fifth semiconductor structure **110**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **45A**), according to some implementations. Fifth and second semiconductor structures **110** and **104** are bonded at bonding interface **4403** therebetween, and fifth semiconductor structure **110** includes two device layers **4514** and a memory stack **4527** (and NAND memory strings **208** therethrough) on opposite sides thereof in the vertical direction (e.g., the y-direction in FIG. **45A**), according to some implementations.

(535) As shown in FIG. **45A**, second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a silicon substrate having single crystalline silicon. Second semiconductor structure **104** can also include a device layer **4502** above and in contact with semiconductor layer **1004**. In some implementations, device layer **4502** includes a first peripheral circuit **4504** and a second peripheral circuit **4506**. First peripheral circuit **4504** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **4506** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **4504** includes a plurality of transistors **4508** in contact with semiconductor layer **1004**, and second peripheral circuit **4506** includes a plurality of transistors **4510** in contact with semiconductor layer **1004**. Transistors **4508** and **4510** can include

any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **4508** and **4510** includes a gate dielectric, and the thickness of the gate dielectric of transistor **4508** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **4510** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **4508** than transistor **4510**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **4508** and **4510**) can be formed on or in semiconductor layer **1004** as well.

(536) In some implementations, second semiconductor structure **104** further includes an interconnect layer **4512** above device layer **4502** to transfer electrical signals to and from peripheral circuits **4506** and **4504**. As shown in FIG. **45A**, interconnect layer **4512** can be disposed vertically between bonding interface **4403** and device layer **4502** (including transistors **4508** and **4510** of peripheral circuits **4504** and **4506**). Interconnect layer **4512** can include a plurality of interconnects. The interconnects in interconnect layer **4512** can be coupled to transistors **4508** and **4510** of peripheral circuits **4504** and **4506** in device layer **4502**. Interconnect layer **4512** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **4512** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **4502** are coupled to one another through the interconnects in interconnect layer **4512**. For example, peripheral circuit **4504** may be coupled to peripheral circuit **4506** through interconnect layer **4512**. The interconnects in interconnect layer **4512** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4512** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(537) As shown in FIG. **45A**, second semiconductor structure **104** can further include a bonding layer **1010** at bonding interface **4403** and above and in contact with interconnect layer **4512**. Bonding layer **1010** can include a plurality of bonding contacts **1011** and dielectrics electrically isolating the bonding contacts. Bonding contacts **1011** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, bonding contacts **1011** of bonding layer **1010** include Cu. The remaining area of bonding layer **1010** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **1011** and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO<sub>2</sub>-to-SiO<sub>2</sub>) bonding simultaneously.

(538) As shown in FIG. **45A**, fifth semiconductor structure **110** can also include a bonding layer **4406** at bonding interface **4403**, e.g., on the opposite side of bonding interface **4403** with respect to bonding layer **1010** in second semiconductor structure **104**. Bonding layer **4406** can include a plurality of bonding contacts **4407** and dielectrics electrically isolating bonding contacts **4407**. Bonding contacts **4407** can include conductive materials, such as Cu. The remaining area of bonding layer **4406** can be formed with dielectric materials, such as silicon oxide. Bonding contacts **4407** and surrounding dielectrics in bonding layer **4406** can be used for hybrid bonding. In some implementations, bonding interface **4403** is the place at which bonding layers **4406** and **1010** are met and bonded. In practice, bonding interface **4403** can be a layer with a certain thickness that includes the top surface of bonding layer **1010** of second semiconductor structure **104** and the bottom surface of bonding layer **4406** of fifth semiconductor structure **110**.

(539) As shown in FIG. **45A**, fifth semiconductor structure **110** can further include an interconnect layer **4528** above and in contact with bonding layer **4406** to transfer electrical signals. Interconnect layer **4528** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **4528** also include

local interconnects, such as bit line contacts and word line contacts. Interconnect layer **4528** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **4528** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4528** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(540) As shown in FIG. **45A**, fifth semiconductor structure **110** can further include a memory cell array, such as an array of NAND memory strings **208** above and in contact with interconnect layer **4528**. In some implementations, interconnect layer **4528** is vertically between NAND memory strings **208** and bonding interface **4403**. Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **4527**. Memory stack **4527** may be an example of memory stack **804** in FIGS. **8A-8C**, and the conductive layer and dielectric layer in memory stack **4527** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **4527** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **4527**.

(541) In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C**, described above in detail with respect to FIGS. **8A-8C**. It is understood that NAND memory strings **208** are not limited to the “charge trap” type of NAND memory strings and may be “floating gate” type of NAND memory strings in other examples.

(542) As shown in FIG. **45A**, fifth semiconductor structure **110** can further include semiconductor layer **4404** disposed above memory stack **4527** and in contact with the sources of NAND memory strings **208** on one side thereof. Semiconductor layer **1002** can include semiconductor materials. Devices, such as NAND memory strings **208** and transistors, can be formed on both sides of semiconductor layer **4404**. The sources of NAND memory strings **208** can be in contact with a first side (e.g., toward the negative y-direction in FIG. **45A**) of semiconductor layer **4404**. In some implementations, semiconductor layer **1002** is a thinned silicon substrate having single crystalline silicon on which memory stack **3627** and NAND memory strings **208** (e.g., including bottom plug channel structure **812A** or sidewall plug channel structure **812B**) are formed on the first side thereof. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed on one side of semiconductor layer **4404** as well.

(543) As shown in FIG. **45A**, fifth semiconductor structure **110** can also include another device layer **4514** above and in contact with a second side (e.g., toward the positive y-direction in FIG. **45A**) of semiconductor layer **4404** opposite to the first side. Device layer **4514** and memory stack **4527** and NAND memory strings **208** can thus be disposed in different planes in the vertical direction, i.e., stacked over one another on opposite sides of semiconductor layer **4404** in fifth semiconductor structure **110**. Further, device layers **4514** and **4502** can also be disposed in different planes in the vertical direction, i.e., stacked over one another, and separated by semiconductor layer **4404** and memory stack **4527** and NAND memory strings **208** in the vertical direction. In some implementations, device layer **4514** includes a first peripheral circuit **4516** and a second peripheral circuit **4518**. First peripheral circuit **4516** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and second peripheral circuit **4518** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **4516** includes a

plurality of transistors **4520** in contact with the second side of semiconductor layer **4404**, and second peripheral circuit **4518** includes a plurality of transistors **4522** in contact with the second side of semiconductor layer **4404**. Transistors **4520** and **4522** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **4520** or **4522** includes a gate dielectric, and the thickness of the gate dielectric of transistor **4520** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **4522** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **4520** than transistor **4522**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **4520** and **4522**) can be formed on the second side of semiconductor layer **3904** as well.

(544) Moreover, the different voltages applied to different transistors **4520**, **4522**, **4508**, and **4510** in fifth and second semiconductor structures **110** and **104** can lead to differences of device dimensions between fifth and second semiconductor structures **110** and **104**. In some implementations, the thickness of the gate dielectric of transistor **4508** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **4520** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **4508** than transistor **4520**. In some implementations, the thickness of the gate dielectric of transistor **4522** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **4510** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **4522** and transistor **4510**. In some implementations, the thickness of semiconductor layer **1004** in which transistor **4508** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **4404** in which transistor **4520** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **4508** than transistor **4520**.

(545) In some implementations, fifth semiconductor structure **110** further includes an interconnect layer **4526** above device layer **4514** to transfer electrical signals to and from peripheral circuits **4516** and **4518**. As shown in FIG. **45A**, device layer **4514** (including transistors **4520** and **4522** of peripheral circuits **4516** and **4518**) can be disposed vertically between semiconductor layer **4404** and interconnect layer **4526**. Interconnect layer **4526** can include a plurality of interconnects. The interconnects in interconnect layer **4012** can be coupled to transistors **4520** and **4522** of peripheral circuits **4518** and **4518** in device layer **4514**. Interconnect layer **4526** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **4526** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **4514** are coupled to one another through the interconnects in interconnect layer **4526**. For example, peripheral circuit **4516** may be coupled to peripheral circuit **4518** through interconnect layer **4526**. The interconnects in interconnect layer **4526** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **4526** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(546) In some implementations, the interconnects in interconnect layer **4526** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **4526** can occur after the high-temperature processes in forming device layer **4514** and memory stack **4527** and NAND memory strings **208** in fifth semiconductor structure **110**, as well as being separated from the high-temperature processes in forming second semiconductor structure **104**, the interconnects of interconnect layer **4526** having Cu can become feasible.

(547) As shown in FIG. **45A**, fifth semiconductor structure **110** can further include one or more contacts **4524** extending vertically through semiconductor layer **4404**. In some implementations, contacts **4524** couple the interconnects in interconnect layer **4526** and the interconnects in interconnect layer **4528**. Contact **4524** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **4524**

includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **4404**. Depending on the thickness of semiconductor layer **4404**, contact **4524** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1  $\mu\text{m}$ ), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1  $\mu\text{m}$  and 100  $\mu\text{m}$ ). (548) As shown in FIG. **45A**, fifth semiconductor structure **110** can further include a pad-out interconnect layer **902** above and in contact with interconnect layer **4526**. In some implementations, device layer **4514** having transistors **4520** and **4522** is disposed vertically between pad-out interconnect layer **902** and semiconductor layer **4404**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **4532**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **4526** can be formed on the same side of semiconductor layer **4404**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **4500** and external devices, e.g., for pad-out purposes.

(549) As a result, peripheral circuits **4516** and **4518** and NAND memory strings **208** on different sides of semiconductor layer **4404** in fifth semiconductor structure **110** can be coupled to peripheral circuits **4504** and **4506** in second semiconductor structure **104** through various interconnection structures, including interconnect layers **4512**, **4526**, and **4528**, bonding layers **1010** and **4406**, and contacts **4524**. Moreover, peripheral circuits **4504**, **4506**, **4516**, and **4518** and NAND memory strings **208** in 3D memory device **4500** can be further coupled to external devices through pad-out interconnect layer **902**.

(550) It is understood that the pad-out of 3D memory devices is not limited to from fifth semiconductor structure **110** having transistors **4520** and **4522** as shown in FIG. **45A** (corresponding to FIG. **44A**) and may be from second semiconductor structure **104** having transistors **4508** and **4510** (corresponding to FIG. **44B**) as described above in detail. It is also understood that in some examples, since transistors **4520** and **4522** are formed on semiconductor layer **4404**, semiconductor layer **4404** may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. In those examples, the channel structures of NAND memory string **208**, which are in contact with semiconductor layer **4404** as well, may include channel structures that are suitable to be formed on single crystalline silicon, but not polysilicon, such as bottom plug channel structure **812A** and sidewall plug channel structure **812B**, described above in detail with respect to FIGS. **8A** and **8B**.

(551) It is also understood that in some examples, a dielectric layer (e.g., silicon oxide layer) may be formed in semiconductor layer **4404**. For example, as shown in FIG. **45B**, semiconductor layer **4404** in a 3D memory device **4501** may include a dielectric layer **4550** (e.g., a silicon oxide layer). Dielectric layer **4550** can extend laterally and be disposed vertically between device layer **4514** and memory stack **4527** and NAND memory strings **208**, which can serve as a shielding layer between the components formed on opposite sides of semiconductor layer **4404**, for example, for reducing the impact across semiconductor layer **4404** on the threshold voltages of transistors **4520** and **4522** caused by memory stack **4527** and NAND memory strings **208**. As shown in FIG. **45B**, semiconductor layer **4404** may include multiple sublayers **4552** and **4554** on opposite sides of dielectric layer **4550**. In some implementations, sublayers **4552** and **4554** are two single crystalline silicon sublayers on opposite sides of dielectric layer **4550** (e.g., semiconductor layer **4404** being an SOI substrate). In some implementations, sublayers **4554** and **4552** are a single crystalline silicon sublayer and a polysilicon sublayer, respectively, on opposite sides of dielectric layer **4550** (e.g., by sequentially depositing a silicon oxide layer and a polysilicon layer on a silicon substrate or by transfer bonding). For example, sublayer **4554** may be a single crystalline silicon sublayer, sublayer **4552** may be a polysilicon sublayer, NAND memory strings **208** may be in contact with sublayer **4552**, and transistors **4520** and **4522** may be in contact with sublayer **4554**.

(552) FIGS. **46A-46G** illustrate a fabrication process for forming the 3D memory devices in FIGS.

**44A** and **44B**, according to some aspects of the present disclosure. FIG. **47** illustrates a flowchart of a method **4700** for forming the 3D memory devices in FIGS. **44A** and **44B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **46A-46G** and **47** include 3D memory devices **4500** and **4501** depicted in FIGS. **45A** and **45B**. FIGS. **46A-46G** and **47** will be described together. It is understood that the operations shown in method **4700** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **47**. For example, operation **4702** and **4704** may be performed in parallel.

(553) Referring to FIG. **47**, method **4700** starts at operation **4702**, in which an array of NAND memory strings is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the first substrate.

(554) As illustrated in FIG. **46A**, a stack structure, such as a memory stack **4604** including interleaved conductive layers and dielectric layers, is formed on a silicon substrate **4602**. To form memory stack **4604**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on silicon substrate **4602**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **4604** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **4604** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **4604** and silicon substrate **4102**.

(555) As illustrated in FIG. **41A**, NAND memory strings **4606** are formed above silicon substrate **4602**, each of which extends vertically through memory stack **4604** to be in contact with silicon substrate **4602**. In some implementations, fabrication processes to form NAND memory string **4606** include forming a channel hole through memory stack **4604** (or the dielectric stack) and into silicon substrate **4602** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **4606** may vary depending on the types of channel structures of NAND memory strings **4606** (e.g., bottom plug channel structure **812A**, sidewall plug channel structure **812B**, or bottom open channel structure **812C** in FIGS. **8A-8C**) and thus, are not elaborated for case of description.

(556) In some implementations, an interconnect layer is formed above the array of NAND memory strings on the first substrate. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **41A**, an interconnect layer **4608** is formed above memory stack **4604** and NAND memory strings **4606**. Interconnect layer **4608** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **4606**. In some implementations, interconnect layer **4608** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4608** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can

also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **46A** can be collectively referred to as interconnect layer **4608**.

(557) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **46A**, a bonding layer **4610** is formed above interconnect layer **4608**. Bonding layer **4610** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **4608** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **4608** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(558) Method **4700** proceeds to operation **4704**, as illustrated in FIG. **47**, in which a first transistor is formed on a first side of a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **46B**, a plurality of transistors **4614** and **4616** are formed on one side of a silicon substrate **4612**. Transistors **4614** and **4616** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **4612** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **4614** and **4616**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **4612** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **4614** is different from the thickness of gate dielectric of transistor **4616**, for example, by depositing a thicker silicon oxide film in the region of transistor **4614** than the region of transistor **4616**, or by etching back part of the silicon oxide film deposited in the region of transistor **4616**. It is understood that the details of fabricating transistors **4614** and **4616** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for case of description.

(559) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **41B**, an interconnect layer **4618** can be formed above transistors **4614** and **4616**. Interconnect layer **4618** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **4614** and **4616**. In some implementations, interconnect layer **4618** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4618** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **46B** can be collectively referred to as interconnect layer **4618**.

(560) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **46B**, a bonding layer **4620** is formed above interconnect layer **4618**. Bonding layer **4620** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a

dielectric layer is deposited on the top surface of interconnect layer **4618** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **4618** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(561) Method **4700** proceeds to operation **4706**, as illustrated in FIG. **47**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(562) As illustrated in FIG. **46C**, silicon substrate **4602** and components formed thereon (e.g., memory stack **4604** and NAND memory strings **4606**) are flipped upside down. Bonding layer **4610** facing down is bonded with bonding layer **4620** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **4632**. That is, silicon substrate **4602** and components formed thereon can be bonded with silicon substrate **4612** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **4610** are in contact with the bonding contacts in bonding layer **4620** at bonding interface **4632**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **46C**, it is understood that in some examples, silicon substrate **4612** and components formed thereon (e.g., transistors **4614** and **4616**) can be flipped upside down, and bonding layer **4620** facing down can be bonded with bonding layer **4610** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **4632** as well.

(563) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **4632** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **4610** and the bonding contacts in bonding layer **4620** are aligned and in contact with one another, such that memory stack **4604** and NAND memory strings **4606** formed therethrough can be coupled to transistors **4614** and **4616** through the bonded bonding contacts across bonding interface **4632**, according to some implementations.

(564) In some implementations, the first substrate is thinned after the bonding from the second side opposite to the first side. As illustrated in FIG. **46D**, silicon substrate **4602** (shown in FIG. **46C**) is thinned from another side opposite to the side on which transistors **4614** and **4616** are formed to become a semiconductor layer **4634** having single crystalline silicon. Silicon substrate **4602** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(565) Method **4700** proceeds to operation **4708**, as illustrated in FIG. **43**, in which a second transistor is formed on a second side of the first substrate opposite to the first side. As illustrated in FIG. **46D**, a plurality of transistors **4624** and **4626** are formed on the other side of thinned silicon substrate **4602** (i.e., semiconductor layer **4634**) opposite to the side on which transistors **4614** and **4616** are formed. Transistors **4624** and **4626** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed on the other side of semiconductor layer **4634** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **4624** and **4626**. In some implementations, isolation regions (e.g., STIs) are also formed on the other side of semiconductor layer **4634** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **4624** is different from the thickness of gate dielectric of transistor **4626**, for example, by depositing a thicker silicon oxide film in the region of transistor



**4624** than the region of transistor **4626**, or by etching back part of the silicon oxide film deposited in the region of transistor **4626**. It is understood that the details of fabricating transistors **4624** and **4626** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for case of description. (566) In some implementations, an interconnect layer is formed above the transistor. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 46D, an interconnect layer **4642** can be formed above transistors **4624** and **4626**. Interconnect layer **4642** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **4624** and **4626**. In some implementations, interconnect layer **4642** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **4642** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 46D can be collectively referred to as interconnect layer **4642**.

(567) In some implementations, the interconnects in interconnect layer **4642** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **4642** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **4642**.

(568) In some implementations, a contact through the thinned first substrate is formed. As illustrated in FIG. 46D, one or more contacts **4636** each extending vertically through semiconductor layer **4634** (i.e., the thinned silicon substrate **4602**) are formed. Contacts **4636** can couple the interconnects in interconnect layer **4608** and the interconnects in interconnect layer **4642**. Contact **4636** can be formed by first patterning contact holes through semiconductor layer **4634** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(569) Method **4700** proceeds to operation **4710**, as illustrated in FIG. 47, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. 46E, a pad-out interconnect layer **4646** is formed above interconnect layer **4642** and transistors **4626** and **4624** on semiconductor layer **4634**. Pad-out interconnect layer **4646** can include interconnects, such as contact pads **4648**, formed in one or more ILD layers. Contact pads **4648** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(570) It is understood that in some examples, the sequence of operation **4706** and **4708** in method **4700** may be switched. In some implementations, after operation **4704**, method **4700** skips operation **4706** and proceeds to operation **4708**, as illustrated in FIG. 47, in which a second transistor is formed on a second side of the first substrate opposite to the first side. After operation **4708**, method **4700** returns to operation **4706**, as illustrated in FIG. 47, in which the first substrate and the second substrate are bonded in a face-to-face manner.

(571) In some implementations, to form 3D memory device **4501** in FIG. 45B, after bonding the first and second substrates in a face-to-face manner at operation **4706**, a semiconductor layer including a dielectric layer vertically between two semiconductor sublayers is formed to replace the

first substrate, such that at operation **4708**, the second transistor is formed on the semiconductor layer, as opposed to the first substrate. As illustrated in FIG. **46F**, silicon substrate **4602** (shown in FIG. **46C**) is replaced by a semiconductor layer **4660** having a first sublayer **4635**, a dielectric layer **4637**, and a second sublayer **4639**. In some implementations, sublayer **4635** is formed by thinning silicon substrate **4602** and thus have the same material as silicon substrate **4602**, i.e., single crystalline silicon. In some implementations, sublayer **4635** is formed by removing silicon substrate **4602** and depositing another layer of semiconductor material, such as polysilicon, in contact with the sources of NAND memory strings **4606**. Dielectric layer **4637** can be formed by depositing a layer of dielectric material, such as silicon oxide, on sublayer **4635** or by oxidizing part of sublayer **4635** (e.g., having single crystalline silicon). Sublayer **4639** can be formed on dielectric layer **4637** using transfer bonding as described above in detail. It is understood that in some examples, dielectric layer **4637** and sublayer **4639** may be transferred together and bonded onto sublayer **4635** by transfer bonding. As illustrated in FIG. **46G**, transistors **4624** and **4626** can be formed on sublayer **4639** of semiconductor layer **4660** using the similar processes as described above in detail. Contacts **4636** can be formed to extend vertically through sublayers **4639**, dielectric layer **4637**, and sublayer **4635** of semiconductor layer **4660** to be coupled to the interconnects of interconnect layer **4608**.

(572) FIG. **50** illustrates a block diagram of a system **5000** having a memory device, according to some aspects of the present disclosure. System **5000** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an augmented reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **50**, system **5000** can include a host **5008** and a memory system **5002** having one or more memory devices **5004** and a memory controller **5006**. Host **5008** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **5008** can be configured to send or receive the data to or from memory devices **5004**.

(573) Memory device **5004** can be any memory devices disclosed herein, such as 3D memory devices **100**, **101**, **120**, and **121**. In some implementations, each memory device **5004** includes an array of memory cells, a first peripheral circuit of the array of memory cells, and a second peripheral circuit of the array of memory cells, which are stacked over one another in different planes, as described above in detail.

(574) Memory controller **5006** is coupled to memory device **5004** and host **5008** and is configured to control memory device **5004**, according to some implementations. Memory controller **5006** can manage the data stored in memory device **5004** and communicate with host **5008**. In some implementations, memory controller **5006** is designed for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, memory controller **5006** is designed for operating in a high duty-cycle environment SSDs or embedded multi-media-cards (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. Memory controller **5006** can be configured to control operations of memory device **5004**, such as read, erase, and program operations. In some implementations, memory controller **5006** is configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit. Memory controller **5006** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **5004** including, but not limited to bad-block management, garbage collection, logical-to-physical address conversion, wear leveling, etc. In some implementations, memory controller **5006** is further configured to process error correction codes (ECCs) with respect to the data read from or written to memory device **5004**. Any other suitable functions may be performed by memory controller **5006**.

as well, for example, formatting memory device **5004**. Memory controller **5006** can communicate with an external device (e.g., host **5008**) according to a particular communication protocol. For example, memory controller **5006** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

(575) Memory controller **5006** and one or more memory devices **5004** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an cMMC package. That is, memory system **5002** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. **51A**, memory controller **5006** and a single memory device **5004** may be integrated into a memory card **5102**. Memory card **5102** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc. Memory card **5102** can further include a memory card connector **5104** coupling memory card **5102** with a host (e.g., host **5008** in FIG. **50**). In another example as shown in FIG. **51B**, memory controller **5006** and multiple memory devices **5004** may be integrated into an SSD **5106**. SSD **5106** can further include an SSD connector **5108** coupling SSD **5106** with a host (e.g., host **5008** in FIG. **50**). In some implementations, the storage capacity and/or the operation speed of SSD **5106** is greater than those of memory card **5102**.

(576) According to one aspect of the present disclosure, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes an array of memory cells and a first semiconductor layer in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a first peripheral circuit of the array of memory cells including a first transistor, and a second semiconductor layer in contact with the first transistor. A third semiconductor structure includes a second peripheral circuit of the array of memory cells including a second transistor, and a third semiconductor layer in contact with the second transistor. The second semiconductor layer is between the first bonding interface and the first peripheral circuit. The third semiconductor layer is between the second bonding interface and the second peripheral circuit.

(577) In some implementations, the first semiconductor layer includes single crystalline silicon.

(578) In some implementations a thickness of the second semiconductor layer is greater than a thickness of the third semiconductor layer.

(579) In some implementations, the first transistor includes a first gate dielectric, the second transistor includes a second gate dielectric, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(580) In some implementations, a difference between the thicknesses of the first and second gate dielectrics is at least 5-fold.

(581) In some implementations, the second semiconductor structure further includes a third peripheral circuit of the array of memory cells, and the third peripheral circuit includes a third transistor including a third gate dielectric. In some implementations, the third semiconductor structure further includes a fourth peripheral circuit of the array of memory cells, and the fourth peripheral circuit including a fourth transistor including a fourth gate dielectric. In some implementations, the third and fourth gate dielectrics have a same thickness.

(582) In some implementations, the thickness of the third and fourth gate dielectrics is between the

thicknesses of the first and second gate dielectrics.

(583) In some implementations, the third and fourth peripheral circuits include at least one of a page buffer circuit or a logic circuit.

(584) In some implementations, the second semiconductor structure further includes a first interconnect layer including a first interconnect coupled to the first transistor and between the second bonding interface and the first peripheral circuit. In some implementations, the third semiconductor structure further includes a second interconnect layer including a second interconnect coupled to the second transistor such that the second peripheral circuit is between the second interconnect layer and the third semiconductor layer.

(585) In some implementations, the second interconnect includes copper, and the first interconnect includes tungsten.

(586) In some implementations, the second semiconductor structure further includes a first contact through the second semiconductor layer, and the third semiconductor structure further includes a second contact through the third semiconductor layer and coupled to the first contact.

(587) In some implementations, the second contact includes copper, and the first contact includes tungsten.

(588) In some implementations, the first contact extends further through the first bonding interface, and the second contact extends further through the second bonding interface.

(589) In some implementations, the third semiconductor structure further includes a pad-out interconnect layer such that the second peripheral circuit is between the pad-out interconnect layer and the third semiconductor layer.

(590) In some implementations, the second peripheral circuit includes an I/O circuit, and the first peripheral circuit includes a driving circuit.

(591) In some implementations, the 3D memory device further includes a first voltage source coupled to the first peripheral circuit and configured to provide a first voltage to the first peripheral circuit, and a second voltage source coupled to the second peripheral circuit and configured to provide a second voltage to the second peripheral circuit. In some implementations, the first voltage is greater than the second voltage.

(592) In some implementations, the array of NAND memory strings is between the first bonding interface and the first semiconductor layer.

(593) According to another aspect of the present disclosure, a system includes a memory device configured to store data. The memory device includes a first semiconductor structure, a second semiconductor structure, a third semiconductor structure, a first bonding interface between the first semiconductor structure and the second semiconductor structure, and a second bonding interface between the second semiconductor structure and the third semiconductor structure. The first semiconductor structure includes an array of memory cells and a first semiconductor layer in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a first peripheral circuit of the array of memory cells including a first transistor, and a second semiconductor layer in contact with the first transistor. A third semiconductor structure includes a second peripheral circuit of the array of memory cells including a second transistor, and a third semiconductor layer in contact with the second transistor. The second semiconductor layer is between the first bonding interface and the first peripheral circuit. The third semiconductor layer is between the second bonding interface and the second peripheral circuit. The system also includes a memory controller coupled to the memory device and configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit.

(594) According to still another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. An array of NAND memory strings is formed on a first substrate. A first semiconductor layer is formed above the array of NAND memory strings. The first semiconductor layer includes single crystalline silicon. A first transistor is formed on the first semiconductor layer. A second semiconductor layer is formed above the first transistor. The second

semiconductor layer includes single crystalline silicon. A second transistor is formed on the second semiconductor layer.

(595) In some implementations, a pad-out interconnect layer is formed above the second transistor.

(596) In some implementations, the first substrate is thinned after forming the second transistor, and a pad-out interconnect layer is formed above the array of NAND memory strings.

(597) In some implementations, a first contact through the first semiconductor layer is formed before forming the second semiconductor layer, and a second contact through the second semiconductor layer is formed.

(598) In some implementations, to form the first semiconductor layer, a second substrate and the first substrate are bonded, and the second substrate is thinned to leave the first semiconductor layer.

(599) In some implementations, to form the second semiconductor layer, a third substrate and the first substrate are bonded, and the third substrate is thinned to leave the second semiconductor layer.

(600) In some implementations, bonding the second and first substrates and bonding the third and first substrates each includes transfers bonding.

(601) In some implementations, to form the first transistor, a first gate dielectric is formed, to form the second transistor, a second gate dielectric is formed, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(602) According to yet another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. An array of NAND memory strings is formed on a first substrate. A first transistor is formed on a second substrate. A second transistor is formed on a third substrate. The first substrate and second substrate are bonded in a face-to-back manner. The second substrate and the third substrate are bonded in a face-to-back manner.

(603) In some implementations, a pad-out interconnect layer is formed above the second transistor.

(604) In some implementations, the first substrate is thinned after bonding the first and second substrates and bonding the second and third substrates, and a pad-out interconnect layer is formed above the array of NAND memory strings.

(605) In some implementations, the second substrate is thinned before bonding the first and second substrates, and a first contact is formed through the thinned second substrate.

(606) In some implementations, the third substrate is thinned before bonding the second and third substrates, and a second contact is formed through the thinned third substrate such that the second contact is coupled to the first contact after bonding the thinned second and third substrates.

(607) In some implementations, to form the first transistor, a first gate dielectric is formed, to form the second transistor, a second gate dielectric is formed, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(608) The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.

(609) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

## Claims

1. A method for forming a three-dimensional (3D) memory device, comprising: forming an array of NAND memory strings on a first substrate; forming a first semiconductor layer above the array of NAND memory strings, wherein the first semiconductor layer comprises single crystalline silicon; forming a first transistor on the first semiconductor layer; forming a second semiconductor layer above the first transistor, wherein the second semiconductor layer comprises single crystalline

silicon; forming a second transistor on the second semiconductor layer; thinning the first substrate after forming the second transistor; and forming a pad-out interconnect layer above the array of NAND memory strings.

2. The method of claim 1, further comprising: forming a first contact through the first semiconductor layer before forming the second semiconductor layer; and forming a second contact through the second semiconductor layer.

3. The method of claim 1, wherein forming the first semiconductor layer comprises: bonding a second substrate and the first substrate; and thinning the second substrate to leave the first semiconductor layer.

4. The method of claim 3, wherein forming the second semiconductor layer comprises: bonding a third substrate and the first substrate; and thinning the third substrate to leave the second semiconductor layer.

5. The method of claim 4, wherein bonding the second and first substrates and bonding the third and first substrates each comprises transfers bonding.

6. The method of claim 1, wherein forming the first transistor comprises forming a first gate dielectric; and forming the second transistor comprises forming a second gate dielectric, wherein a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

7. The method of claim 6, wherein a difference between the thicknesses of the first and second gate dielectrics is at least 5-fold.

8. The method of claim 6, further comprising: forming a third transistor on the first semiconductor layer; and forming a fourth transistor on the second semiconductor layer, wherein forming the third transistor comprises forming a third gate dielectric, forming the fourth transistor comprises forming a fourth gate dielectric, and the third and fourth gate dielectrics have a same thickness.

9. The method of claim 8, wherein the thickness of the third and fourth gate dielectrics is between the thicknesses of the first and second gate dielectrics.

10. The method of claim 1, wherein a thickness of the first semiconductor layer is greater than a thickness of the second semiconductor layer.

11. The method of claim 1, further comprising forming contact pads in the pad-out interconnect layer.

12. The method of claim 11, wherein contact pads comprise copper.

13. A method for forming a three-dimensional (3D) memory device, comprising: forming an array of NAND memory strings on a first substrate; forming a first transistor on a second substrate; forming a second transistor on a third substrate; bonding the first substrate and second substrate in a face-to-back manner; bonding the second substrate and the third substrate in a face-to-back manner; thinning the first substrate after bonding the first and second substrates and bonding the second and third substrates; and forming a pad-out interconnect layer above the array of NAND memory strings.

14. The method of claim 13, further comprising: thinning the second substrate before bonding the first and second substrates; and forming a first contact through the thinned second substrate.

15. The method of claim 14, further comprising: thinning the third substrate before bonding the second and third substrates; and forming a second contact through the thinned third substrate such that the second contact is coupled to the first contact after bonding the thinned second and third substrates.

16. The method of claim 13, wherein forming the first transistor comprises forming a first gate dielectric; and forming the second transistor comprises forming a second gate dielectric, wherein a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

17. A method for forming a three-dimensional (3D) memory device, comprising: forming a first semiconductor structure including an array of NAND memory strings on a first substrate; forming a second semiconductor structure, comprising: forming a first peripheral circuit of the array of NAND memory strings on a second substrate, thinning the second substrate, and forming a first

contact through the thinned second substrate; forming a third semiconductor structure including a second peripheral circuit of the array of NAND memory strings on a third substrate; bonding the first semiconductor structure and second semiconductor structure, such that the second substrate is between the array of NAND memory strings and the first peripheral circuit; and bonding the second substrate and the third substrate, such that the third substrate is between the first peripheral circuit and the second peripheral circuit.

18. The method of claim 17, forming the third semiconductor structure further comprises: thinning the third substrate; and forming a second contact through the thinned third substrate.

19. The method of claim 18, further comprising: thinning the first substrate; and forming a pad-out interconnect layer on the thinned first substrate and coupled with the second peripheral circuit through the first contact and the second contact.

20. The method of claim 17, further comprising: forming a pad-out interconnect layer on the second peripheral circuit and coupled with the array of NAND memory strings through the first contact and the second contact.

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